

FX Direction - Strategy & Backtest Report

End date (cap): 2026-05-01

Last N observations: 1000

WF window (bars): 200

Train / test split: 70% / 30%

PCA components: 10

Probability threshold: 51%

Target modes: binary, ternary

Leads (days): 1-10 (10 values)

Pairs: EUR/USD, GBP/USD, USD/JPY, USD/CHF, AUD/USD, NZD/USD, USD/CAD, EUR/JPY, EUR/GBP, GBP/JPY

Cross-pair features: yes (all others merged)

GARCH features: yes

Walk-forward: rolling window with inner train/test split; OOS predictions pooled. Backtest: binary = long/short from two classes; ternary = long / flat / short from three classes. Threshold mode: trade only when max class probability meets the threshold (else flat). Rankings: Sharpe ratio on stitched OOS strategy returns.

Run summary

Successful runs: 200 / 200

EUR/USD (EUR_USD) - binary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | MLP neural net | Logit (statsmodels, PCA) | Probit (statsmodels, PCA) | Random forest

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.5048	0.353	0.0188	-0.1049	0.468	0.5061	0.8774	840
MLP neural net	0.4988	0.322	0.0173	-0.0943	0.434	0.5034	0.8726	840
Logit (statsmodels	0.4881	-0.130	-0.0072	-0.1133	-0.178	0.4881	0.9464	840
Probit (statsmodel	0.4893	-0.133	-0.0073	-0.1094	-0.182	0.4875	0.9500	840
Random forest	0.4917	-0.149	-0.0081	-0.1315	-0.205	0.4886	0.9381	840
Logistic (sklearn,	0.4929	-0.194	-0.0106	-0.1064	-0.264	0.4856	0.9488	840
Gradient boosting	0.4917	-0.397	-0.0221	-0.1380	-0.547	0.4916	0.9905	840
Ensemble (weighted	0.4786	-0.406	-0.0220	-0.1277	-0.556	0.4815	0.9667	840
Ensemble (mean pro	0.4738	-0.416	-0.0227	-0.1339	-0.573	0.4772	0.9655	840
Hist gradient boos	0.4750	-0.919	-0.0503	-0.2153	-1.242	0.4746	0.9857	840
XGBoost	0.4679	-0.988	-0.0539	-0.2150	-1.335	0.4692	0.9869	840

EUR/USD (EUR_USD) - binary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | Gradient boosting (sklearn) | Ensemble (weighted train acc) | Ensemble (mean proba) | SVM (RBF)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.5190	0.914	0.0725	-0.1726	1.465	0.5181	0.8869	840
Gradient boosting	0.5202	0.767	0.0643	-0.1556	1.099	0.5199	0.9869	840
Ensemble (weighted	0.5131	0.288	0.0234	-0.1781	0.402	0.5158	0.9786	840
Ensemble (mean pro	0.5083	0.058	0.0047	-0.2070	0.080	0.5087	0.9619	840
SVM (RBF)	0.5179	0.047	0.0037	-0.2011	0.062	0.5157	0.9464	840
Random forest	0.5083	-0.029	-0.0024	-0.2314	-0.040	0.5056	0.9512	840
XGBoost	0.5048	-0.166	-0.0134	-0.2477	-0.231	0.5036	0.9833	840
Probit (statsmodel	0.4988	-0.205	-0.0163	-0.1772	-0.283	0.5025	0.9595	840
Logit (statsmodels	0.4940	-0.301	-0.0239	-0.1772	-0.417	0.4982	0.9679	840
Logistic (sklearn,	0.5048	-0.390	-0.0306	-0.1904	-0.533	0.5061	0.9738	840
Hist gradient boos	0.4976	-0.459	-0.0365	-0.3257	-0.644	0.4952	0.9881	840

EUR/USD (EUR_USD) - binary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | Logit (statsmodels, PCA) | Probit (statsmodels, PCA) | Hist gradient boosting | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.5119	-0.388	-0.0393	-0.4472	-0.516	0.5102	0.9917	840
Logit (statsmodels	0.4952	-0.469	-0.0471	-0.3252	-0.634	0.4988	0.9714	840
Probit (statsmodel	0.4952	-0.589	-0.0582	-0.3463	-0.788	0.4963	0.9714	840
Hist gradient boos	0.4917	-0.636	-0.0636	-0.4536	-0.884	0.4898	0.9917	840
Logistic (sklearn,	0.4905	-0.695	-0.0679	-0.3524	-0.941	0.4907	0.9631	840
Ensemble (weighted	0.4940	-0.808	-0.0798	-0.4840	-1.070	0.4957	0.9702	840
Ensemble (mean pro	0.4964	-0.823	-0.0813	-0.4720	-1.097	0.4927	0.9762	840
XGBoost	0.4857	-0.955	-0.0940	-0.5113	-1.270	0.4868	0.9905	840
Random forest	0.4964	-0.975	-0.0953	-0.5103	-1.281	0.4963	0.9690	840
MLP neural net	0.4798	-1.148	-0.1075	-0.4709	-1.525	0.4813	0.8905	840
SVM (RBF)	0.4821	-1.233	-0.1188	-0.4662	-1.658	0.4809	0.9655	840

EUR/USD (EUR_USD) - binary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | MLP neural net | Gradient boosting (sklearn) | Logistic (sklearn, on global PCA) | Logit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.4905	-0.315	-0.0374	-0.3696	-0.481	0.4904	0.9952	840
MLP neural net	0.5024	-0.768	-0.0872	-0.4967	-1.049	0.5006	0.9345	840
Gradient boosting	0.4821	-0.801	-0.0922	-0.3959	-1.166	0.4803	0.9964	840
Logistic (sklearn,	0.4857	-0.824	-0.0940	-0.4683	-1.165	0.4842	0.9786	840
Logit (statsmodels	0.4810	-0.936	-0.1064	-0.4864	-1.321	0.4849	0.9845	840
Probit (statsmodel	0.4786	-0.985	-0.1115	-0.4871	-1.390	0.4825	0.9869	840
Ensemble (weighted	0.4881	-1.043	-0.1178	-0.5106	-1.470	0.4848	0.9774	840
Ensemble (mean pro	0.4798	-1.187	-0.1324	-0.5357	-1.662	0.4818	0.9786	840
XGBoost	0.4786	-1.238	-0.1389	-0.5199	-1.759	0.4791	0.9964	840
SVM (RBF)	0.4643	-1.304	-0.1453	-0.4761	-1.863	0.4648	0.9810	840
Random forest	0.4738	-1.319	-0.1459	-0.5219	-1.865	0.4732	0.9762	840

EUR/USD (EUR_USD) - binary, lead 5d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | Logit (statsmodels, PCA) | Probit (statsmodels, PCA) | MLP neural net | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.4881	-0.420	-0.0556	-0.4560	-0.643	0.4881	0.9976	840
Logit (statsmodels	0.5131	-0.645	-0.0837	-0.5197	-0.911	0.5103	0.9821	840
Probit (statsmodel	0.5095	-0.700	-0.0907	-0.5255	-0.994	0.5067	0.9845	840
MLP neural net	0.5214	-0.836	-0.1049	-0.5804	-1.086	0.5137	0.9107	840
Ensemble (mean pro	0.4881	-0.936	-0.1192	-0.5554	-1.311	0.4854	0.9786	840
Logistic (sklearn,	0.4976	-1.001	-0.1265	-0.5794	-1.430	0.4964	0.9857	840
Ensemble (weighted	0.4798	-1.014	-0.1285	-0.5615	-1.430	0.4807	0.9857	840
Gradient boosting	0.4679	-1.134	-0.1426	-0.5501	-1.603	0.4689	0.9952	840
SVM (RBF)	0.4893	-1.154	-0.1443	-0.5333	-1.634	0.4859	0.9726	840
Random forest	0.4774	-1.402	-0.1722	-0.5768	-1.960	0.4743	0.9738	840
XGBoost	0.4679	-1.542	-0.1888	-0.6347	-2.167	0.4647	0.9940	840

EUR/USD (EUR_USD) - binary, lead 6d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Probit (statsmodels, PCA) | Logit (statsmodels, PCA) | MLP neural net | Logistic (sklearn, on global PCA) | SVM (RBF)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Probit (statsmodel	0.5226	-0.146	-0.0214	-0.5350	-0.209	0.5241	0.9881	840
Logit (statsmodels	0.5214	-0.218	-0.0317	-0.5389	-0.313	0.5199	0.9869	840
MLP neural net	0.5357	-0.232	-0.0327	-0.6089	-0.297	0.5338	0.9167	840
Logistic (sklearn,	0.5048	-0.290	-0.0419	-0.5336	-0.423	0.5079	0.9821	840
SVM (RBF)	0.4976	-0.969	-0.1327	-0.5470	-1.384	0.4988	0.9738	840
Gradient boosting	0.4786	-1.205	-0.1623	-0.5993	-1.848	0.4760	0.9929	840
Ensemble (mean pro	0.4821	-1.306	-0.1755	-0.6608	-1.888	0.4837	0.9869	840
Ensemble (weighted	0.4821	-1.336	-0.1787	-0.6572	-1.932	0.4812	0.9821	840
Hist gradient boos	0.4679	-1.449	-0.1928	-0.6574	-2.151	0.4701	0.9952	840
Random forest	0.4702	-1.554	-0.2049	-0.6557	-2.241	0.4711	0.9881	840
XGBoost	0.4655	-1.572	-0.2072	-0.6993	-2.350	0.4659	0.9940	840

EUR/USD (EUR_USD) - binary, lead 7d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Probit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | Logit (statsmodels, PCA) | MLP neural net | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Probit (statsmodel	0.5202	0.261	0.0424	-0.5232	0.408	0.5207	0.9786	840
Logistic (sklearn,	0.5179	0.206	0.0334	-0.5034	0.319	0.5199	0.9869	840
Logit (statsmodels	0.5214	0.087	0.0140	-0.5231	0.135	0.5176	0.9821	840
MLP neural net	0.5083	-0.707	-0.1035	-0.5657	-0.903	0.5187	0.9226	840
Hist gradient boos	0.4810	-0.934	-0.1381	-0.6531	-1.462	0.4809	0.9952	840
Gradient boosting	0.4869	-0.948	-0.1400	-0.6259	-1.443	0.4874	0.9917	840
Ensemble (weighted	0.4905	-0.973	-0.1435	-0.6216	-1.442	0.4897	0.9869	840
Ensemble (mean pro	0.4905	-0.985	-0.1451	-0.6238	-1.461	0.4897	0.9869	840
SVM (RBF)	0.4750	-1.124	-0.1624	-0.6103	-1.572	0.4731	0.9738	840
XGBoost	0.4631	-1.238	-0.1785	-0.6818	-1.938	0.4639	0.9905	840
Random forest	0.4619	-1.479	-0.2078	-0.6508	-2.251	0.4584	0.9738	840

EUR/USD (EUR_USD) - binary, lead 8d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Probit (statsmodels, PCA) | Gradient boosting (sklearn) | Logistic (sklearn, on global PCA) | Hist gradient boosting | Logit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Probit (statsmodel	0.5321	0.938	0.1693	-0.4867	1.625	0.5351	0.9833	840
Gradient boosting	0.5119	0.828	0.1490	-0.3438	1.368	0.5131	0.9976	840
Logistic (sklearn,	0.5333	0.827	0.1481	-0.4305	1.399	0.5349	0.9881	840
Hist gradient boos	0.5167	0.677	0.1203	-0.4581	1.130	0.5179	0.9952	840
Logit (statsmodels	0.5286	0.627	0.1111	-0.4792	1.054	0.5306	0.9917	840
Ensemble (mean pro	0.5155	0.465	0.0806	-0.5007	0.786	0.5175	0.9845	840
XGBoost	0.5167	0.417	0.0719	-0.4345	0.716	0.5150	0.9940	840
Ensemble (weighted	0.5143	0.415	0.0711	-0.4887	0.698	0.5163	0.9869	840
MLP neural net	0.5095	0.348	0.0589	-0.4967	0.471	0.5206	0.9262	840
Random forest	0.4964	-0.100	-0.0165	-0.5583	-0.157	0.5012	0.9738	840
SVM (RBF)	0.5083	-0.387	-0.0624	-0.5841	-0.531	0.5073	0.9810	840

EUR/USD (EUR_USD) - binary, lead 9d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Probit (statsmodels, PCA) | Logit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | XGBoost | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Probit (statsmodel	0.5655	1.565	0.3127	-0.3457	2.711	0.5654	0.9917	840
Logit (statsmodels	0.5679	1.518	0.3019	-0.3379	2.634	0.5665	0.9940	840
Logistic (sklearn,	0.5488	1.062	0.2040	-0.3639	1.838	0.5495	0.9857	840
XGBoost	0.5119	0.785	0.1473	-0.3944	1.366	0.5119	0.9976	840
Ensemble (mean pro	0.5274	0.693	0.1281	-0.4309	1.244	0.5224	0.9845	840
Ensemble (weighted	0.5190	0.623	0.1147	-0.4233	1.090	0.5200	0.9798	840
Gradient boosting	0.5119	0.572	0.1055	-0.4543	1.005	0.5114	0.9940	840
Hist gradient boos	0.5048	0.257	0.0461	-0.4564	0.448	0.5036	0.9929	840
Random forest	0.5202	0.099	0.0173	-0.5143	0.154	0.5164	0.9798	840
SVM (RBF)	0.5464	-0.024	-0.0041	-0.5741	-0.031	0.5470	0.9881	840
MLP neural net	0.5190	-0.145	-0.0246	-0.4661	-0.209	0.5144	0.9095	840

EUR/USD (EUR_USD) - binary, lead 10d

Rows used: 1000 | Effective end: 2021-12-18

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Probit (statsmodels, PCA) | Logit (statsmodels, PCA) | Hist gradient boosting | Ensemble (mean proba) | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Probit (statsmodel	0.5310	1.066	0.2112	-0.3687	1.922	0.5322	0.9798	840
Logit (statsmodels	0.5286	0.946	0.1878	-0.3959	1.672	0.5283	0.9893	840
Hist gradient boos	0.4952	0.775	0.1518	-0.3896	1.432	0.4964	0.9976	840
Ensemble (mean pro	0.5214	0.728	0.1416	-0.4373	1.248	0.5176	0.9821	840
Logistic (sklearn,	0.5202	0.717	0.1396	-0.4740	1.315	0.5187	0.9893	840
Ensemble (weighted	0.5131	0.703	0.1364	-0.4320	1.205	0.5152	0.9774	840
XGBoost	0.5071	0.547	0.1050	-0.4716	0.902	0.5084	0.9952	840
Gradient boosting	0.5214	0.480	0.0915	-0.5578	0.781	0.5215	0.9952	840
SVM (RBF)	0.5417	0.402	0.0757	-0.5609	0.529	0.5412	0.9833	840
MLP neural net	0.5321	0.308	0.0562	-0.6143	0.404	0.5348	0.9571	840
Random forest	0.5095	0.112	0.0206	-0.5310	0.171	0.5067	0.9821	840

EUR/USD (EUR_USD) - ternary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 13 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Ensemble (weighted train acc) | Ensemble (mean proba) | MNLogit (statsmodels, PCA) | Gradient boosting (sklearn) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Ensemble (weighted	0.8667	0.568	0.0011	0.0000	NA	1.0000	0.0013	780
Ensemble (mean pro	0.8654	0.568	0.0011	0.0000	NA	1.0000	0.0013	780
MNLogit (statsmode	0.8256	0.224	0.0027	-0.0126	0.094	0.5000	0.0538	780
Gradient boosting	0.7936	0.118	0.0018	-0.0150	0.053	0.5139	0.0923	780
Hist gradient boos	0.8256	-0.228	-0.0023	-0.0247	-0.083	0.4615	0.0500	780
Logistic (sklearn,	0.4808	-0.341	-0.0107	-0.0702	-0.338	0.4644	0.4141	780
XGBoost	0.8538	-0.646	-0.0030	-0.0114	-0.128	0.4444	0.0115	780
Random forest	0.8679	NA	0.0000	0.0000	NA	NA	0.0000	780
MLP neural net	0.8667	NA	0.0000	0.0000	NA	NA	0.0000	780
SVM (RBF)	0.8679	NA	0.0000	0.0000	NA	NA	0.0000	780

EUR/USD (EUR_USD) - ternary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Hist gradient boosting | SVM (RBF) | MLP neural net | MNLogit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.6250	-0.327	-0.0116	-0.1344	-0.199	0.5111	0.1607	840
Hist gradient boos	0.5631	-0.432	-0.0227	-0.1391	-0.312	0.4941	0.3012	840
SVM (RBF)	0.7119	-0.548	-0.0025	-0.0083	NA	0.0000	0.0012	840
MLP neural net	0.7095	-0.548	-0.0024	-0.0080	NA	0.0000	0.0012	840
MNLogit (statsmode	0.5845	-0.700	-0.0341	-0.1652	-0.434	0.5217	0.2464	840
Logistic (sklearn,	0.3464	-1.024	-0.0605	-0.2880	-0.952	0.4941	0.5083	840
Ensemble (weighted	0.6274	-1.479	-0.0591	-0.2279	-0.752	0.4017	0.1393	840
Gradient boosting	0.5512	-1.628	-0.0816	-0.2912	-1.219	0.4398	0.3167	840
Ensemble (mean pro	0.6298	-1.671	-0.0650	-0.2246	-0.835	0.3947	0.1357	840
Random forest	0.6833	-1.764	-0.0308	-0.1012	-0.495	0.1667	0.0214	840

EUR/USD (EUR_USD) - ternary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | MLP neural net | SVM (RBF) | MNLogit (statsmodels, PCA) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.3202	0.175	0.0145	-0.2001	0.178	0.5102	0.5810	840
MLP neural net	0.5893	0.021	0.0003	-0.0232	0.003	0.5000	0.0071	840
SVM (RBF)	0.5655	-0.035	-0.0008	-0.0585	-0.015	0.5179	0.0667	840
MNLogit (statsmode	0.4750	-0.057	-0.0040	-0.1576	-0.041	0.5344	0.3119	840
Hist gradient boos	0.4631	-0.061	-0.0044	-0.1835	-0.048	0.5116	0.3583	840
XGBoost	0.5060	-0.171	-0.0119	-0.1921	-0.120	0.4958	0.2810	840
Ensemble (mean pro	0.5071	-0.388	-0.0242	-0.1787	-0.231	0.4862	0.2155	840
Ensemble (weighted	0.5036	-0.430	-0.0271	-0.1831	-0.265	0.4789	0.2262	840
Gradient boosting	0.4690	-0.502	-0.0377	-0.2738	-0.406	0.5060	0.3952	840
Random forest	0.5631	-0.968	-0.0517	-0.1961	-0.393	0.3467	0.0893	840

EUR/USD (EUR_USD) - ternary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | MNLogit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | Hist gradient boosting | MLP neural net

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.4940	0.652	0.0240	-0.0696	0.363	0.5268	0.1333	840
MNLogit (statsmode	0.4417	0.391	0.0351	-0.1788	0.319	0.5528	0.3833	840
Logistic (sklearn,	0.3095	0.272	0.0271	-0.1954	0.279	0.5422	0.5929	840
Hist gradient boos	0.4095	0.034	0.0031	-0.2584	0.032	0.5180	0.4619	840
MLP neural net	0.5155	-0.110	-0.0026	-0.0537	-0.023	0.5000	0.0381	840
Ensemble (mean pro	0.4571	-0.251	-0.0197	-0.2153	-0.165	0.5219	0.2714	840
Ensemble (weighted	0.4464	-0.470	-0.0366	-0.2340	-0.318	0.5105	0.2821	840
XGBoost	0.4405	-0.581	-0.0480	-0.2922	-0.448	0.4965	0.3357	840
Random forest	0.4988	-0.928	-0.0459	-0.2309	-0.387	0.4727	0.1310	840
Gradient boosting	0.4119	-1.451	-0.1263	-0.4428	-1.264	0.4656	0.4500	840

EUR/USD (EUR_USD) - ternary, lead 5d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | MNLogit (statsmodels, PCA) | Hist gradient boosting | MLP neural net | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.3619	0.401	0.0467	-0.2532	0.423	0.5435	0.6024	840
MNLogit (statsmode	0.4024	0.297	0.0327	-0.2006	0.282	0.5318	0.4679	840
Hist gradient boos	0.3798	0.156	0.0158	-0.1799	0.162	0.5035	0.5131	840
MLP neural net	0.4690	0.117	0.0076	-0.1257	0.040	0.5455	0.0524	840
Ensemble (weighted	0.3976	-0.071	-0.0070	-0.2246	-0.066	0.4913	0.4095	840
Ensemble (mean pro	0.4024	-0.074	-0.0071	-0.2226	-0.067	0.4955	0.4012	840
Gradient boosting	0.3905	-0.731	-0.0767	-0.3290	-0.814	0.4589	0.5500	840
SVM (RBF)	0.4083	-0.739	-0.0423	-0.2655	-0.526	0.4889	0.2143	840
XGBoost	0.4012	-0.783	-0.0763	-0.3824	-0.759	0.4737	0.4524	840
Random forest	0.4167	-1.116	-0.0896	-0.3311	-0.756	0.4216	0.2429	840

EUR/USD (EUR_USD) - ternary, lead 6d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | MNLogit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | SVM (RBF) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.4274	0.419	0.0275	-0.1121	0.184	0.4762	0.0750	840
MNLogit (statsmode	0.3440	0.318	0.0400	-0.3135	0.324	0.5446	0.5333	840
Logistic (sklearn,	0.3167	0.317	0.0408	-0.3133	0.352	0.5410	0.6250	840
SVM (RBF)	0.3821	-0.256	-0.0176	-0.2116	-0.217	0.5067	0.2679	840
Ensemble (mean pro	0.3357	-1.070	-0.1081	-0.4448	-1.014	0.4497	0.4262	840
Ensemble (weighted	0.3310	-1.242	-0.1253	-0.4787	-1.181	0.4417	0.4393	840
Gradient boosting	0.3571	-1.416	-0.1554	-0.5268	-1.536	0.4353	0.5333	840
Hist gradient boos	0.3143	-1.643	-0.1722	-0.5865	-1.808	0.4489	0.5702	840
Random forest	0.3512	-1.667	-0.1468	-0.4631	-1.356	0.4138	0.3107	840
XGBoost	0.3190	-1.989	-0.2050	-0.6081	-2.127	0.4130	0.5131	840

EUR/USD (EUR_USD) - ternary, lead 7d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MNLogit (statsmodels, PCA) | MLP neural net | SVM (RBF) | Logistic (sklearn, on global PCA) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MNLogit (statsmode	0.3512	-0.257	-0.0339	-0.4164	-0.273	0.5178	0.5679	840
MLP neural net	0.3952	-0.300	-0.0267	-0.1576	-0.167	0.4820	0.1655	840
SVM (RBF)	0.3821	-0.324	-0.0228	-0.2768	-0.254	0.4929	0.2512	840
Logistic (sklearn,	0.3369	-0.494	-0.0663	-0.4505	-0.560	0.5114	0.6286	840
Hist gradient boos	0.3357	-0.849	-0.1009	-0.5699	-0.977	0.4712	0.5583	840
XGBoost	0.3310	-1.275	-0.1445	-0.6060	-1.477	0.4365	0.4964	840
Ensemble (mean pro	0.3345	-1.346	-0.1487	-0.5502	-1.367	0.4462	0.4643	840
Gradient boosting	0.3440	-1.483	-0.1680	-0.5868	-1.662	0.4339	0.5405	840
Ensemble (weighted	0.3357	-1.486	-0.1626	-0.5715	-1.517	0.4333	0.4643	840
Random forest	0.3464	-1.857	-0.1626	-0.5217	-1.490	0.4353	0.3310	840

EUR/USD (EUR_USD) - ternary, lead 8d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | MLP neural net | MNLogit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.3560	0.359	0.0502	-0.4017	0.436	0.5211	0.5369	840
MLP neural net	0.4452	0.347	0.0361	-0.3521	0.255	0.5105	0.1702	840
MNLogit (statsmode	0.3583	0.202	0.0289	-0.4028	0.216	0.5353	0.5738	840
Logistic (sklearn,	0.3440	0.172	0.0254	-0.3875	0.206	0.5237	0.6274	840
Ensemble (mean pro	0.3690	0.076	0.0097	-0.4371	0.085	0.5149	0.4393	840
Gradient boosting	0.3833	0.011	0.0015	-0.4091	0.014	0.4870	0.5036	840
Ensemble (weighted	0.3643	-0.121	-0.0157	-0.4336	-0.132	0.5000	0.4452	840
Random forest	0.3607	-0.511	-0.0495	-0.3634	-0.463	0.4730	0.2869	840
XGBoost	0.3548	-0.537	-0.0690	-0.5394	-0.625	0.4844	0.4964	840
SVM (RBF)	0.3905	-0.623	-0.0532	-0.4044	-0.509	0.4922	0.3048	840

EUR/USD (EUR_USD) - ternary, lead 9d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | Logistic (sklearn, on global PCA) | MNLogit (statsmodels, PCA) | Ensemble (weighted train acc) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.3655	1.446	0.2249	-0.2170	2.132	0.5430	0.5262	840
Logistic (sklearn,	0.3286	1.234	0.1974	-0.3559	1.830	0.5638	0.6250	840
MNLogit (statsmode	0.3595	1.198	0.1912	-0.3967	1.629	0.5688	0.5798	840
Ensemble (weighted	0.3440	1.111	0.1582	-0.3035	1.378	0.5636	0.4583	840
Hist gradient boos	0.3738	1.092	0.1757	-0.3218	1.332	0.5522	0.5583	840
Ensemble (mean pro	0.3488	1.008	0.1404	-0.3358	1.230	0.5524	0.4548	840
XGBoost	0.3560	0.751	0.1127	-0.3483	0.855	0.5467	0.5226	840
SVM (RBF)	0.3464	-0.048	-0.0047	-0.4047	-0.040	0.5269	0.3095	840
MLP neural net	0.3631	-0.242	-0.0193	-0.2367	-0.152	0.4746	0.1405	840
Random forest	0.3369	-0.365	-0.0391	-0.3844	-0.336	0.5128	0.3714	840

EUR/USD (EUR_USD) - ternary, lead 10d

Rows used: 1000 | Effective end: 2021-12-18

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | MNLogit (statsmodels, PCA) | Ensemble (mean proba) | Hist gradient boosting | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.3524	1.115	0.1911	-0.4258	1.640	0.5451	0.6333	840
MNLogit (statsmode	0.3583	0.887	0.1483	-0.4609	1.201	0.5431	0.6071	840
Ensemble (mean pro	0.3548	0.518	0.0752	-0.4679	0.611	0.5269	0.4655	840
Hist gradient boos	0.3631	0.370	0.0588	-0.5204	0.462	0.5021	0.5667	840
Ensemble (weighted	0.3464	0.369	0.0543	-0.4809	0.415	0.5200	0.4762	840
Random forest	0.3417	0.290	0.0356	-0.3707	0.280	0.5169	0.3524	840
XGBoost	0.3345	0.185	0.0273	-0.5192	0.213	0.5067	0.5333	840
Gradient boosting	0.3357	0.100	0.0160	-0.4747	0.136	0.4906	0.6333	840
MLP neural net	0.3631	-0.077	-0.0080	-0.3274	-0.063	0.4080	0.1488	840
SVM (RBF)	0.3655	-0.193	-0.0208	-0.4220	-0.157	0.5228	0.2869	840

GBP/USD (GBP_USD) - binary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | SVM (RBF) | Logistic (sklearn, on global PCA) | Logit (statsmodels, PCA) | Probit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.4679	-0.043	-0.0033	-0.1509	-0.070	0.4713	0.9345	840
SVM (RBF)	0.5179	-0.291	-0.0189	-0.1495	-0.357	0.5185	0.8036	840
Logistic (sklearn,	0.4810	-0.309	-0.0239	-0.1257	-0.416	0.4827	0.9643	840
Logit (statsmodels	0.4881	-0.328	-0.0255	-0.1577	-0.436	0.4895	0.9679	840
Probit (statsmodel	0.4881	-0.384	-0.0299	-0.1568	-0.513	0.4865	0.9714	840
Gradient boosting	0.4833	-0.787	-0.0606	-0.2096	-1.009	0.4818	0.9833	840
Ensemble (mean pro	0.4845	-0.907	-0.0693	-0.2282	-1.155	0.4753	0.9643	840
Ensemble (weighted	0.4738	-0.908	-0.0691	-0.2179	-1.149	0.4741	0.9643	840
XGBoost	0.4857	-0.913	-0.0701	-0.2212	-1.148	0.4855	0.9857	840
Random forest	0.4762	-1.038	-0.0781	-0.2400	-1.314	0.4745	0.9583	840
Hist gradient boos	0.4738	-1.108	-0.0848	-0.2789	-1.399	0.4736	0.9905	840

GBP/USD (GBP_USD) - binary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | Logistic (sklearn, on global PCA) | Logit (statsmodels, PCA) | Probit (statsmodels, PCA) | Random forest

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.4988	0.193	0.0220	-0.2293	0.293	0.5019	0.9250	840
Logistic (sklearn,	0.5071	0.103	0.0121	-0.2607	0.136	0.5081	0.9560	840
Logit (statsmodels	0.5155	0.083	0.0099	-0.2918	0.104	0.5161	0.9619	840
Probit (statsmodel	0.5143	0.073	0.0087	-0.2918	0.092	0.5179	0.9631	840
Random forest	0.5060	0.008	0.0009	-0.2580	0.010	0.5117	0.9631	840
Ensemble (weighted	0.5143	-0.188	-0.0221	-0.2450	-0.239	0.5115	0.9821	840
Ensemble (mean pro	0.5179	-0.198	-0.0232	-0.2501	-0.248	0.5141	0.9679	840
Gradient boosting	0.4976	-0.256	-0.0300	-0.2271	-0.341	0.4976	0.9929	840
SVM (RBF)	0.5000	-0.266	-0.0304	-0.3414	-0.328	0.4981	0.9440	840
XGBoost	0.5000	-0.283	-0.0331	-0.2369	-0.366	0.5018	0.9940	840
Hist gradient boos	0.4762	-0.755	-0.0856	-0.3019	-0.986	0.4764	0.9845	840

GBP/USD (GBP_USD) - binary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | Probit (statsmodels, PCA) | Logit (statsmodels, PCA) | SVM (RBF) | MLP neural net

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.5369	1.305	0.2145	-0.2531	1.806	0.5420	0.9774	840
Probit (statsmodel	0.5321	0.929	0.1489	-0.2745	1.242	0.5316	0.9786	840
Logit (statsmodels	0.5321	0.864	0.1379	-0.2745	1.161	0.5273	0.9798	840
SVM (RBF)	0.5060	0.211	0.0317	-0.3051	0.280	0.5067	0.9726	840
MLP neural net	0.5095	0.177	0.0265	-0.4234	0.226	0.5112	0.9524	840
Gradient boosting	0.5036	0.058	0.0087	-0.3126	0.078	0.5036	0.9929	840
Ensemble (mean pro	0.5060	0.056	0.0084	-0.3793	0.075	0.5030	0.9821	840
Hist gradient boos	0.5000	0.010	0.0015	-0.3288	0.014	0.5012	0.9952	840
Ensemble (weighted	0.5048	0.008	0.0013	-0.3761	0.011	0.5030	0.9798	840
XGBoost	0.5000	-0.022	-0.0032	-0.3369	-0.029	0.4994	0.9917	840
Random forest	0.4988	-0.324	-0.0470	-0.3658	-0.419	0.4920	0.9631	840

GBP/USD (GBP_USD) - binary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | Logit (statsmodels, PCA) | Probit (statsmodels, PCA) | MLP neural net | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.5321	0.896	0.1680	-0.2930	1.243	0.5298	0.9774	840
Logit (statsmodels	0.5321	0.808	0.1509	-0.3296	1.111	0.5323	0.9774	840
Probit (statsmodel	0.5321	0.800	0.1500	-0.3365	1.106	0.5321	0.9821	840
MLP neural net	0.5083	-0.026	-0.0045	-0.5095	-0.033	0.5109	0.9250	840
XGBoost	0.5024	-0.235	-0.0402	-0.4196	-0.308	0.5030	0.9869	840
Ensemble (mean pro	0.5083	-0.336	-0.0570	-0.4879	-0.435	0.5103	0.9798	840
SVM (RBF)	0.5024	-0.484	-0.0810	-0.5020	-0.636	0.5012	0.9857	840
Gradient boosting	0.5048	-0.548	-0.0916	-0.4887	-0.735	0.5060	0.9929	840
Hist gradient boos	0.4940	-0.571	-0.0953	-0.3940	-0.757	0.4946	0.9964	840
Ensemble (weighted	0.5024	-0.575	-0.0950	-0.4996	-0.745	0.5000	0.9786	840
Random forest	0.5060	-0.622	-0.1026	-0.4667	-0.801	0.5042	0.9821	840

GBP/USD (GBP_USD) - binary, lead 5d

Rows used: 1000 | Effective end: 2021-12-25

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | Probit (statsmodels, PCA) | Logit (statsmodels, PCA) | Hist gradient boosting | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.5083	0.676	0.1438	-0.3514	0.966	0.5109	0.9857	840
Probit (statsmodel	0.5095	0.533	0.1116	-0.4549	0.734	0.5109	0.9833	840
Logit (statsmodels	0.5119	0.483	0.1003	-0.4615	0.669	0.5084	0.9881	840
Hist gradient boos	0.5310	0.418	0.0867	-0.3678	0.569	0.5324	0.9929	840
XGBoost	0.5262	0.306	0.0608	-0.3095	0.426	0.5269	0.9940	840
MLP neural net	0.5107	0.116	0.0232	-0.6594	0.146	0.5095	0.9417	840
Random forest	0.5012	0.029	0.0056	-0.3902	0.040	0.5006	0.9750	840
Ensemble (mean pro	0.5214	-0.114	-0.0222	-0.5504	-0.140	0.5177	0.9774	840
Ensemble (weighted	0.5155	-0.122	-0.0239	-0.5155	-0.151	0.5177	0.9750	840
Gradient boosting	0.5024	-0.399	-0.0763	-0.4732	-0.499	0.5012	0.9929	840
SVM (RBF)	0.4798	-0.687	-0.1270	-0.5708	-0.882	0.4823	0.9750	840

GBP/USD (GBP_USD) - binary, lead 6d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Probit (statsmodels, PCA) | Logit (statsmodels, PCA) | Hist gradient boosting | Logistic (sklearn, on global PCA) | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Probit (statsmodel	0.5202	0.789	0.1888	-0.5052	1.025	0.5235	0.9893	840
Logit (statsmodels	0.5155	0.694	0.1642	-0.5052	0.904	0.5193	0.9881	840
Hist gradient boos	0.5202	0.658	0.1557	-0.3589	0.863	0.5197	0.9964	840
Logistic (sklearn,	0.5119	0.570	0.1333	-0.4356	0.767	0.5096	0.9881	840
XGBoost	0.5131	0.419	0.0963	-0.3586	0.552	0.5138	0.9893	840
Gradient boosting	0.5071	0.319	0.0727	-0.3586	0.426	0.5084	0.9929	840
Ensemble (weighted	0.5048	0.117	0.0260	-0.4506	0.155	0.5060	0.9929	840
Ensemble (mean pro	0.5060	0.062	0.0134	-0.4905	0.080	0.5042	0.9821	840
Random forest	0.4964	-0.469	-0.0960	-0.4975	-0.589	0.4976	0.9762	840
SVM (RBF)	0.4810	-0.518	-0.1071	-0.6414	-0.649	0.4805	0.9786	840
MLP neural net	0.4583	-1.062	-0.2044	-0.7529	-1.324	0.4542	0.9226	840

GBP/USD (GBP_USD) - binary, lead 7d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | Ensemble (mean proba) | XGBoost | Ensemble (weighted train acc) | Probit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.5155	0.751	0.1932	-0.5783	0.958	0.5163	0.9845	840
Ensemble (mean pro	0.5179	0.483	0.1214	-0.4825	0.624	0.5145	0.9857	840
XGBoost	0.4964	0.454	0.1126	-0.4616	0.800	0.4946	0.9869	840
Ensemble (weighted	0.5119	0.306	0.0751	-0.4609	0.397	0.5072	0.9857	840
Probit (statsmodel	0.5071	0.267	0.0653	-0.6027	0.333	0.5061	0.9833	840
Logit (statsmodels	0.5060	0.234	0.0569	-0.6080	0.291	0.5036	0.9810	840
Gradient boosting	0.5024	0.030	0.0072	-0.5118	0.040	0.5012	0.9952	840
Hist gradient boos	0.4869	-0.148	-0.0345	-0.4890	-0.215	0.4892	0.9929	840
MLP neural net	0.5131	-0.201	-0.0458	-0.7097	-0.234	0.5150	0.9155	840
SVM (RBF)	0.4976	-0.291	-0.0664	-0.6867	-0.361	0.4940	0.9881	840
Random forest	0.4750	-0.739	-0.1604	-0.5574	-1.014	0.4703	0.9821	840

GBP/USD (GBP_USD) - binary, lead 8d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | Probit (statsmodels, PCA) | MLP neural net | Ensemble (mean proba) | Logit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.5036	0.434	0.1154	-0.6082	0.542	0.5042	0.9845	840
Probit (statsmodel	0.5119	0.386	0.1018	-0.6623	0.469	0.5085	0.9857	840
MLP neural net	0.5226	0.384	0.0986	-0.6214	0.444	0.5236	0.9071	840
Ensemble (mean pro	0.5024	0.277	0.0720	-0.5435	0.384	0.5006	0.9893	840
Logit (statsmodels	0.5060	0.239	0.0620	-0.6648	0.291	0.5048	0.9857	840
Gradient boosting	0.4964	0.193	0.0499	-0.4745	0.269	0.4976	0.9976	840
XGBoost	0.5083	0.180	0.0463	-0.4930	0.251	0.5060	0.9929	840
Ensemble (weighted	0.4988	0.150	0.0380	-0.5462	0.204	0.4988	0.9833	840
SVM (RBF)	0.5000	0.118	0.0226	-0.4625	0.160	0.5064	0.9238	840
Hist gradient boos	0.4845	-0.464	-0.1104	-0.5566	-0.656	0.4845	0.9976	840
Random forest	0.4750	-0.628	-0.1446	-0.5192	-0.879	0.4725	0.9750	840

GBP/USD (GBP_USD) - binary, lead 9d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | Probit (statsmodels, PCA) | Logit (statsmodels, PCA) | Ensemble (weighted train acc) | SVM (RBF)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.5071	0.558	0.1578	-0.6091	0.683	0.5067	0.9798	840
Probit (statsmodel	0.5083	0.297	0.0814	-0.6587	0.355	0.5084	0.9881	840
Logit (statsmodels	0.4988	0.140	0.0376	-0.6587	0.168	0.4994	0.9845	840
Ensemble (weighted	0.4964	-0.106	-0.0273	-0.7356	-0.127	0.4976	0.9857	840
SVM (RBF)	0.4929	-0.125	-0.0247	-0.4321	-0.176	0.4942	0.9202	840
Ensemble (mean pro	0.4988	-0.133	-0.0345	-0.7362	-0.160	0.4945	0.9821	840
MLP neural net	0.5214	-0.217	-0.0545	-0.7337	-0.241	0.5184	0.9369	840
XGBoost	0.5024	-0.235	-0.0598	-0.7359	-0.278	0.5000	0.9881	840
Hist gradient boos	0.5071	-0.687	-0.1654	-0.7454	-0.800	0.5042	0.9940	840
Gradient boosting	0.4762	-0.699	-0.1684	-0.7680	-0.851	0.4761	0.9952	840
Random forest	0.4667	-1.399	-0.3053	-0.7780	-1.640	0.4644	0.9869	840

GBP/USD (GBP_USD) - binary, lead 10d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | SVM (RBF) | Logit (statsmodels, PCA) | Probit (statsmodels, PCA) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.5179	1.098	0.3392	-0.4717	1.523	0.5168	0.9905	840
SVM (RBF)	0.5155	0.906	0.2788	-0.4442	1.430	0.5181	0.9881	840
Logit (statsmodels	0.5095	0.584	0.1720	-0.5414	0.754	0.5097	0.9857	840
Probit (statsmodel	0.5048	0.239	0.0673	-0.6394	0.286	0.5072	0.9881	840
Ensemble (weighted	0.5107	-0.021	-0.0056	-0.6527	-0.024	0.5090	0.9917	840
Ensemble (mean pro	0.5083	-0.038	-0.0104	-0.6617	-0.045	0.5090	0.9893	840
Hist gradient boos	0.5000	-0.039	-0.0107	-0.6184	-0.047	0.5000	0.9952	840
XGBoost	0.5119	-0.256	-0.0675	-0.7123	-0.294	0.5120	0.9881	840
Random forest	0.4952	-0.356	-0.0918	-0.7131	-0.432	0.4975	0.9714	840
MLP neural net	0.5238	-0.566	-0.1398	-0.8263	-0.610	0.5203	0.9405	840
Gradient boosting	0.5012	-0.856	-0.2083	-0.8060	-0.986	0.5018	0.9940	840

GBP/USD (GBP_USD) - ternary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MNLogit (statsmodels, PCA) | Ensemble (weighted train acc) | Logistic (sklearn, on global PCA) | XGBoost | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MNLogit (statsmode	0.7369	1.069	0.0223	-0.0071	0.882	0.5625	0.0381	840
Ensemble (weighted	0.7274	0.166	0.0020	-0.0163	0.044	0.4231	0.0310	840
Logistic (sklearn,	0.3464	0.126	0.0061	-0.1081	0.117	0.5088	0.4071	840
XGBoost	0.7036	0.064	0.0011	-0.0252	0.027	0.4918	0.0726	840
Hist gradient boos	0.5917	-0.012	-0.0004	-0.0473	-0.008	0.5095	0.2500	840
Ensemble (mean pro	0.7333	-0.419	-0.0040	-0.0197	-0.085	0.3333	0.0250	840
Gradient boosting	0.6202	-0.640	-0.0268	-0.1078	-0.401	0.4573	0.2369	840
Random forest	0.7726	-0.872	-0.0011	-0.0041	-0.306	0.2500	0.0048	840
MLP neural net	0.7774	NA	0.0000	0.0000	NA	NA	0.0000	840
SVM (RBF)	0.7798	NA	0.0000	0.0000	NA	NA	0.0000	840

GBP/USD (GBP_USD) - ternary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | Logistic (sklearn, on global PCA) | Ensemble (mean proba) | SVM (RBF) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.4429	0.790	0.0632	-0.1069	0.714	0.5457	0.3905	840
Logistic (sklearn,	0.3452	0.625	0.0565	-0.2225	0.458	0.5540	0.4190	840
Ensemble (mean pro	0.4917	0.484	0.0311	-0.1015	0.287	0.5430	0.1798	840
SVM (RBF)	0.5810	0.434	0.0035	-0.0057	NA	0.6667	0.0036	840
Ensemble (weighted	0.4786	0.304	0.0181	-0.1015	0.167	0.5466	0.1917	840
Random forest	0.5488	0.183	0.0065	-0.0870	0.035	0.6154	0.0464	840
XGBoost	0.4619	0.060	0.0038	-0.1268	0.040	0.5277	0.2798	840
MNLogit (statsmode	0.5060	-0.126	-0.0098	-0.2225	-0.062	0.5197	0.1810	840
MLP neural net	0.5869	-0.843	-0.0021	-0.0071	-0.089	0.0000	0.0036	840
Gradient boosting	0.4119	-0.919	-0.0741	-0.2957	-0.685	0.4637	0.4262	840

GBP/USD (GBP_USD) - ternary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | Hist gradient boosting | SVM (RBF) | Ensemble (mean proba) | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.4786	0.868	0.0545	-0.0459	0.670	0.5556	0.0429	840
Hist gradient boos	0.3607	0.625	0.0796	-0.2804	0.550	0.5511	0.5357	840
SVM (RBF)	0.4369	0.419	0.0124	-0.0616	0.219	0.5517	0.0690	840
Ensemble (mean pro	0.3440	0.261	0.0294	-0.3051	0.185	0.5537	0.3655	840
Logistic (sklearn,	0.2762	0.050	0.0060	-0.2743	0.048	0.5114	0.5750	840
XGBoost	0.3536	0.011	0.0013	-0.2922	0.009	0.5315	0.4726	840
Ensemble (weighted	0.3440	0.005	0.0005	-0.3051	0.003	0.5315	0.3964	840
MNLogit (statsmode	0.3500	-0.350	-0.0398	-0.3051	-0.277	0.5043	0.4131	840
Random forest	0.3821	-0.549	-0.0530	-0.3165	-0.291	0.5000	0.2000	840
Gradient boosting	0.3333	-0.784	-0.0957	-0.3326	-0.783	0.4846	0.6190	840

GBP/USD (GBP_USD) - ternary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MNLogit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | MLP neural net | SVM (RBF) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MNLogit (statsmode	0.3131	0.447	0.0662	-0.2930	0.449	0.5280	0.5321	840
Logistic (sklearn,	0.2905	0.371	0.0565	-0.2930	0.403	0.5155	0.6143	840
MLP neural net	0.4083	0.247	0.0173	-0.1198	0.093	0.5476	0.1000	840
SVM (RBF)	0.3464	-0.041	-0.0023	-0.1480	-0.037	0.5113	0.2631	840
Hist gradient boos	0.3131	-0.285	-0.0432	-0.3365	-0.314	0.5131	0.6821	840
Ensemble (mean pro	0.3321	-0.309	-0.0420	-0.3587	-0.279	0.5140	0.5095	840
Random forest	0.3381	-0.400	-0.0449	-0.3127	-0.298	0.5076	0.3143	840
XGBoost	0.3286	-0.412	-0.0605	-0.4029	-0.404	0.5188	0.6333	840
Gradient boosting	0.3690	-0.457	-0.0685	-0.3996	-0.472	0.5055	0.6476	840
Ensemble (weighted	0.3298	-0.496	-0.0695	-0.4114	-0.450	0.5109	0.5476	840

GBP/USD (GBP_USD) - ternary, lead 5d

Rows used: 1000 | Effective end: 2021-12-25

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MNLogit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | Hist gradient boosting | SVM (RBF) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MNLogit (statsmode	0.3179	0.639	0.1125	-0.4639	0.595	0.5316	0.6024	840
Logistic (sklearn,	0.3071	0.585	0.1015	-0.4639	0.548	0.5352	0.6250	840
Hist gradient boos	0.3214	0.402	0.0727	-0.2209	0.466	0.5121	0.6417	840
SVM (RBF)	0.3381	0.267	0.0222	-0.1639	0.257	0.5017	0.3560	840
Ensemble (mean pro	0.3452	0.241	0.0379	-0.3701	0.221	0.5169	0.5274	840
XGBoost	0.3452	0.202	0.0352	-0.3304	0.210	0.5095	0.6238	840
Ensemble (weighted	0.3429	0.168	0.0268	-0.3856	0.160	0.5097	0.5536	840
Random forest	0.3417	-0.215	-0.0299	-0.2888	-0.178	0.4986	0.4202	840
Gradient boosting	0.3214	-0.406	-0.0646	-0.3474	-0.439	0.4773	0.6560	840
MLP neural net	0.3536	-0.608	-0.0769	-0.4466	-0.237	0.4643	0.1000	840

GBP/USD (GBP_USD) - ternary, lead 6d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | MNLogit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | Hist gradient boosting | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.3464	0.408	0.0810	-0.4639	0.416	0.5351	0.7119	840
MNLogit (statsmode	0.3429	0.230	0.0445	-0.4943	0.222	0.5187	0.6381	840
Logistic (sklearn,	0.3345	0.138	0.0257	-0.4885	0.129	0.5155	0.6143	840
Hist gradient boos	0.3607	0.067	0.0135	-0.3712	0.075	0.4958	0.7083	840
XGBoost	0.3369	-0.132	-0.0253	-0.4182	-0.144	0.5083	0.7143	840
Random forest	0.3405	-0.256	-0.0406	-0.3854	-0.220	0.5025	0.4738	840
MLP neural net	0.3381	-0.296	-0.0455	-0.5145	-0.136	0.5489	0.1583	840
Ensemble (mean pro	0.3429	-0.309	-0.0560	-0.4566	-0.295	0.5068	0.6083	840
Ensemble (weighted	0.3369	-0.389	-0.0704	-0.4369	-0.381	0.5000	0.6310	840
SVM (RBF)	0.3262	-0.411	-0.0425	-0.2907	-0.446	0.4665	0.4619	840

GBP/USD (GBP_USD) - ternary, lead 7d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | Hist gradient boosting | SVM (RBF) | Gradient boosting (sklearn) | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.3500	0.460	0.0763	-0.2267	0.512	0.4940	0.4940	840
Hist gradient boos	0.3655	0.246	0.0538	-0.3954	0.298	0.4948	0.6833	840
SVM (RBF)	0.3393	-0.052	-0.0067	-0.2661	-0.058	0.4867	0.4917	840
Gradient boosting	0.3536	-0.052	-0.0103	-0.4955	-0.060	0.5051	0.7024	840
XGBoost	0.3679	-0.154	-0.0300	-0.4135	-0.186	0.4923	0.6917	840
Logistic (sklearn,	0.3190	-0.194	-0.0395	-0.5384	-0.191	0.4917	0.6440	840
Ensemble (weighted	0.3536	-0.365	-0.0713	-0.4980	-0.344	0.4911	0.6036	840
Ensemble (mean pro	0.3583	-0.399	-0.0765	-0.5011	-0.371	0.4869	0.5893	840
MNLogit (statsmode	0.3286	-0.562	-0.1120	-0.5609	-0.556	0.4771	0.6488	840
MLP neural net	0.3190	-0.603	-0.1035	-0.5517	-0.398	0.4875	0.3345	840

GBP/USD (GBP_USD) - ternary, lead 8d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | MLP neural net | SVM (RBF) | XGBoost | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.3214	0.239	0.0568	-0.3950	0.295	0.5057	0.7369	840
MLP neural net	0.3512	-0.009	-0.0016	-0.5668	-0.005	0.5364	0.2619	840
SVM (RBF)	0.3119	-0.019	-0.0025	-0.3281	-0.025	0.4712	0.5583	840
XGBoost	0.3024	-0.079	-0.0180	-0.3816	-0.095	0.4841	0.7131	840
Ensemble (weighted	0.3024	-0.133	-0.0277	-0.4764	-0.143	0.4870	0.6429	840
Random forest	0.2988	-0.172	-0.0285	-0.3545	-0.209	0.4735	0.5381	840
Gradient boosting	0.3107	-0.390	-0.0857	-0.4652	-0.456	0.4796	0.7298	840
Ensemble (mean pro	0.3060	-0.454	-0.0887	-0.5267	-0.473	0.4793	0.6333	840
Logistic (sklearn,	0.3179	-0.459	-0.0959	-0.5919	-0.463	0.4767	0.6893	840
MNLogit (statsmode	0.3202	-0.749	-0.1560	-0.6368	-0.779	0.4740	0.7333	840

GBP/USD (GBP_USD) - ternary, lead 9d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | MLP neural net | MNLogit (statsmodels, PCA) | XGBoost | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.3393	0.832	0.1857	-0.4168	0.964	0.5146	0.6917	840
MLP neural net	0.3369	0.761	0.0966	-0.2665	0.686	0.4792	0.2286	840
MNLogit (statsmode	0.3250	0.467	0.1052	-0.4730	0.550	0.5033	0.7286	840
XGBoost	0.3119	0.230	0.0497	-0.4303	0.328	0.4769	0.7214	840
Ensemble (mean pro	0.3298	0.091	0.0179	-0.4521	0.108	0.4755	0.6310	840
Hist gradient boos	0.3298	0.054	0.0114	-0.4078	0.067	0.4912	0.7464	840
Ensemble (weighted	0.3262	-0.003	-0.0007	-0.4640	-0.004	0.4779	0.6476	840
SVM (RBF)	0.3143	-0.004	-0.0007	-0.3031	-0.005	0.4844	0.5726	840
Random forest	0.2988	-0.783	-0.1129	-0.4388	-0.783	0.4650	0.5274	840
Gradient boosting	0.2988	-0.895	-0.1802	-0.5812	-1.119	0.4466	0.7250	840

GBP/USD (GBP_USD) - ternary, lead 10d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | MNLogit (statsmodels, PCA) | MLP neural net | SVM (RBF) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.3226	1.333	0.2909	-0.4341	2.126	0.4981	0.6429	840
MNLogit (statsmode	0.3155	0.902	0.1983	-0.4679	1.341	0.4860	0.6810	840
MLP neural net	0.3702	0.657	0.1180	-0.4936	0.478	0.5468	0.4071	840
SVM (RBF)	0.3690	0.274	0.0472	-0.3502	0.338	0.5072	0.5821	840
Ensemble (mean pro	0.3298	-0.164	-0.0334	-0.5325	-0.182	0.4887	0.6310	840
Ensemble (weighted	0.3238	-0.343	-0.0696	-0.5601	-0.391	0.4838	0.6619	840
Hist gradient boos	0.2988	-0.674	-0.1439	-0.6508	-0.864	0.4634	0.7476	840
XGBoost	0.3155	-0.893	-0.1683	-0.6746	-1.021	0.4792	0.7429	840
Random forest	0.3321	-1.247	-0.2075	-0.6408	-1.210	0.4525	0.5393	840
Gradient boosting	0.3238	-1.422	-0.2578	-0.7940	-1.549	0.4814	0.7667	840

USD/JPY (USD_JPY) - binary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Ensemble (mean proba) | SVM (RBF) | Ensemble (weighted train acc) | Logit (statsmodels, PCA) | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Ensemble (mean pro	0.4988	0.811	0.0514	-0.0542	1.214	0.5006	0.9417	840
SVM (RBF)	0.5202	0.695	0.0370	-0.0528	0.989	0.5172	0.8631	840
Ensemble (weighted	0.4988	0.533	0.0339	-0.0721	0.749	0.4969	0.9488	840
Logit (statsmodels	0.5036	0.460	0.0298	-0.0874	0.700	0.5056	0.9631	840
XGBoost	0.4976	0.432	0.0283	-0.0666	0.639	0.4952	0.9857	840
Probit (statsmodel	0.5024	0.411	0.0263	-0.0865	0.620	0.5062	0.9619	840
Logistic (sklearn,	0.5012	0.303	0.0189	-0.0596	0.474	0.5037	0.9595	840
Gradient boosting	0.5000	0.268	0.0174	-0.1145	0.345	0.5006	0.9917	840
Random forest	0.4952	0.190	0.0116	-0.0850	0.245	0.5006	0.9512	840
MLP neural net	0.4893	-0.161	-0.0101	-0.1230	-0.224	0.4879	0.8833	840
Hist gradient boos	0.4905	-0.228	-0.0146	-0.1090	-0.299	0.4910	0.9940	840

USD/JPY (USD_JPY) - binary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | Ensemble (mean proba) | Probit (statsmodels, PCA) | Ensemble (weighted train acc) | Logit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.5131	0.613	0.0526	-0.2031	0.836	0.5195	0.8548	840
Ensemble (mean pro	0.5119	0.540	0.0470	-0.1511	0.788	0.5129	0.9679	840
Probit (statsmodel	0.5107	0.447	0.0402	-0.1326	0.665	0.5141	0.9679	840
Ensemble (weighted	0.5095	0.429	0.0384	-0.1449	0.590	0.5129	0.9679	840
Logit (statsmodels	0.5131	0.425	0.0382	-0.1303	0.632	0.5129	0.9679	840
Logistic (sklearn,	0.5131	0.332	0.0297	-0.1414	0.491	0.5128	0.9750	840
Random forest	0.5226	0.283	0.0251	-0.1401	0.353	0.5181	0.9512	840
SVM (RBF)	0.5155	0.094	0.0083	-0.1880	0.122	0.5120	0.9440	840
Hist gradient boos	0.5048	0.067	0.0059	-0.1417	0.090	0.5030	0.9821	840
XGBoost	0.5095	-0.069	-0.0061	-0.1407	-0.090	0.5067	0.9821	840
Gradient boosting	0.4976	-0.622	-0.0536	-0.2313	-0.781	0.4976	0.9881	840

USD/JPY (USD_JPY) - binary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Ensemble (mean proba) | Logit (statsmodels, PCA) | Probit (statsmodels, PCA) | Ensemble (weighted train acc) | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Ensemble (mean pro	0.5405	1.470	0.1691	-0.1578	2.291	0.5375	0.9679	840
Logit (statsmodels	0.5333	1.334	0.1534	-0.1833	2.053	0.5358	0.9798	840
Probit (statsmodel	0.5345	1.319	0.1520	-0.1864	2.041	0.5326	0.9857	840
Ensemble (weighted	0.5238	1.307	0.1502	-0.1578	1.999	0.5325	0.9702	840
Logistic (sklearn,	0.5274	1.218	0.1390	-0.1788	1.886	0.5296	0.9845	840
Random forest	0.5286	0.602	0.0658	-0.1980	0.788	0.5275	0.9524	840
MLP neural net	0.4917	0.584	0.0628	-0.2174	0.883	0.4987	0.9095	840
XGBoost	0.5190	0.569	0.0629	-0.1953	0.778	0.5187	0.9845	840
Hist gradient boos	0.5167	0.488	0.0540	-0.1821	0.660	0.5156	0.9929	840
SVM (RBF)	0.5083	0.449	0.0488	-0.1885	0.617	0.5156	0.9512	840
Gradient boosting	0.5167	-0.025	-0.0027	-0.3239	-0.031	0.5169	0.9857	840

USD/JPY (USD_JPY) - binary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Logit (statsmodels, PCA) | Probit (statsmodels, PCA) | Ensemble (weighted train acc) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.5417	1.758	0.2477	-0.1810	2.557	0.5418	0.9821	840
Logit (statsmodels	0.5500	1.635	0.2294	-0.1956	2.365	0.5516	0.9798	840
Probit (statsmodel	0.5488	1.572	0.2200	-0.1956	2.286	0.5504	0.9798	840
Ensemble (weighted	0.5310	1.482	0.2057	-0.1956	2.146	0.5311	0.9750	840
Ensemble (mean pro	0.5321	1.465	0.2022	-0.1956	2.114	0.5342	0.9738	840
Gradient boosting	0.5310	1.448	0.2013	-0.1821	2.144	0.5312	0.9905	840
Logistic (sklearn,	0.5464	1.380	0.1912	-0.1956	2.011	0.5461	0.9810	840
SVM (RBF)	0.5321	1.310	0.1801	-0.2342	1.773	0.5344	0.9690	840
Hist gradient boos	0.5167	1.024	0.1389	-0.1908	1.467	0.5192	0.9929	840
Random forest	0.5155	1.015	0.1364	-0.2082	1.433	0.5184	0.9714	840
MLP neural net	0.5107	0.640	0.0829	-0.3129	0.925	0.5129	0.9262	840

USD/JPY (USD_JPY) - binary, lead 5d

Rows used: 1000 | Effective end: 2021-12-25

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | XGBoost | Ensemble (weighted train acc) | Gradient boosting (sklearn) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.5643	2.124	0.3445	-0.2106	3.177	0.5653	0.9940	840
XGBoost	0.5690	2.031	0.3276	-0.2267	2.947	0.5701	0.9940	840
Ensemble (weighted	0.5690	1.998	0.3211	-0.2339	2.881	0.5688	0.9857	840
Gradient boosting	0.5524	1.985	0.3188	-0.2106	3.126	0.5523	0.9893	840
Ensemble (mean pro	0.5619	1.921	0.3063	-0.2339	2.771	0.5650	0.9798	840
Logit (statsmodels	0.5571	1.766	0.2784	-0.2339	2.549	0.5614	0.9881	840
Probit (statsmodel	0.5595	1.752	0.2765	-0.2339	2.533	0.5601	0.9905	840
Logistic (sklearn,	0.5560	1.702	0.2669	-0.2339	2.457	0.5557	0.9833	840
SVM (RBF)	0.5583	1.387	0.2124	-0.2781	1.853	0.5591	0.9774	840
Random forest	0.5333	1.222	0.1862	-0.2597	1.754	0.5334	0.9798	840
MLP neural net	0.5214	1.220	0.1814	-0.3213	1.969	0.5242	0.9333	840

USD/JPY (USD_JPY) - binary, lead 6d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logit (statsmodels, PCA) | Probit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | Random forest | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logit (statsmodels	0.5786	2.221	0.3994	-0.2762	3.243	0.5847	0.9833	840
Probit (statsmodel	0.5821	2.125	0.3803	-0.2762	3.127	0.5800	0.9893	840
Logistic (sklearn,	0.5750	1.948	0.3422	-0.2762	2.829	0.5754	0.9786	840
Random forest	0.5476	1.841	0.3194	-0.2772	2.958	0.5496	0.9833	840
Gradient boosting	0.5500	1.774	0.3097	-0.3009	2.783	0.5502	0.9952	840
Ensemble (weighted	0.5452	1.746	0.3009	-0.2604	2.758	0.5458	0.9881	840
Hist gradient boos	0.5286	1.694	0.2940	-0.2662	2.818	0.5306	0.9917	840
SVM (RBF)	0.5548	1.657	0.2849	-0.3147	2.280	0.5577	0.9798	840
XGBoost	0.5333	1.588	0.2727	-0.2624	2.543	0.5349	0.9905	840
Ensemble (mean pro	0.5429	1.583	0.2691	-0.2642	2.423	0.5425	0.9810	840
MLP neural net	0.5560	1.220	0.1979	-0.2762	1.743	0.5538	0.9071	840

USD/JPY (USD_JPY) - binary, lead 7d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Probit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | SVM (RBF) | Logit (statsmodels, PCA) | Random forest

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Probit (statsmodel	0.5667	1.722	0.3223	-0.3028	2.591	0.5669	0.9869	840
Logistic (sklearn,	0.5643	1.617	0.3007	-0.3028	2.429	0.5653	0.9940	840
SVM (RBF)	0.5560	1.611	0.2997	-0.3381	2.199	0.5534	0.9917	840
Logit (statsmodels	0.5643	1.590	0.2951	-0.3028	2.366	0.5651	0.9881	840
Random forest	0.5310	1.508	0.2772	-0.3242	2.503	0.5311	0.9774	840
Gradient boosting	0.5250	1.314	0.2386	-0.2589	2.117	0.5239	0.9952	840
Ensemble (mean pro	0.5321	1.185	0.2125	-0.3474	1.796	0.5322	0.9798	840
XGBoost	0.5155	1.151	0.2062	-0.3282	1.835	0.5181	0.9857	840
Ensemble (weighted	0.5310	1.139	0.2040	-0.3625	1.732	0.5312	0.9905	840
Hist gradient boos	0.5214	1.036	0.1850	-0.4101	1.619	0.5209	0.9964	840
MLP neural net	0.5369	0.705	0.1197	-0.3190	1.030	0.5314	0.9095	840

USD/JPY (USD_JPY) - binary, lead 8d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logit (statsmodels, PCA) | Gradient boosting (sklearn) | SVM (RBF) | Probit (statsmodels, PCA) | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logit (statsmodels	0.5476	1.646	0.3252	-0.3154	2.435	0.5487	0.9786	840
Gradient boosting	0.5357	1.632	0.3232	-0.3044	2.621	0.5341	0.9940	840
SVM (RBF)	0.5512	1.612	0.3171	-0.3425	2.186	0.5543	0.9750	840
Probit (statsmodel	0.5405	1.539	0.3013	-0.3154	2.287	0.5443	0.9798	840
Logistic (sklearn,	0.5405	1.513	0.2963	-0.3077	2.251	0.5447	0.9857	840
XGBoost	0.5214	1.435	0.2787	-0.3111	2.299	0.5222	0.9917	840
Hist gradient boos	0.5262	1.418	0.2719	-0.2765	2.283	0.5264	0.9905	840
Ensemble (weighted	0.5274	1.309	0.2511	-0.3231	1.964	0.5309	0.9821	840
Ensemble (mean pro	0.5286	1.254	0.2398	-0.3231	1.888	0.5296	0.9869	840
Random forest	0.5131	1.020	0.1911	-0.3855	1.578	0.5152	0.9774	840
MLP neural net	0.4833	0.197	0.0338	-0.3231	0.311	0.4787	0.9226	840

USD/JPY (USD_JPY) - binary, lead 9d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Hist gradient boosting | Logistic (sklearn, on global PCA) | Ensemble (mean proba) | Logit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.5631	1.773	0.3739	-0.2381	2.650	0.5620	0.9893	840
Hist gradient boos	0.5571	1.750	0.3691	-0.3084	2.633	0.5605	0.9940	840
Logistic (sklearn,	0.5607	1.615	0.3353	-0.3385	2.385	0.5586	0.9845	840
Ensemble (mean pro	0.5560	1.588	0.3286	-0.3092	2.328	0.5596	0.9786	840
Logit (statsmodels	0.5595	1.550	0.3203	-0.3446	2.262	0.5614	0.9881	840
Probit (statsmodel	0.5583	1.505	0.3091	-0.3446	2.197	0.5576	0.9821	840
Ensemble (weighted	0.5560	1.502	0.3092	-0.3458	2.221	0.5564	0.9821	840
Gradient boosting	0.5560	1.410	0.2885	-0.3758	2.102	0.5568	0.9964	840
Random forest	0.5345	1.054	0.2073	-0.4330	1.568	0.5362	0.9702	840
SVM (RBF)	0.5595	1.029	0.1970	-0.3496	1.253	0.5614	0.9798	840
MLP neural net	0.5131	0.332	0.0593	-0.4254	0.486	0.5148	0.9226	840

USD/JPY (USD_JPY) - binary, lead 10d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | Logit (statsmodels, PCA) | Probit (statsmodels, PCA) | SVM (RBF) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.5452	1.514	0.3235	-0.2979	2.276	0.5462	0.9917	840
Logit (statsmodels	0.5440	1.489	0.3175	-0.2979	2.247	0.5469	0.9905	840
Probit (statsmodel	0.5405	1.436	0.3035	-0.3131	2.157	0.5455	0.9821	840
SVM (RBF)	0.5524	1.391	0.2927	-0.3633	1.997	0.5542	0.9774	840
Ensemble (mean pro	0.5452	1.335	0.2810	-0.3138	2.017	0.5445	0.9905	840
Ensemble (weighted	0.5440	1.317	0.2732	-0.3138	1.965	0.5466	0.9845	840
Hist gradient boos	0.5452	1.058	0.2171	-0.3355	1.542	0.5461	0.9940	840
Random forest	0.5345	1.008	0.2045	-0.3206	1.469	0.5359	0.9774	840
XGBoost	0.5190	0.919	0.1864	-0.3441	1.380	0.5179	0.9952	840
Gradient boosting	0.5298	0.839	0.1684	-0.3558	1.235	0.5300	0.9929	840
MLP neural net	0.5095	0.387	0.0743	-0.3537	0.599	0.5019	0.9536	840

USD/JPY (USD_JPY) - ternary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Ensemble (weighted train acc) | Ensemble (mean proba) | Gradient boosting (sklearn) | XGBoost | MNLogit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Ensemble (weighted	0.8083	0.667	0.0120	-0.0034	NA	0.8333	0.0071	840
Ensemble (mean pro	0.8095	0.667	0.0120	-0.0034	NA	0.8333	0.0071	840
Gradient boosting	0.8012	0.661	0.0248	-0.0496	0.260	0.6071	0.1000	840
XGBoost	0.8667	0.495	0.0089	-0.0080	0.618	0.6000	0.0060	840
MNLogit (statsmode	0.7107	0.432	0.0173	-0.0407	0.271	0.5439	0.1357	840
Logistic (sklearn,	0.5095	0.118	0.0060	-0.0889	0.126	0.5120	0.4464	840
Hist gradient boos	0.8214	-0.607	-0.0163	-0.0772	-0.128	0.5741	0.0643	840
Random forest	0.8690	NA	0.0000	0.0000	NA	NA	0.0000	840
MLP neural net	0.8690	NA	0.0000	0.0000	NA	NA	0.0000	840
SVM (RBF)	0.8702	NA	0.0000	0.0000	NA	NA	0.0000	840

USD/JPY (USD_JPY) - ternary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Ensemble (weighted train acc) | Ensemble (mean proba) | MNLogit (statsmodels, PCA) | SVM (RBF) | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Ensemble (weighted	0.7071	0.667	0.0146	-0.0197	0.265	0.6667	0.0214	840
Ensemble (mean pro	0.7060	0.666	0.0146	-0.0197	0.264	0.6471	0.0202	840
MNLogit (statsmode	0.6607	0.564	0.0262	-0.0725	0.306	0.5429	0.1250	840
SVM (RBF)	0.7298	0.548	0.0003	0.0000	NA	1.0000	0.0012	840
Logistic (sklearn,	0.3940	0.487	0.0361	-0.1375	0.561	0.4977	0.5071	840
XGBoost	0.6833	0.211	0.0037	-0.0329	0.078	0.5882	0.0607	840
Gradient boosting	0.6500	0.006	0.0003	-0.0883	0.003	0.4800	0.1488	840
Hist gradient boos	0.6571	-0.251	-0.0111	-0.1432	-0.105	0.5347	0.1202	840
Random forest	0.7226	-0.442	-0.0024	-0.0143	-0.050	0.5000	0.0071	840
MLP neural net	0.7274	NA	0.0000	0.0000	NA	NA	0.0000	840

USD/JPY (USD_JPY) - ternary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | Random forest | MNLogit (statsmodels, PCA) | Gradient boosting (sklearn) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.4012	0.548	0.0453	-0.1786	0.610	0.5074	0.4833	840
Random forest	0.6190	0.253	0.0046	-0.0423	0.078	0.6000	0.0238	840
MNLogit (statsmode	0.5298	0.229	0.0155	-0.1722	0.176	0.5161	0.2583	840
Gradient boosting	0.5440	-0.068	-0.0038	-0.1003	-0.040	0.5353	0.2024	840
Hist gradient boos	0.5357	-0.078	-0.0043	-0.0930	-0.055	0.5490	0.2429	840
XGBoost	0.5560	-0.232	-0.0096	-0.1087	-0.154	0.5357	0.1667	840
Ensemble (weighted	0.5690	-0.288	-0.0122	-0.0972	-0.132	0.5500	0.1190	840
SVM (RBF)	0.6214	-0.372	-0.0049	-0.0272	-0.066	0.4286	0.0167	840
Ensemble (mean pro	0.5679	-0.540	-0.0232	-0.1236	-0.258	0.5149	0.1202	840
MLP neural net	0.6310	NA	0.0000	0.0000	NA	NA	0.0000	840

USD/JPY (USD_JPY) - ternary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MNLogit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | Ensemble (mean proba) | Ensemble (weighted train acc) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MNLogit (statsmode	0.4536	1.714	0.1830	-0.1435	1.856	0.5409	0.3786	840
Logistic (sklearn,	0.3774	1.513	0.1804	-0.1521	2.031	0.5139	0.6000	840
Ensemble (mean pro	0.4821	0.828	0.0720	-0.1435	0.676	0.5205	0.2607	840
Ensemble (weighted	0.4833	0.725	0.0623	-0.1435	0.579	0.5223	0.2667	840
Hist gradient boos	0.4524	0.598	0.0534	-0.1390	0.541	0.5223	0.3738	840
XGBoost	0.4583	0.226	0.0210	-0.1536	0.170	0.5153	0.3119	840
Gradient boosting	0.4369	-0.107	-0.0100	-0.1817	-0.083	0.4916	0.3536	840
Random forest	0.4798	-0.332	-0.0167	-0.1235	-0.139	0.4953	0.1274	840
SVM (RBF)	0.5286	-0.394	-0.0153	-0.1808	-0.187	0.4737	0.0905	840
MLP neural net	0.5405	NA	0.0000	0.0000	NA	NA	0.0000	840

USD/JPY (USD_JPY) - ternary, lead 5d

Rows used: 1000 | Effective end: 2021-12-25

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MNLogit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | SVM (RBF) | Ensemble (mean proba) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MNLogit (statsmode	0.4286	1.763	0.2113	-0.1724	2.122	0.5595	0.4405	840
Logistic (sklearn,	0.3857	1.608	0.2106	-0.1815	2.093	0.5486	0.5881	840
SVM (RBF)	0.4881	1.166	0.0637	-0.1228	0.600	0.5854	0.0976	840
Ensemble (mean pro	0.4595	1.148	0.1158	-0.1998	0.986	0.5524	0.2952	840
Ensemble (weighted	0.4595	1.139	0.1156	-0.1915	0.995	0.5543	0.3071	840
Gradient boosting	0.4202	0.965	0.1098	-0.2150	1.002	0.5293	0.4679	840
XGBoost	0.4536	0.948	0.0990	-0.1744	0.830	0.5407	0.3655	840
Hist gradient boos	0.4202	0.933	0.1100	-0.2122	0.948	0.5356	0.4845	840
Random forest	0.4786	0.561	0.0416	-0.1153	0.347	0.5077	0.1548	840
MLP neural net	0.4917	0.543	0.0410	-0.1488	0.251	0.4516	0.0738	840

USD/JPY (USD_JPY) - ternary, lead 6d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Gradient boosting (sklearn) | Hist gradient boosting | Ensemble (weighted train acc) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.4286	2.532	0.1785	-0.1167	1.885	0.6554	0.1762	840
Gradient boosting	0.3750	2.213	0.3034	-0.1553	2.383	0.5866	0.5155	840
Hist gradient boos	0.4190	2.198	0.3081	-0.1775	2.367	0.5769	0.5262	840
Ensemble (weighted	0.4131	2.100	0.2496	-0.1553	1.880	0.6090	0.3714	840
Ensemble (mean pro	0.4071	2.034	0.2407	-0.1553	1.807	0.6111	0.3643	840
XGBoost	0.4179	2.006	0.2544	-0.1775	1.911	0.5894	0.4262	840
MNLogit (statsmode	0.3881	1.598	0.2066	-0.1742	1.826	0.5529	0.4952	840
Logistic (sklearn,	0.3655	1.417	0.1884	-0.1815	1.771	0.5344	0.5881	840
Random forest	0.4214	1.336	0.1317	-0.1553	0.853	0.5973	0.1774	840
MLP neural net	0.4429	0.272	0.0231	-0.1553	0.115	0.4833	0.0714	840

USD/JPY (USD_JPY) - ternary, lead 7d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MNLogit (statsmodels, PCA) | Ensemble (mean proba) | Logistic (sklearn, on global PCA) | SVM (RBF) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MNLogit (statsmode	0.4012	1.824	0.2601	-0.2224	2.047	0.5795	0.4643	840
Ensemble (mean pro	0.3940	1.639	0.2147	-0.1967	1.765	0.5680	0.3940	840
Logistic (sklearn,	0.3726	1.573	0.2318	-0.2224	1.990	0.5572	0.5726	840
SVM (RBF)	0.4440	1.564	0.1109	-0.1189	1.112	0.5873	0.1500	840
Ensemble (weighted	0.3976	1.552	0.2039	-0.1967	1.684	0.5612	0.3988	840
Random forest	0.4226	1.550	0.1720	-0.1817	1.148	0.5738	0.2179	840
XGBoost	0.3893	1.501	0.2045	-0.2344	1.657	0.5576	0.4548	840
Hist gradient boos	0.3869	1.417	0.2105	-0.2850	1.793	0.5244	0.5357	840
Gradient boosting	0.3762	1.409	0.1940	-0.2207	1.595	0.5635	0.5155	840
MLP neural net	0.4524	0.470	0.0454	-0.1961	0.236	0.5714	0.0917	840

USD/JPY (USD_JPY) - ternary, lead 8d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Ensemble (mean proba) | Hist gradient boosting | XGBoost | MLP neural net

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.4262	1.834	0.1536	-0.1176	1.390	0.6304	0.1643	840
Ensemble (mean pro	0.3869	1.456	0.2019	-0.2024	1.476	0.5584	0.4179	840
Hist gradient boos	0.3833	1.447	0.2080	-0.2154	1.829	0.5563	0.5500	840
XGBoost	0.3893	1.437	0.2037	-0.2325	1.621	0.5420	0.4679	840
MLP neural net	0.3952	1.424	0.1445	-0.1215	0.789	0.6477	0.1048	840
Ensemble (weighted	0.3893	1.371	0.1919	-0.2159	1.438	0.5541	0.4405	840
MNLogit (statsmode	0.3738	1.319	0.1998	-0.2569	1.439	0.5664	0.5024	840
Gradient boosting	0.3893	1.103	0.1669	-0.3452	1.315	0.5305	0.5655	840
Logistic (sklearn,	0.3619	1.045	0.1639	-0.3055	1.213	0.5405	0.5881	840
Random forest	0.4119	1.027	0.1151	-0.1937	0.758	0.5510	0.2333	840

USD/JPY (USD_JPY) - ternary, lead 9d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | Ensemble (weighted train acc) | Ensemble (mean proba) | XGBoost | SVM (RBF)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.3952	2.082	0.3560	-0.2292	2.375	0.5817	0.5464	840
Ensemble (weighted	0.3893	1.982	0.3097	-0.1573	2.034	0.6074	0.4155	840
Ensemble (mean pro	0.3964	1.908	0.2906	-0.1573	1.886	0.6071	0.4000	840
XGBoost	0.3750	1.869	0.2883	-0.1521	2.014	0.5864	0.4548	840
SVM (RBF)	0.4238	1.855	0.1625	-0.1145	1.450	0.6463	0.1750	840
Random forest	0.4012	1.743	0.2245	-0.1481	1.370	0.6022	0.2155	840
Hist gradient boos	0.3893	1.508	0.2494	-0.1733	1.877	0.5684	0.5655	840
Logistic (sklearn,	0.3702	1.162	0.1909	-0.3186	1.341	0.5587	0.5476	840
MNLogit (statsmode	0.3810	0.996	0.1573	-0.3186	1.065	0.5542	0.4726	840
MLP neural net	0.4143	0.582	0.0634	-0.2118	0.309	0.5119	0.1000	840

USD/JPY (USD_JPY) - ternary, lead 10d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | Random forest | SVM (RBF) | Gradient boosting (sklearn) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.3726	1.882	0.3040	-0.2040	2.617	0.5694	0.4976	840
Random forest	0.3893	1.741	0.2242	-0.1263	1.736	0.5979	0.2250	840
SVM (RBF)	0.4143	1.670	0.1771	-0.1893	1.019	0.6364	0.2357	840
Gradient boosting	0.3881	1.508	0.2607	-0.3749	1.791	0.5508	0.5619	840
Ensemble (weighted	0.3786	1.480	0.2377	-0.2515	1.527	0.5766	0.4274	840
XGBoost	0.3738	1.395	0.2236	-0.2587	1.581	0.5590	0.4940	840
Ensemble (mean pro	0.3786	1.364	0.2207	-0.2662	1.423	0.5686	0.4250	840
MLP neural net	0.4071	0.831	0.0890	-0.2135	0.636	0.5481	0.1238	840
Logistic (sklearn,	0.3917	0.795	0.1371	-0.3064	0.911	0.5302	0.5524	840
MNLogit (statsmode	0.4036	0.661	0.1093	-0.3064	0.706	0.5332	0.4845	840

USD/CHF (USD_CHF) - binary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Ensemble (mean proba) | Ensemble (weighted train acc) | Hist gradient boosting | MLP neural net

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.5143	0.702	0.0428	-0.0619	1.088	0.5169	0.9833	840
Ensemble (mean pro	0.5083	0.522	0.0312	-0.0639	0.775	0.5106	0.9583	840
Ensemble (weighted	0.5071	0.484	0.0291	-0.0639	0.719	0.5074	0.9595	840
Hist gradient boos	0.5095	0.483	0.0295	-0.0651	0.730	0.5085	0.9810	840
MLP neural net	0.5024	0.290	0.0169	-0.0962	0.408	0.5013	0.8976	840
Random forest	0.5202	0.284	0.0168	-0.0719	0.414	0.5200	0.9524	840
Gradient boosting	0.5083	0.141	0.0085	-0.1142	0.198	0.5121	0.9833	840
SVM (RBF)	0.5167	0.040	0.0022	-0.0819	0.052	0.5251	0.8524	840
Logistic (sklearn,	0.5012	0.037	0.0021	-0.0950	0.053	0.4955	0.9274	840
Probit (statsmodel	0.5060	-0.053	-0.0031	-0.0967	-0.075	0.5050	0.9476	840
Logit (statsmodels	0.5048	-0.113	-0.0066	-0.1002	-0.160	0.5038	0.9524	840

USD/CHF (USD_CHF) - binary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | Ensemble (weighted train acc) | XGBoost | Ensemble (mean proba) | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.5488	1.915	0.1771	-0.1040	3.067	0.5507	0.9857	840
Ensemble (weighted	0.5500	1.780	0.1619	-0.1093	2.595	0.5556	0.9536	840
XGBoost	0.5405	1.614	0.1469	-0.1110	2.433	0.5406	0.9821	840
Ensemble (mean pro	0.5512	1.572	0.1431	-0.1181	2.318	0.5501	0.9631	840
Gradient boosting	0.5452	1.208	0.1091	-0.1447	1.728	0.5425	0.9810	840
Random forest	0.5345	0.931	0.0811	-0.1395	1.313	0.5354	0.9429	840
Logistic (sklearn,	0.5274	0.907	0.0804	-0.1465	1.334	0.5275	0.9750	840
Logit (statsmodels	0.5274	0.902	0.0801	-0.1572	1.319	0.5309	0.9619	840
Probit (statsmodel	0.5274	0.820	0.0724	-0.1517	1.196	0.5296	0.9643	840
MLP neural net	0.5155	0.721	0.0611	-0.1225	1.026	0.5228	0.9131	840
SVM (RBF)	0.5167	0.675	0.0592	-0.1368	1.009	0.5201	0.9476	840

USD/CHF (USD_CHF) - binary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | XGBoost | Gradient boosting (sklearn) | Ensemble (weighted train acc) | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.5488	1.105	0.1249	-0.1841	1.533	0.5536	0.9548	840
XGBoost	0.5440	1.066	0.1229	-0.1574	1.523	0.5450	0.9917	840
Gradient boosting	0.5440	1.028	0.1183	-0.2311	1.452	0.5475	0.9893	840
Ensemble (weighted	0.5369	0.936	0.1057	-0.2359	1.344	0.5366	0.9762	840
Logistic (sklearn,	0.5262	0.931	0.1058	-0.1805	1.352	0.5290	0.9655	840
Logit (statsmodels	0.5310	0.891	0.1005	-0.2051	1.282	0.5313	0.9702	840
SVM (RBF)	0.5345	0.842	0.0939	-0.1938	1.232	0.5390	0.9452	840
Ensemble (mean pro	0.5321	0.832	0.0936	-0.2602	1.197	0.5334	0.9798	840
Probit (statsmodel	0.5262	0.745	0.0834	-0.2073	1.074	0.5264	0.9702	840
Hist gradient boos	0.5238	0.566	0.0634	-0.1944	0.838	0.5234	0.9940	840
MLP neural net	0.5250	0.354	0.0377	-0.2655	0.489	0.5179	0.9286	840

USD/CHF (USD_CHF) - binary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | MLP neural net | XGBoost | Gradient boosting (sklearn) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.5202	0.882	0.1176	-0.2466	1.290	0.5176	0.9798	840
MLP neural net	0.5440	0.845	0.1087	-0.3059	1.086	0.5401	0.9214	840
XGBoost	0.5238	0.757	0.1008	-0.2466	1.107	0.5246	0.9940	840
Gradient boosting	0.5238	0.635	0.0835	-0.2393	0.920	0.5246	0.9940	840
Hist gradient boos	0.5262	0.614	0.0809	-0.2148	0.894	0.5263	0.9952	840
Ensemble (weighted	0.5107	0.417	0.0542	-0.2902	0.610	0.5103	0.9869	840
Ensemble (mean pro	0.5071	0.385	0.0498	-0.2941	0.562	0.5085	0.9786	840
SVM (RBF)	0.5071	0.272	0.0345	-0.2582	0.392	0.5061	0.9690	840
Logistic (sklearn,	0.4988	0.180	0.0228	-0.2753	0.257	0.4994	0.9821	840
Logit (statsmodels	0.4964	0.053	0.0068	-0.2757	0.076	0.4982	0.9821	840
Probit (statsmodel	0.4964	0.017	0.0021	-0.2741	0.024	0.4964	0.9881	840

USD/CHF (USD_CHF) - binary, lead 5d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | XGBoost | Ensemble (weighted train acc) | Gradient boosting (sklearn) | Random forest

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.5357	1.535	0.2414	-0.2474	2.435	0.5364	0.9964	840
XGBoost	0.5488	1.387	0.2143	-0.3057	1.996	0.5506	0.9881	840
Ensemble (weighted	0.5345	1.294	0.1979	-0.3534	2.038	0.5311	0.9774	840
Gradient boosting	0.5310	0.987	0.1490	-0.3938	1.433	0.5313	0.9881	840
Random forest	0.5238	0.911	0.1339	-0.3402	1.286	0.5242	0.9583	840
Ensemble (mean pro	0.5238	0.815	0.1207	-0.3594	1.221	0.5239	0.9726	840
SVM (RBF)	0.5214	0.655	0.0961	-0.2839	0.980	0.5190	0.9726	840
Logit (statsmodels	0.5024	0.376	0.0542	-0.3641	0.560	0.4988	0.9714	840
MLP neural net	0.4964	0.298	0.0416	-0.3468	0.435	0.5065	0.9167	840
Logistic (sklearn,	0.5024	0.262	0.0375	-0.3634	0.394	0.5012	0.9833	840
Probit (statsmodel	0.4940	0.171	0.0243	-0.3641	0.254	0.4951	0.9738	840

USD/CHF (USD_CHF) - binary, lead 6d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Random forest | Hist gradient boosting | Gradient boosting (sklearn) | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.5155	1.013	0.1685	-0.3619	1.640	0.5162	0.9940	840
Random forest	0.5095	0.971	0.1596	-0.3777	1.575	0.5097	0.9833	840
Hist gradient boos	0.5095	0.884	0.1451	-0.4545	1.465	0.5096	0.9929	840
Gradient boosting	0.5131	0.833	0.1368	-0.4150	1.298	0.5131	1.0000	840
Logistic (sklearn,	0.5083	0.731	0.1174	-0.2854	1.114	0.5055	0.9702	840
Ensemble (weighted	0.5060	0.660	0.1058	-0.4111	1.032	0.5061	0.9833	840
SVM (RBF)	0.5107	0.651	0.1042	-0.2808	0.986	0.5171	0.9738	840
Ensemble (mean pro	0.5048	0.561	0.0899	-0.4025	0.869	0.5036	0.9810	840
Logit (statsmodels	0.5095	0.544	0.0869	-0.3012	0.819	0.5061	0.9810	840
Probit (statsmodel	0.4988	0.438	0.0692	-0.3208	0.658	0.5036	0.9810	840
MLP neural net	0.5155	0.352	0.0547	-0.3994	0.505	0.5169	0.9536	840

USD/CHF (USD_CHF) - binary, lead 7d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | Hist gradient boosting | XGBoost | Ensemble (weighted train acc) | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.5690	2.031	0.3896	-0.3135	3.071	0.5695	0.9762	840
Hist gradient boos	0.5429	1.557	0.2909	-0.2865	2.556	0.5432	0.9929	840
XGBoost	0.5429	1.537	0.2860	-0.3063	2.418	0.5434	0.9881	840
Ensemble (weighted	0.5345	1.273	0.2327	-0.3321	2.000	0.5350	0.9857	840
Gradient boosting	0.5429	1.247	0.2270	-0.2707	1.931	0.5450	0.9917	840
Ensemble (mean pro	0.5238	1.157	0.2080	-0.3248	1.813	0.5292	0.9786	840
MLP neural net	0.5571	0.964	0.1689	-0.3006	1.386	0.5505	0.9429	840
SVM (RBF)	0.5476	0.650	0.1128	-0.3317	0.918	0.5475	0.9893	840
Probit (statsmodel	0.5000	0.488	0.0829	-0.3812	0.755	0.5024	0.9810	840
Logit (statsmodels	0.5036	0.401	0.0677	-0.3896	0.623	0.5018	0.9845	840
Logistic (sklearn,	0.4905	0.398	0.0674	-0.3992	0.629	0.4880	0.9905	840

USD/CHF (USD_CHF) - binary, lead 8d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | XGBoost | Gradient boosting (sklearn) | Hist gradient boosting | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.5762	1.903	0.3806	-0.2462	2.794	0.5765	0.9726	840
XGBoost	0.5667	1.726	0.3453	-0.2493	2.649	0.5654	0.9917	840
Gradient boosting	0.5488	1.566	0.3098	-0.2341	2.459	0.5496	0.9964	840
Hist gradient boos	0.5274	1.240	0.2383	-0.2335	2.049	0.5281	0.9940	840
Ensemble (weighted	0.5393	1.200	0.2287	-0.2539	1.854	0.5389	0.9786	840
Ensemble (mean pro	0.5321	1.053	0.1986	-0.2679	1.633	0.5329	0.9762	840
Logit (statsmodels	0.4976	0.578	0.1048	-0.3749	0.915	0.5006	0.9821	840
Probit (statsmodel	0.4917	0.399	0.0710	-0.4105	0.632	0.4896	0.9726	840
SVM (RBF)	0.5274	0.314	0.0558	-0.4871	0.436	0.5289	0.9881	840
Logistic (sklearn,	0.4881	0.046	0.0079	-0.3676	0.072	0.4855	0.9833	840
MLP neural net	0.4905	-0.309	-0.0511	-0.4025	-0.454	0.4869	0.9512	840

USD/CHF (USD_CHF) - binary, lead 9d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | Hist gradient boosting | XGBoost | Ensemble (weighted train acc) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.5750	2.137	0.4598	-0.2359	3.341	0.5737	0.9774	840
Hist gradient boos	0.5429	1.996	0.4275	-0.2164	3.337	0.5418	0.9976	840
XGBoost	0.5524	1.790	0.3771	-0.3001	2.875	0.5522	0.9917	840
Ensemble (weighted	0.5357	1.524	0.3130	-0.2457	2.438	0.5380	0.9869	840
Ensemble (mean pro	0.5310	1.390	0.2824	-0.2457	2.223	0.5338	0.9857	840
Gradient boosting	0.5655	1.377	0.2771	-0.2926	1.962	0.5653	0.9940	840
MLP neural net	0.5298	0.990	0.1891	-0.3217	1.530	0.5295	0.9488	840
Probit (statsmodel	0.5190	0.912	0.1778	-0.3198	1.475	0.5192	0.9905	840
SVM (RBF)	0.5321	0.842	0.1625	-0.4657	1.166	0.5352	0.9810	840
Logit (statsmodels	0.5095	0.751	0.1444	-0.3186	1.223	0.5096	0.9905	840
Logistic (sklearn,	0.4881	0.465	0.0864	-0.2891	0.773	0.4843	0.9833	840

USD/CHF (USD_CHF) - binary, lead 10d

Rows used: 1000 | Effective end: 2021-12-18

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | XGBoost | Ensemble (mean proba) | Hist gradient boosting | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.5881	2.572	0.5997	-0.2603	4.049	0.5877	0.9702	840
XGBoost	0.5560	1.921	0.4268	-0.3122	3.088	0.5574	0.9952	840
Ensemble (mean pro	0.5476	1.869	0.4117	-0.3368	3.102	0.5429	0.9845	840
Hist gradient boos	0.5429	1.838	0.4040	-0.3698	3.076	0.5425	0.9940	840
Gradient boosting	0.5429	1.819	0.4001	-0.3806	3.034	0.5425	0.9940	840
Ensemble (weighted	0.5464	1.809	0.3938	-0.3365	2.987	0.5458	0.9881	840
Probit (statsmodel	0.5226	1.620	0.3462	-0.3402	2.695	0.5275	0.9750	840
Logit (statsmodels	0.5214	1.398	0.2946	-0.3582	2.353	0.5216	0.9905	840
Logistic (sklearn,	0.5024	1.236	0.2565	-0.3731	2.117	0.5079	0.9821	840
MLP neural net	0.5143	0.858	0.1633	-0.4338	1.373	0.5110	0.9202	840
SVM (RBF)	0.5310	0.855	0.1717	-0.5119	1.174	0.5297	0.9821	840

USD/CHF (USD_CHF) - ternary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | MNLogit (statsmodels, PCA) | Gradient boosting (sklearn) | Ensemble (mean proba) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.4833	0.798	0.0351	-0.0629	0.904	0.5013	0.4655	840
MNLogit (statsmode	0.6869	0.230	0.0054	-0.0409	0.109	0.5063	0.0940	840
Gradient boosting	0.7202	0.118	0.0032	-0.0663	0.071	0.5214	0.1667	840
Ensemble (mean pro	0.7298	-0.176	-0.0020	-0.0240	-0.045	0.5000	0.0262	840
Ensemble (weighted	0.7321	-0.176	-0.0020	-0.0240	-0.045	0.5000	0.0262	840
Hist gradient boos	0.7321	-0.320	-0.0072	-0.0577	-0.159	0.5172	0.1381	840
XGBoost	0.7988	-0.354	-0.0060	-0.0379	-0.115	0.4524	0.0500	840
Random forest	0.8369	NA	0.0000	0.0000	NA	NA	0.0000	840
MLP neural net	0.8321	NA	0.0000	0.0000	NA	NA	0.0000	840
SVM (RBF)	0.8369	NA	0.0000	0.0000	NA	NA	0.0000	840

USD/CHF (USD_CHF) - ternary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Ensemble (mean proba) | MNLogit (statsmodels, PCA) | Ensemble (weighted train acc) | Random forest | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Ensemble (mean pro	0.6440	1.059	0.0451	-0.0705	0.878	0.5398	0.1345	840
MNLogit (statsmode	0.6286	0.845	0.0428	-0.1057	0.560	0.5578	0.1750	840
Ensemble (weighted	0.6417	0.778	0.0332	-0.0745	0.627	0.5120	0.1488	840
Random forest	0.6750	0.597	0.0060	-0.0142	0.192	0.5833	0.0143	840
Hist gradient boos	0.5357	0.592	0.0361	-0.1303	0.610	0.5224	0.3714	840
XGBoost	0.6048	0.577	0.0221	-0.0717	0.446	0.5157	0.1893	840
Gradient boosting	0.5738	0.387	0.0225	-0.1199	0.374	0.4848	0.3143	840
Logistic (sklearn,	0.3750	0.067	0.0044	-0.1380	0.073	0.4816	0.4845	840
SVM (RBF)	0.6810	-0.777	-0.0018	-0.0062	-0.072	0.2500	0.0048	840
MLP neural net	0.6845	NA	0.0000	0.0000	NA	NA	0.0000	840

USD/CHF (USD_CHF) - ternary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | Ensemble (weighted train acc) | XGBoost | Hist gradient boosting | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.5095	1.261	0.0841	-0.1298	1.220	0.5657	0.3262	840
Ensemble (weighted	0.5393	0.961	0.0602	-0.1326	0.695	0.5503	0.2250	840
XGBoost	0.5298	0.929	0.0637	-0.1181	0.767	0.5391	0.2738	840
Hist gradient boos	0.4762	0.855	0.0661	-0.1665	0.867	0.5552	0.4202	840
Ensemble (mean pro	0.5381	0.805	0.0498	-0.1326	0.561	0.5506	0.2119	840
SVM (RBF)	0.5869	0.558	0.0052	-0.0093	0.136	0.4286	0.0083	840
MNLogit (statsmode	0.4893	0.173	0.0135	-0.1696	0.139	0.4926	0.3214	840
Logistic (sklearn,	0.3631	0.030	0.0027	-0.2026	0.032	0.4916	0.5643	840
Random forest	0.5690	-0.060	-0.0020	-0.1174	-0.023	0.4923	0.0774	840
MLP neural net	0.5845	NA	0.0000	0.0000	NA	NA	0.0000	840

USD/CHF (USD_CHF) - ternary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Ensemble (mean proba) | Ensemble (weighted train acc) | Hist gradient boosting | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.4762	1.703	0.1490	-0.1758	1.479	0.5644	0.3143	840
Ensemble (mean pro	0.4845	1.548	0.1243	-0.1617	1.204	0.5727	0.2619	840
Ensemble (weighted	0.4798	1.344	0.1025	-0.1698	1.014	0.5586	0.2643	840
Hist gradient boos	0.4452	1.293	0.1147	-0.1930	1.238	0.5457	0.4167	840
Logistic (sklearn,	0.3964	0.832	0.0908	-0.2270	0.844	0.5483	0.5298	840
Gradient boosting	0.4905	0.744	0.0680	-0.1855	0.646	0.5308	0.3476	840
MLP neural net	0.5250	0.548	0.0001	0.0000	NA	1.0000	0.0012	840
MNLogit (statsmode	0.4643	0.510	0.0481	-0.2089	0.425	0.5261	0.3417	840
Random forest	0.5095	0.011	0.0005	-0.1773	0.005	0.4756	0.0976	840
SVM (RBF)	0.5107	-0.167	-0.0071	-0.1777	-0.079	0.4324	0.0881	840

USD/CHF (USD_CHF) - ternary, lead 5d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Hist gradient boosting | Ensemble (weighted train acc) | MNLogit (statsmodels, PCA) | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.4298	1.145	0.1294	-0.2039	1.179	0.5459	0.4536	840
Hist gradient boos	0.4333	0.832	0.1009	-0.2738	0.920	0.5416	0.5298	840
Ensemble (weighted	0.4369	0.803	0.0890	-0.2319	0.769	0.5452	0.4214	840
MNLogit (statsmode	0.4274	0.771	0.0862	-0.2751	0.726	0.5301	0.4155	840
Gradient boosting	0.4071	0.710	0.0782	-0.1753	0.693	0.5530	0.4607	840
Random forest	0.4357	0.697	0.0458	-0.1209	0.421	0.5590	0.1917	840
Ensemble (mean pro	0.4369	0.684	0.0756	-0.2249	0.652	0.5362	0.4107	840
SVM (RBF)	0.4333	0.589	0.0353	-0.2393	0.434	0.5294	0.1821	840
Logistic (sklearn,	0.3774	0.287	0.0335	-0.3057	0.307	0.5079	0.5274	840
MLP neural net	0.4512	-1.197	-0.0274	-0.0897	-0.186	0.2143	0.0167	840

USD/CHF (USD_CHF) - ternary, lead 6d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Hist gradient boosting | Ensemble (mean proba) | Ensemble (weighted train acc) | Random forest

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.3964	1.672	0.2053	-0.2890	1.876	0.5539	0.4750	840
Hist gradient boos	0.3667	1.374	0.1732	-0.3195	1.602	0.5496	0.5524	840
Ensemble (mean pro	0.3988	1.369	0.1576	-0.2649	1.369	0.5526	0.3964	840
Ensemble (weighted	0.4012	1.363	0.1570	-0.2737	1.376	0.5484	0.4060	840
Random forest	0.3964	1.178	0.0697	-0.1117	0.888	0.5486	0.2083	840
SVM (RBF)	0.3488	1.166	0.0881	-0.1799	0.931	0.5571	0.2607	840
Gradient boosting	0.3988	1.029	0.1354	-0.2521	1.033	0.5494	0.5179	840
Logistic (sklearn,	0.3440	0.641	0.0861	-0.2482	0.723	0.5318	0.5798	840
MNLogit (statsmode	0.3679	0.362	0.0452	-0.2670	0.374	0.5148	0.4833	840
MLP neural net	0.4060	0.043	0.0016	-0.0956	0.011	0.5946	0.0440	840

USD/CHF (USD_CHF) - ternary, lead 7d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | Gradient boosting (sklearn) | Ensemble (weighted train acc) | Random forest | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.3690	2.183	0.3234	-0.2949	2.556	0.5882	0.5667	840
Gradient boosting	0.3917	2.094	0.3160	-0.3277	2.427	0.5991	0.5405	840
Ensemble (weighted	0.3702	2.002	0.2624	-0.3287	2.088	0.5972	0.4286	840
Random forest	0.3726	1.946	0.1660	-0.1718	1.794	0.5972	0.2512	840
XGBoost	0.3798	1.719	0.2295	-0.2858	1.994	0.5590	0.4940	840
Ensemble (mean pro	0.3667	1.617	0.2073	-0.3451	1.643	0.5764	0.4131	840
SVM (RBF)	0.3488	0.983	0.0846	-0.2322	0.749	0.5616	0.2607	840
Logistic (sklearn,	0.3476	0.669	0.0939	-0.3529	0.780	0.5266	0.5810	840
MNLogit (statsmode	0.3548	0.489	0.0655	-0.3319	0.536	0.5117	0.5071	840
MLP neural net	0.3702	-0.064	-0.0034	-0.2457	-0.028	0.5357	0.1000	840

USD/CHF (USD_CHF) - ternary, lead 8d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | XGBoost | Gradient boosting (sklearn) | SVM (RBF) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.3810	2.536	0.2380	-0.0918	2.336	0.6201	0.2726	840
XGBoost	0.3738	2.209	0.3373	-0.2549	2.514	0.5598	0.4976	840
Gradient boosting	0.3857	1.905	0.2806	-0.2154	1.973	0.6000	0.5238	840
SVM (RBF)	0.3571	1.751	0.1876	-0.2656	1.418	0.5805	0.2810	840
Ensemble (mean pro	0.3583	1.696	0.2404	-0.2854	1.771	0.5500	0.4286	840
Ensemble (weighted	0.3536	1.603	0.2214	-0.2950	1.652	0.5429	0.4298	840
Hist gradient boos	0.3500	1.322	0.1899	-0.3725	1.593	0.5163	0.5488	840
MNLogit (statsmode	0.3643	1.185	0.1804	-0.2878	1.357	0.5355	0.5536	840
Logistic (sklearn,	0.3571	0.901	0.1372	-0.3142	1.082	0.5168	0.6036	840
MLP neural net	0.3690	0.040	0.0023	-0.2102	0.014	0.4231	0.0619	840

USD/CHF (USD_CHF) - ternary, lead 9d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Random forest | Ensemble (weighted train acc) | Hist gradient boosting | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.3798	2.858	0.5047	-0.2547	3.534	0.6152	0.5786	840
Random forest	0.4095	2.496	0.3153	-0.2677	2.216	0.6388	0.3131	840
Ensemble (weighted	0.3905	2.462	0.4084	-0.2748	2.809	0.5901	0.4821	840
Hist gradient boos	0.3833	2.372	0.4240	-0.2970	3.120	0.5652	0.6024	840
Ensemble (mean pro	0.3845	2.286	0.3692	-0.2877	2.541	0.5807	0.4571	840
Gradient boosting	0.4226	2.218	0.3909	-0.2302	2.636	0.6070	0.6119	840
SVM (RBF)	0.3536	1.800	0.1876	-0.2903	1.381	0.5728	0.2536	840
MNLogit (statsmode	0.3548	1.000	0.1608	-0.3833	1.238	0.5030	0.5893	840
Logistic (sklearn,	0.3429	0.839	0.1327	-0.3836	1.063	0.4900	0.5952	840
MLP neural net	0.3464	0.160	0.0169	-0.3294	0.113	0.4026	0.1833	840

USD/CHF (USD_CHF) - ternary, lead 10d

Rows used: 1000 | Effective end: 2021-12-18

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | Hist gradient boosting | Ensemble (weighted train acc) | Random forest | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.4024	2.493	0.4278	-0.2547	2.799	0.5958	0.5155	840
Hist gradient boos	0.3893	2.069	0.3734	-0.2012	2.678	0.5678	0.5619	840
Ensemble (weighted	0.3679	2.043	0.3220	-0.3654	2.044	0.5913	0.4107	840
Random forest	0.3929	1.991	0.2321	-0.2653	1.523	0.6100	0.3083	840
Ensemble (mean pro	0.3667	1.874	0.2896	-0.3697	1.859	0.5778	0.3976	840
XGBoost	0.3833	1.718	0.2708	-0.3044	1.843	0.5728	0.4821	840
SVM (RBF)	0.3655	1.688	0.1984	-0.2882	1.219	0.6151	0.2845	840
Logistic (sklearn,	0.3571	0.985	0.1671	-0.3877	1.283	0.5224	0.6107	840
MNLogit (statsmode	0.3452	0.919	0.1495	-0.3688	1.137	0.5239	0.5476	840
MLP neural net	0.3512	0.541	0.0604	-0.3006	0.364	0.4855	0.1643	840

AUD/USD (AUD_USD) - binary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | Ensemble (mean proba) | Ensemble (weighted train acc) | SVM (RBF) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.5107	0.639	0.0585	-0.1337	0.868	0.5097	0.9833	840
Ensemble (mean pro	0.5048	0.356	0.0321	-0.1539	0.476	0.5025	0.9690	840
Ensemble (weighted	0.5071	0.274	0.0245	-0.1637	0.364	0.5067	0.9726	840
SVM (RBF)	0.5202	0.143	0.0121	-0.1899	0.201	0.5156	0.8750	840
Hist gradient boos	0.4940	0.104	0.0093	-0.1450	0.141	0.4910	0.9869	840
Probit (statsmodel	0.4988	0.094	0.0083	-0.1556	0.121	0.5063	0.9476	840
Logit (statsmodels	0.5024	0.054	0.0048	-0.1668	0.070	0.5050	0.9500	840
Logistic (sklearn,	0.5071	-0.007	-0.0006	-0.1817	-0.009	0.5049	0.9667	840
XGBoost	0.5036	-0.294	-0.0256	-0.1988	-0.383	0.5030	0.9798	840
Random forest	0.5012	-0.517	-0.0444	-0.1850	-0.686	0.4994	0.9607	840
MLP neural net	0.4798	-1.044	-0.0850	-0.3153	-1.255	0.4712	0.8893	840

AUD/USD (AUD_USD) - binary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | SVM (RBF) | Hist gradient boosting | Gradient boosting (sklearn) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.5298	0.393	0.0530	-0.2979	0.498	0.5316	0.9810	840
SVM (RBF)	0.5321	0.350	0.0374	-0.1721	0.469	0.5386	0.8952	840
Hist gradient boos	0.5310	0.284	0.0380	-0.3203	0.367	0.5308	0.9845	840
Gradient boosting	0.5274	0.143	0.0191	-0.2496	0.185	0.5263	0.9952	840
Ensemble (weighted	0.5167	-0.092	-0.0120	-0.3109	-0.119	0.5190	0.9726	840
Ensemble (mean pro	0.5179	-0.125	-0.0163	-0.3233	-0.161	0.5220	0.9738	840
Logit (statsmodels	0.5179	-0.257	-0.0331	-0.3067	-0.322	0.5183	0.9738	840
Probit (statsmodel	0.5167	-0.283	-0.0364	-0.3176	-0.356	0.5153	0.9726	840
Logistic (sklearn,	0.5155	-0.324	-0.0417	-0.2982	-0.404	0.5208	0.9714	840
MLP neural net	0.5238	-0.424	-0.0528	-0.3178	-0.504	0.5201	0.9179	840
Random forest	0.5036	-0.696	-0.0861	-0.3826	-0.840	0.5037	0.9619	840

AUD/USD (AUD_USD) - binary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Gradient boosting (sklearn) | Hist gradient boosting | Probit (statsmodels, PCA) | Logit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.5238	0.429	0.0747	-0.3198	0.547	0.5228	0.9929	840
Gradient boosting	0.5131	0.005	0.0009	-0.3992	0.007	0.5150	0.9917	840
Hist gradient boos	0.5024	-0.117	-0.0194	-0.3631	-0.151	0.5012	0.9857	840
Probit (statsmodel	0.4917	-0.188	-0.0310	-0.4228	-0.243	0.4939	0.9738	840
Logit (statsmodels	0.4905	-0.255	-0.0417	-0.4319	-0.330	0.4915	0.9762	840
Random forest	0.5024	-0.386	-0.0623	-0.4625	-0.467	0.5019	0.9631	840
Logistic (sklearn,	0.4869	-0.406	-0.0652	-0.4651	-0.534	0.4860	0.9774	840
Ensemble (weighted	0.4929	-0.506	-0.0810	-0.3908	-0.647	0.4945	0.9774	840
SVM (RBF)	0.4702	-0.529	-0.0841	-0.5032	-0.781	0.4705	0.9690	840
Ensemble (mean pro	0.4845	-0.570	-0.0906	-0.4107	-0.729	0.4902	0.9762	840
MLP neural net	0.5012	-0.719	-0.1093	-0.5463	-0.851	0.5026	0.9119	840

AUD/USD (AUD_USD) - binary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Random forest | Ensemble (mean proba) | Ensemble (weighted train acc) | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.5119	0.005	0.0010	-0.5000	0.007	0.5109	0.9833	840
Random forest	0.5083	-0.155	-0.0296	-0.4451	-0.198	0.5073	0.9786	840
Ensemble (mean pro	0.4905	-0.397	-0.0748	-0.4868	-0.518	0.4921	0.9845	840
Ensemble (weighted	0.4940	-0.414	-0.0780	-0.4886	-0.540	0.4909	0.9821	840
XGBoost	0.4988	-0.429	-0.0809	-0.4263	-0.558	0.5000	0.9929	840
Probit (statsmodel	0.4952	-0.484	-0.0901	-0.5197	-0.619	0.4951	0.9738	840
Logit (statsmodels	0.4964	-0.524	-0.0977	-0.5292	-0.673	0.4933	0.9750	840
MLP neural net	0.5238	-0.525	-0.0954	-0.5363	-0.616	0.5165	0.9036	840
Gradient boosting	0.4917	-0.541	-0.1009	-0.4644	-0.757	0.4904	0.9952	840
Logistic (sklearn,	0.4905	-0.652	-0.1198	-0.5188	-0.831	0.4872	0.9774	840
Hist gradient boos	0.4940	-0.775	-0.1411	-0.5273	-0.992	0.4958	0.9917	840

AUD/USD (AUD_USD) - binary, lead 5d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | XGBoost | Hist gradient boosting | Random forest | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.5012	0.330	0.0765	-0.5599	0.485	0.5048	0.9857	840
XGBoost	0.4929	0.288	0.0666	-0.3806	0.440	0.4928	0.9905	840
Hist gradient boos	0.4798	-0.148	-0.0326	-0.4667	-0.227	0.4785	0.9952	840
Random forest	0.4869	-0.350	-0.0736	-0.4100	-0.481	0.4867	0.9810	840
Ensemble (mean pro	0.4857	-0.370	-0.0794	-0.5780	-0.480	0.4892	0.9905	840
Ensemble (weighted	0.4893	-0.378	-0.0812	-0.5805	-0.493	0.4874	0.9917	840
Probit (statsmodel	0.4905	-0.462	-0.0979	-0.6104	-0.594	0.4939	0.9833	840
Logit (statsmodels	0.4952	-0.486	-0.1028	-0.6178	-0.628	0.4940	0.9905	840
Gradient boosting	0.4821	-0.669	-0.1389	-0.5780	-0.873	0.4833	0.9952	840
Logistic (sklearn,	0.4929	-0.843	-0.1698	-0.6086	-1.077	0.4879	0.9810	840
MLP neural net	0.5024	-0.921	-0.1815	-0.6836	-1.091	0.4955	0.9202	840

AUD/USD (AUD_USD) - binary, lead 6d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | XGBoost | Random forest | Hist gradient boosting | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.5048	1.177	0.3338	-0.3907	1.766	0.5036	0.9976	840
XGBoost	0.5119	1.106	0.3104	-0.2700	1.718	0.5120	0.9929	840
Random forest	0.5095	1.073	0.2832	-0.3476	1.556	0.5117	0.9655	840
Hist gradient boos	0.5036	0.958	0.2638	-0.2981	1.489	0.5042	0.9917	840
Ensemble (weighted	0.4940	0.406	0.1019	-0.5414	0.529	0.4957	0.9798	840
SVM (RBF)	0.5060	0.292	0.0740	-0.6085	0.427	0.5078	0.9893	840
Ensemble (mean pro	0.4964	0.274	0.0690	-0.5762	0.347	0.4957	0.9726	840
MLP neural net	0.5143	-0.122	-0.0287	-0.5697	-0.141	0.5064	0.9357	840
Logit (statsmodels	0.4774	-0.456	-0.1057	-0.6290	-0.576	0.4825	0.9869	840
Probit (statsmodel	0.4810	-0.472	-0.1091	-0.6287	-0.596	0.4806	0.9810	840
Logistic (sklearn,	0.4690	-0.668	-0.1502	-0.6329	-0.843	0.4697	0.9833	840

AUD/USD (AUD_USD) - binary, lead 7d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | XGBoost | Random forest | SVM (RBF) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.5226	0.743	0.2174	-0.6412	0.933	0.5234	0.9917	840
XGBoost	0.5190	0.634	0.1828	-0.6514	0.789	0.5205	0.9881	840
Random forest	0.5202	0.507	0.1410	-0.6547	0.622	0.5206	0.9833	840
SVM (RBF)	0.5167	0.480	0.1343	-0.6234	0.706	0.5157	0.9857	840
Ensemble (weighted	0.5131	0.270	0.0742	-0.6628	0.330	0.5133	0.9857	840
Ensemble (mean pro	0.5083	0.135	0.0364	-0.6655	0.166	0.5066	0.9893	840
Logit (statsmodels	0.4964	-0.023	-0.0062	-0.6410	-0.029	0.4970	0.9917	840
Logistic (sklearn,	0.4976	-0.046	-0.0122	-0.6531	-0.058	0.4970	0.9917	840
MLP neural net	0.5190	-0.070	-0.0181	-0.6541	-0.081	0.5181	0.9536	840
Probit (statsmodel	0.4940	-0.082	-0.0216	-0.6447	-0.103	0.4946	0.9964	840
Hist gradient boos	0.5095	-0.156	-0.0405	-0.6456	-0.187	0.5084	0.9952	840

AUD/USD (AUD_USD) - binary, lead 8d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | Random forest | SVM (RBF) | XGBoost | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.5274	1.286	0.4345	-0.6913	1.599	0.5269	0.9964	840
Random forest	0.5369	1.214	0.3925	-0.6483	1.480	0.5329	0.9762	840
SVM (RBF)	0.5214	1.190	0.2810	-0.3465	1.773	0.5357	0.9179	840
XGBoost	0.5179	1.002	0.3243	-0.6975	1.239	0.5206	0.9833	840
Ensemble (weighted	0.5214	0.845	0.2675	-0.6786	1.035	0.5181	0.9857	840
Ensemble (mean pro	0.5190	0.805	0.2533	-0.6881	0.984	0.5193	0.9881	840
Hist gradient boos	0.5060	0.434	0.1294	-0.6956	0.532	0.5042	0.9940	840
MLP neural net	0.5333	0.396	0.1164	-0.6919	0.457	0.5358	0.9464	840
Logit (statsmodels	0.5000	0.074	0.0211	-0.7229	0.091	0.5018	0.9893	840
Logistic (sklearn,	0.5095	0.043	0.0120	-0.7272	0.052	0.5072	0.9905	840
Probit (statsmodel	0.4988	-0.058	-0.0162	-0.7272	-0.071	0.4988	0.9881	840

AUD/USD (AUD_USD) - binary, lead 9d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | XGBoost | Ensemble (weighted train acc) | Hist gradient boosting | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.5429	1.629	0.6137	-0.7024	1.963	0.5434	0.9881	840
XGBoost	0.5357	1.524	0.5670	-0.7289	1.859	0.5341	0.9964	840
Ensemble (weighted	0.5238	1.061	0.3659	-0.7335	1.280	0.5261	0.9798	840
Hist gradient boos	0.5155	1.022	0.3525	-0.7446	1.257	0.5168	0.9929	840
Ensemble (mean pro	0.5238	0.999	0.3413	-0.7344	1.210	0.5218	0.9810	840
SVM (RBF)	0.5405	0.997	0.3382	-0.7456	1.184	0.5418	0.9821	840
Gradient boosting	0.5310	0.747	0.2473	-0.6965	0.903	0.5293	0.9964	840
Logit (statsmodels	0.4964	0.075	0.0226	-0.7522	0.092	0.4946	0.9869	840
Logistic (sklearn,	0.4929	0.068	0.0205	-0.7547	0.084	0.4964	0.9905	840
Probit (statsmodel	0.4940	-0.014	-0.0042	-0.7537	-0.017	0.4922	0.9893	840
MLP neural net	0.5024	-0.106	-0.0302	-0.7368	-0.122	0.5076	0.9357	840

AUD/USD (AUD_USD) - binary, lead 10d

Rows used: 1000 | Effective end: 2021-12-18

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | SVM (RBF) | XGBoost | Random forest | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.5262	0.590	0.1936	-0.7905	0.674	0.5198	0.9000	840
SVM (RBF)	0.5000	0.557	0.1860	-0.7689	0.684	0.4982	0.9821	840
XGBoost	0.4976	0.383	0.1245	-0.7875	0.464	0.4958	0.9917	840
Random forest	0.4869	0.143	0.0446	-0.7739	0.172	0.4891	0.9833	840
Hist gradient boos	0.4821	-0.013	-0.0039	-0.7653	-0.015	0.4814	0.9940	840
Ensemble (weighted	0.4798	-0.065	-0.0198	-0.7795	-0.079	0.4825	0.9869	840
Ensemble (mean pro	0.4810	-0.081	-0.0246	-0.7841	-0.098	0.4818	0.9833	840
Probit (statsmodel	0.4857	-0.208	-0.0621	-0.8178	-0.254	0.4868	0.9929	840
Gradient boosting	0.4869	-0.214	-0.0638	-0.7309	-0.256	0.4886	0.9940	840
Logit (statsmodels	0.4881	-0.264	-0.0775	-0.8178	-0.320	0.4868	0.9929	840
Logistic (sklearn,	0.4786	-0.347	-0.0997	-0.8178	-0.416	0.4801	0.9869	840

AUD/USD (AUD_USD) - ternary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | Logistic (sklearn, on global PCA) | Hist gradient boosting | Gradient boosting (sklearn) | MNLogit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.6964	0.750	0.0100	-0.0015	NA	0.7500	0.0048	840
Logistic (sklearn,	0.3917	0.123	0.0070	-0.1492	0.087	0.5105	0.3964	840
Hist gradient boos	0.5762	-0.152	-0.0080	-0.1221	-0.081	0.5051	0.2357	840
Gradient boosting	0.5893	-0.267	-0.0144	-0.1327	-0.145	0.4696	0.2155	840
MNLogit (statsmode	0.6583	-0.337	-0.0139	-0.1306	-0.101	0.5000	0.0929	840
Ensemble (mean pro	0.6702	-0.354	-0.0135	-0.1174	-0.070	0.5294	0.0405	840
Ensemble (weighted	0.6738	-0.376	-0.0143	-0.1219	-0.078	0.5000	0.0429	840
XGBoost	0.6619	-0.693	-0.0298	-0.1472	-0.225	0.4058	0.0821	840
Random forest	0.7012	NA	0.0000	0.0000	NA	NA	0.0000	840
SVM (RBF)	0.7036	NA	0.0000	0.0000	NA	NA	0.0000	840

AUD/USD (AUD_USD) - ternary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | Hist gradient boosting | Ensemble (weighted train acc) | MNLogit (statsmodels, PCA) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.4643	0.255	0.0239	-0.1988	0.204	0.5099	0.3619	840
Hist gradient boos	0.3988	-0.301	-0.0261	-0.2398	-0.221	0.5244	0.4155	840
Ensemble (weighted	0.4655	-0.370	-0.0245	-0.1922	-0.171	0.5067	0.1786	840
MNLogit (statsmode	0.4464	-0.422	-0.0292	-0.2746	-0.183	0.5376	0.2060	840
Ensemble (mean pro	0.4690	-0.447	-0.0284	-0.1930	-0.189	0.5224	0.1595	840
Random forest	0.5143	-0.463	-0.0148	-0.0997	-0.113	0.4643	0.0333	840
SVM (RBF)	0.5369	-0.597	-0.0053	-0.0195	-0.045	0.2500	0.0048	840
Logistic (sklearn,	0.3429	-0.683	-0.0565	-0.2737	-0.453	0.5237	0.4024	840
XGBoost	0.4583	-0.806	-0.0593	-0.2524	-0.401	0.5160	0.2238	840
MLP neural net	0.5321	NA	0.0000	0.0000	NA	NA	0.0000	840

AUD/USD (AUD_USD) - ternary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | XGBoost | MNLogit (statsmodels, PCA) | Hist gradient boosting | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.4333	0.548	0.0050	0.0000	NA	1.0000	0.0012	840
XGBoost	0.3369	-0.226	-0.0288	-0.3876	-0.181	0.5107	0.4452	840
MNLogit (statsmode	0.3738	-0.282	-0.0344	-0.3962	-0.206	0.5135	0.3964	840
Hist gradient boos	0.3369	-0.516	-0.0651	-0.4010	-0.430	0.5080	0.5202	840
Ensemble (mean pro	0.3333	-0.548	-0.0607	-0.3726	-0.351	0.5260	0.3440	840
Logistic (sklearn,	0.3202	-0.637	-0.0808	-0.4186	-0.522	0.5107	0.5012	840
Ensemble (weighted	0.3345	-0.690	-0.0778	-0.3786	-0.471	0.5144	0.3726	840
Gradient boosting	0.3560	-0.694	-0.0895	-0.4082	-0.604	0.4941	0.5060	840
Random forest	0.3583	-0.744	-0.0734	-0.3312	-0.352	0.5215	0.1940	840
SVM (RBF)	0.3750	-1.037	-0.0438	-0.1805	-0.350	0.4211	0.0679	840

AUD/USD (AUD_USD) - ternary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | MNLogit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | SVM (RBF) | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.4060	0.184	0.0078	-0.1024	0.088	0.4884	0.0512	840
MNLogit (statsmode	0.3167	-0.412	-0.0632	-0.5276	-0.379	0.4880	0.5464	840
Logistic (sklearn,	0.3060	-0.574	-0.0879	-0.5615	-0.551	0.4838	0.5881	840
SVM (RBF)	0.3452	-0.823	-0.0740	-0.2771	-0.589	0.4959	0.2905	840
Gradient boosting	0.3405	-0.918	-0.1425	-0.5076	-0.883	0.4743	0.6024	840
Ensemble (mean pro	0.2952	-0.919	-0.1313	-0.4902	-0.787	0.4604	0.4810	840
Ensemble (weighted	0.3036	-0.988	-0.1435	-0.4915	-0.880	0.4654	0.5167	840
Random forest	0.3524	-1.079	-0.1177	-0.3826	-0.717	0.4685	0.3024	840
Hist gradient boos	0.3048	-1.081	-0.1692	-0.5064	-1.149	0.4649	0.6786	840
XGBoost	0.3167	-1.114	-0.1715	-0.5283	-1.113	0.4690	0.6143	840

AUD/USD (AUD_USD) - ternary, lead 5d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | Logistic (sklearn, on global PCA) | SVM (RBF) | MLP neural net | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.3357	0.168	0.0235	-0.3473	0.143	0.5228	0.3917	840
Logistic (sklearn,	0.3167	-0.088	-0.0166	-0.5901	-0.089	0.5055	0.6500	840
SVM (RBF)	0.3488	-0.107	-0.0130	-0.2149	-0.109	0.5097	0.4274	840
MLP neural net	0.3143	-0.164	-0.0140	-0.1985	-0.134	0.4901	0.1798	840
Gradient boosting	0.3357	-0.209	-0.0414	-0.5782	-0.218	0.4991	0.6655	840
Hist gradient boos	0.2774	-0.333	-0.0543	-0.3654	-0.404	0.4853	0.6893	840
Ensemble (mean pro	0.2976	-0.414	-0.0752	-0.6035	-0.397	0.4990	0.5988	840
MNLogit (statsmode	0.3214	-0.495	-0.0902	-0.5859	-0.498	0.4856	0.6202	840
Ensemble (weighted	0.2917	-0.559	-0.0986	-0.6030	-0.529	0.4940	0.5952	840
XGBoost	0.2917	-0.705	-0.1226	-0.5694	-0.753	0.4704	0.6429	840

AUD/USD (AUD_USD) - ternary, lead 6d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | Hist gradient boosting | Random forest | Ensemble (weighted train acc) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.3393	1.303	0.3028	-0.2749	1.897	0.5073	0.7298	840
Hist gradient boos	0.3286	1.076	0.2160	-0.2459	1.677	0.5073	0.7345	840
Random forest	0.3500	0.914	0.1502	-0.3063	1.249	0.4883	0.4583	840
Ensemble (weighted	0.3321	0.900	0.1749	-0.3279	1.301	0.4833	0.6429	840
Ensemble (mean pro	0.3345	0.898	0.1782	-0.3739	1.189	0.4821	0.6321	840
XGBoost	0.3071	0.695	0.1387	-0.3388	1.041	0.4841	0.7131	840
SVM (RBF)	0.3381	0.227	0.0322	-0.3356	0.269	0.4988	0.5107	840
MLP neural net	0.2988	0.199	0.0276	-0.3213	0.184	0.4585	0.2440	840
Logistic (sklearn,	0.3167	0.102	0.0219	-0.6222	0.108	0.4729	0.7024	840
MNLogit (statsmode	0.3214	-0.067	-0.0145	-0.6379	-0.073	0.4723	0.7310	840

AUD/USD (AUD_USD) - ternary, lead 7d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Ensemble (weighted train acc) | Ensemble (mean proba) | XGBoost | SVM (RBF) | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Ensemble (weighted	0.3405	1.100	0.2570	-0.4230	1.274	0.5183	0.6500	840
Ensemble (mean pro	0.3464	0.878	0.1987	-0.4131	1.003	0.5085	0.6274	840
XGBoost	0.3262	0.849	0.1995	-0.4175	1.023	0.5104	0.6881	840
SVM (RBF)	0.3679	0.839	0.1405	-0.2621	0.997	0.5427	0.5155	840
Gradient boosting	0.3488	0.794	0.1900	-0.3866	0.988	0.5000	0.7143	840
Random forest	0.3690	0.762	0.1341	-0.2803	0.975	0.4950	0.4786	840
MNLogit (statsmode	0.3417	0.632	0.1514	-0.5700	0.688	0.5053	0.6714	840
Logistic (sklearn,	0.3321	0.496	0.1171	-0.5923	0.521	0.5036	0.6667	840
Hist gradient boos	0.3190	0.360	0.0775	-0.4121	0.414	0.5042	0.7107	840
MLP neural net	0.3512	-0.145	-0.0116	-0.1703	-0.102	0.4731	0.1107	840

AUD/USD (AUD_USD) - ternary, lead 8d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Logistic (sklearn, on global PCA) | XGBoost | Ensemble (weighted train acc) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.3333	0.949	0.1573	-0.2898	1.202	0.5399	0.5667	840
Logistic (sklearn,	0.3333	0.929	0.2417	-0.6059	1.057	0.5203	0.7345	840
XGBoost	0.3262	0.898	0.2326	-0.6246	1.008	0.5192	0.7131	840
Ensemble (weighted	0.3440	0.848	0.2190	-0.6247	0.903	0.5239	0.6976	840
Hist gradient boos	0.3107	0.815	0.2093	-0.5447	0.981	0.4946	0.7655	840
Random forest	0.3536	0.813	0.1620	-0.4474	0.878	0.5124	0.5298	840
Ensemble (mean pro	0.3393	0.738	0.1844	-0.6143	0.771	0.5248	0.6964	840
MNLogit (statsmode	0.3238	0.607	0.1583	-0.6410	0.688	0.5063	0.7595	840
Gradient boosting	0.3417	0.433	0.1170	-0.6956	0.432	0.5211	0.7333	840
MLP neural net	0.3262	-0.713	-0.0900	-0.4608	-0.301	0.4560	0.1488	840

AUD/USD (AUD_USD) - ternary, lead 9d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | Ensemble (mean proba) | Ensemble (weighted train acc) | Hist gradient boosting | MNLogit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.3500	1.625	0.4162	-0.3212	2.499	0.5363	0.7548	840
Ensemble (mean pro	0.3571	1.539	0.3681	-0.3137	2.179	0.5203	0.6452	840
Ensemble (weighted	0.3524	1.492	0.3610	-0.3204	2.141	0.5207	0.6607	840
Hist gradient boos	0.3595	1.441	0.3818	-0.3800	2.122	0.5150	0.7560	840
MNLogit (statsmode	0.3381	1.408	0.3961	-0.5562	1.657	0.5288	0.7226	840
XGBoost	0.3488	1.391	0.3509	-0.3575	2.063	0.5161	0.7381	840
Logistic (sklearn,	0.3405	1.148	0.3109	-0.5749	1.404	0.5025	0.7274	840
SVM (RBF)	0.3393	1.054	0.2043	-0.3646	1.425	0.5299	0.5976	840
Random forest	0.3702	0.916	0.1902	-0.3338	1.123	0.5077	0.5440	840
MLP neural net	0.3381	0.034	0.0035	-0.3426	0.031	0.4796	0.2631	840

AUD/USD (AUD_USD) - ternary, lead 10d

Rows used: 1000 | Effective end: 2021-12-18

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | MLP neural net | Ensemble (mean proba) | Hist gradient boosting | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.3786	1.373	0.3102	-0.3561	2.002	0.5135	0.6190	840
MLP neural net	0.4167	1.305	0.3928	-0.6938	0.982	0.5585	0.4988	840
Ensemble (mean pro	0.3762	0.605	0.1861	-0.7483	0.608	0.4917	0.7167	840
Hist gradient boos	0.3714	0.577	0.1815	-0.7891	0.617	0.4916	0.7821	840
Ensemble (weighted	0.3750	0.509	0.1545	-0.7562	0.515	0.4877	0.7250	840
XGBoost	0.3774	0.401	0.1172	-0.7590	0.408	0.4914	0.7655	840
Random forest	0.3738	0.213	0.0504	-0.6102	0.205	0.4884	0.6167	840
MNLogit (statsmode	0.3607	0.192	0.0545	-0.7652	0.190	0.4839	0.7036	840
Logistic (sklearn,	0.3500	-0.018	-0.0049	-0.7652	-0.017	0.4611	0.6429	840
Gradient boosting	0.3500	-0.100	-0.0262	-0.6469	-0.105	0.4971	0.8167	840

NZD/USD (NZD_USD) - binary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logit (statsmodels, PCA) | Probit (statsmodels, PCA) | MLP neural net | SVM (RBF) | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logit (statsmodels	0.4952	0.088	0.0077	-0.1961	0.121	0.4969	0.9488	840
Probit (statsmodel	0.4976	0.027	0.0023	-0.1993	0.037	0.4931	0.9440	840
MLP neural net	0.4917	-0.201	-0.0171	-0.1757	-0.289	0.4960	0.9000	840
SVM (RBF)	0.5131	-0.314	-0.0266	-0.1905	-0.435	0.5020	0.8988	840
Logistic (sklearn,	0.4738	-0.342	-0.0295	-0.1879	-0.489	0.4701	0.9345	840
Hist gradient boos	0.4905	-0.525	-0.0453	-0.2779	-0.696	0.4886	0.9893	840
Random forest	0.5036	-0.615	-0.0525	-0.2415	-0.808	0.5000	0.9548	840
Ensemble (mean pro	0.4845	-0.710	-0.0603	-0.2420	-0.944	0.4834	0.9679	840
Ensemble (weighted	0.4821	-0.731	-0.0621	-0.2463	-0.969	0.4846	0.9655	840
Gradient boosting	0.5048	-0.775	-0.0648	-0.2866	-1.040	0.5043	0.9798	840
XGBoost	0.4738	-1.108	-0.0937	-0.3231	-1.470	0.4765	0.9869	840

NZD/USD (NZD_USD) - binary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logit (statsmodels, PCA) | MLP neural net | Logistic (sklearn, on global PCA) | Probit (statsmodels, PCA) | SVM (RBF)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logit (statsmodels	0.4917	0.258	0.0333	-0.1786	0.392	0.4914	0.9643	840
MLP neural net	0.4821	0.201	0.0253	-0.2436	0.305	0.4772	0.9381	840
Logistic (sklearn,	0.4833	0.169	0.0215	-0.2297	0.264	0.4858	0.9655	840
Probit (statsmodel	0.4869	0.032	0.0040	-0.2243	0.048	0.4827	0.9619	840
SVM (RBF)	0.4821	0.029	0.0037	-0.1894	0.046	0.4782	0.9560	840
Hist gradient boos	0.4869	-0.007	-0.0009	-0.2644	-0.010	0.4850	0.9940	840
Random forest	0.4881	-0.183	-0.0222	-0.3350	-0.263	0.4840	0.9643	840
Gradient boosting	0.5012	-0.269	-0.0340	-0.3119	-0.369	0.5030	0.9917	840
Ensemble (mean pro	0.4762	-0.348	-0.0432	-0.3433	-0.508	0.4736	0.9679	840
Ensemble (weighted	0.4702	-0.547	-0.0675	-0.3520	-0.796	0.4681	0.9690	840
XGBoost	0.4702	-0.687	-0.0835	-0.3809	-0.953	0.4677	0.9774	840

NZD/USD (NZD_USD) - binary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | MLP neural net | Gradient boosting (sklearn) | Probit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logit (statsmodels	0.4679	0.123	0.0198	-0.3650	0.197	0.4720	0.9762	840
Logistic (sklearn,	0.4738	0.116	0.0186	-0.2448	0.190	0.4707	0.9762	840
MLP neural net	0.4857	0.109	0.0170	-0.2984	0.176	0.4864	0.9202	840
Gradient boosting	0.4869	0.048	0.0078	-0.3771	0.070	0.4862	0.9917	840
Probit (statsmodel	0.4714	0.008	0.0013	-0.3801	0.013	0.4682	0.9738	840
Ensemble (mean pro	0.4619	-0.400	-0.0622	-0.3644	-0.598	0.4600	0.9810	840
Ensemble (weighted	0.4619	-0.600	-0.0909	-0.3930	-0.865	0.4592	0.9774	840
SVM (RBF)	0.4548	-0.688	-0.1042	-0.4034	-1.080	0.4509	0.9690	840
Hist gradient boos	0.4595	-1.034	-0.1537	-0.5341	-1.431	0.4589	0.9988	840
XGBoost	0.4548	-1.115	-0.1640	-0.5224	-1.546	0.4530	0.9881	840
Random forest	0.4607	-1.368	-0.1961	-0.5979	-1.788	0.4568	0.9643	840

NZD/USD (NZD_USD) - binary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | Logit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | Probit (statsmodels, PCA) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.5095	0.809	0.1598	-0.4872	1.347	0.5142	0.9238	840
Logit (statsmodels	0.5119	0.792	0.1596	-0.2775	1.279	0.5116	0.9774	840
Logistic (sklearn,	0.5107	0.768	0.1552	-0.3419	1.278	0.5109	0.9857	840
Probit (statsmodel	0.5060	0.663	0.1322	-0.2836	1.067	0.5061	0.9738	840
Ensemble (mean pro	0.4905	0.450	0.0877	-0.2830	0.736	0.4896	0.9750	840
Ensemble (weighted	0.4857	0.290	0.0561	-0.3279	0.476	0.4849	0.9869	840
Hist gradient boos	0.4821	-0.239	-0.0441	-0.3704	-0.358	0.4815	0.9964	840
SVM (RBF)	0.4560	-0.605	-0.1072	-0.5016	-0.938	0.4628	0.9750	840
Gradient boosting	0.4952	-0.638	-0.1131	-0.4967	-0.879	0.4958	0.9917	840
XGBoost	0.4571	-0.878	-0.1523	-0.5343	-1.315	0.4567	0.9905	840
Random forest	0.4762	-1.335	-0.2134	-0.6327	-1.752	0.4699	0.9500	840

NZD/USD (NZD_USD) - binary, lead 5d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | Logit (statsmodels, PCA) | Probit (statsmodels, PCA) | Ensemble (mean proba) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.5298	0.770	0.1763	-0.4100	1.161	0.5268	0.9786	840
Logit (statsmodels	0.5131	0.384	0.0858	-0.4991	0.568	0.5144	0.9905	840
Probit (statsmodel	0.5095	0.328	0.0728	-0.5213	0.486	0.5103	0.9869	840
Ensemble (mean pro	0.4905	0.196	0.0428	-0.4809	0.300	0.4909	0.9845	840
Ensemble (weighted	0.4881	0.165	0.0359	-0.4657	0.253	0.4909	0.9845	840
XGBoost	0.4964	0.110	0.0240	-0.3945	0.167	0.4964	0.9881	840
Random forest	0.4869	0.010	0.0021	-0.4776	0.015	0.4883	0.9702	840
Hist gradient boos	0.4929	-0.008	-0.0016	-0.3888	-0.011	0.4940	0.9929	840
MLP neural net	0.4810	-0.472	-0.0935	-0.4951	-0.658	0.4733	0.9155	840
Gradient boosting	0.4690	-0.589	-0.1189	-0.4892	-0.914	0.4707	0.9964	840
SVM (RBF)	0.4750	-0.622	-0.1247	-0.5520	-0.973	0.4714	0.9798	840

NZD/USD (NZD_USD) - binary, lead 6d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | Probit (statsmodels, PCA) | Logit (statsmodels, PCA) | Hist gradient boosting | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.5024	0.131	0.0309	-0.5021	0.214	0.5030	0.9893	840
Probit (statsmodel	0.4952	0.128	0.0301	-0.5021	0.209	0.4970	0.9798	840
Logit (statsmodels	0.4988	0.108	0.0251	-0.5046	0.180	0.4976	0.9810	840
Hist gradient boos	0.4893	-0.201	-0.0458	-0.4033	-0.323	0.4898	0.9964	840
XGBoost	0.4798	-0.392	-0.0870	-0.4370	-0.612	0.4784	0.9929	840
Ensemble (mean pro	0.4738	-0.452	-0.0994	-0.5595	-0.738	0.4746	0.9833	840
Ensemble (weighted	0.4750	-0.538	-0.1174	-0.5574	-0.879	0.4711	0.9881	840
Gradient boosting	0.4869	-0.597	-0.1300	-0.5262	-0.883	0.4863	0.9964	840
Random forest	0.4488	-1.323	-0.2559	-0.6992	-2.063	0.4501	0.9655	840
SVM (RBF)	0.4524	-1.555	-0.3022	-0.7715	-2.379	0.4510	0.9845	840
MLP neural net	0.4488	-1.600	-0.2981	-0.7961	-2.481	0.4488	0.9417	840

NZD/USD (NZD_USD) - binary, lead 7d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Gradient boosting (sklearn) | Hist gradient boosting | Random forest | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.5202	1.079	0.3100	-0.4042	1.789	0.5203	0.9952	840
Gradient boosting	0.5179	0.909	0.2556	-0.4089	1.381	0.5179	0.9952	840
Hist gradient boos	0.5048	0.876	0.2455	-0.3887	1.525	0.5042	0.9964	840
Random forest	0.5107	0.760	0.2058	-0.5124	1.231	0.5092	0.9750	840
Ensemble (weighted	0.4976	0.588	0.1576	-0.5400	1.015	0.4988	0.9810	840
Ensemble (mean pro	0.4988	0.476	0.1241	-0.5570	0.813	0.4982	0.9893	840
MLP neural net	0.5131	0.364	0.0916	-0.4875	0.561	0.5052	0.9095	840
SVM (RBF)	0.4738	-0.551	-0.1287	-0.6607	-0.861	0.4716	0.9845	840
Logistic (sklearn,	0.4821	-0.555	-0.1295	-0.6668	-0.869	0.4782	0.9833	840
Probit (statsmodel	0.4714	-0.622	-0.1433	-0.7093	-0.979	0.4724	0.9905	840
Logit (statsmodels	0.4750	-0.718	-0.1645	-0.7293	-1.121	0.4736	0.9929	840

NZD/USD (NZD_USD) - binary, lead 8d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Ensemble (weighted train acc) | Logistic (sklearn, on global PCA) | Ensemble (mean proba) | Probit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.5381	1.486	0.4619	-0.5177	2.340	0.5380	0.9869	840
Ensemble (weighted	0.5012	0.465	0.1277	-0.6807	0.813	0.5037	0.9762	840
Logistic (sklearn,	0.4976	0.362	0.0993	-0.6761	0.622	0.4982	0.9893	840
Ensemble (mean pro	0.5000	0.346	0.0938	-0.6976	0.606	0.5000	0.9810	840
Probit (statsmodel	0.4905	0.321	0.0881	-0.6981	0.555	0.4934	0.9917	840
Logit (statsmodels	0.4929	0.310	0.0847	-0.6945	0.530	0.4958	0.9893	840
Random forest	0.5226	0.156	0.0406	-0.6843	0.215	0.5207	0.9786	840
Hist gradient boos	0.5131	0.013	0.0034	-0.7040	0.017	0.5119	0.9976	840
SVM (RBF)	0.4964	-0.297	-0.0634	-0.6130	-0.453	0.4994	0.9298	840
MLP neural net	0.4905	-0.540	-0.1266	-0.7337	-0.816	0.4805	0.8845	840
Gradient boosting	0.4786	-0.785	-0.1862	-0.7775	-1.113	0.4785	0.9952	840

NZD/USD (NZD_USD) - binary, lead 9d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Random forest | Hist gradient boosting | Gradient boosting (sklearn) | SVM (RBF)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.5190	1.092	0.3494	-0.5302	1.861	0.5204	0.9905	840
Random forest	0.5036	0.886	0.2753	-0.5953	1.562	0.5048	0.9857	840
Hist gradient boos	0.5167	0.668	0.2012	-0.6240	1.086	0.5191	0.9952	840
Gradient boosting	0.4738	-0.204	-0.0545	-0.5886	-0.347	0.4731	0.9940	840
SVM (RBF)	0.4857	-0.268	-0.0709	-0.7512	-0.450	0.4856	0.9905	840
Ensemble (weighted	0.4833	-0.345	-0.0905	-0.7673	-0.557	0.4850	0.9893	840
Ensemble (mean pro	0.4810	-0.426	-0.1105	-0.7761	-0.688	0.4819	0.9881	840
Probit (statsmodel	0.4821	-0.556	-0.1422	-0.7967	-0.912	0.4814	0.9917	840
Logistic (sklearn,	0.4702	-0.560	-0.1421	-0.7923	-0.927	0.4697	0.9810	840
Logit (statsmodels	0.4726	-0.801	-0.1977	-0.8306	-1.293	0.4753	0.9869	840
MLP neural net	0.4726	-1.151	-0.2635	-0.8105	-1.796	0.4726	0.9345	840

NZD/USD (NZD_USD) - binary, lead 10d

Rows used: 1000 | Effective end: 2021-12-18

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | SVM (RBF) | Ensemble (weighted train acc) | MLP neural net | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.5190	0.631	0.1978	-0.6533	1.096	0.5179	0.9976	840
SVM (RBF)	0.5083	0.009	0.0026	-0.7105	0.016	0.5072	0.9881	840
Ensemble (weighted	0.4917	-0.153	-0.0424	-0.7250	-0.262	0.4922	0.9869	840
MLP neural net	0.5060	-0.157	-0.0426	-0.6709	-0.248	0.5077	0.9286	840
Ensemble (mean pro	0.4893	-0.302	-0.0823	-0.7469	-0.518	0.4872	0.9798	840
Hist gradient boos	0.4905	-0.660	-0.1724	-0.8543	-0.864	0.4905	1.0000	840
Random forest	0.5036	-0.721	-0.1856	-0.8709	-0.930	0.5012	0.9881	840
Logit (statsmodels	0.4679	-1.234	-0.2967	-0.9134	-2.013	0.4682	0.9917	840
XGBoost	0.4821	-1.328	-0.3160	-0.8805	-1.719	0.4833	0.9976	840
Logistic (sklearn,	0.4548	-1.415	-0.3310	-0.9089	-2.380	0.4544	0.9929	840
Probit (statsmodel	0.4571	-1.432	-0.3355	-0.9166	-2.344	0.4587	0.9940	840

NZD/USD (NZD_USD) - ternary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Ensemble (weighted train acc) | MNLogit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.6238	0.505	0.0124	-0.0221	0.225	0.5663	0.0988	840
Ensemble (weighted	0.6274	0.041	0.0007	-0.0288	0.013	0.6182	0.0655	840
MNLogit (statsmode	0.5905	-0.009	-0.0003	-0.0509	-0.006	0.5156	0.1524	840
Logistic (sklearn,	0.3488	-0.164	-0.0088	-0.0926	-0.156	0.5230	0.4143	840
Ensemble (mean pro	0.6202	-0.227	-0.0037	-0.0295	-0.065	0.5957	0.0560	840
Hist gradient boos	0.5595	-0.357	-0.0131	-0.0830	-0.200	0.5367	0.2107	840
Gradient boosting	0.5286	-1.115	-0.0545	-0.1888	-0.800	0.4650	0.2893	840
Random forest	0.6881	NA	0.0000	0.0000	NA	NA	0.0000	840
MLP neural net	0.6929	NA	0.0000	0.0000	NA	NA	0.0000	840
SVM (RBF)	0.6964	NA	0.0000	0.0000	NA	NA	0.0000	840

NZD/USD (NZD_USD) - ternary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | Hist gradient boosting | XGBoost | MLP neural net | SVM (RBF)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.4417	0.706	0.0595	-0.1058	0.824	0.4872	0.3714	840
Hist gradient boos	0.4048	0.606	0.0438	-0.1413	0.721	0.4839	0.4060	840
XGBoost	0.4321	0.503	0.0303	-0.1331	0.502	0.4781	0.2714	840
MLP neural net	0.5167	0.404	0.0106	-0.0527	0.473	0.5000	0.0119	840
SVM (RBF)	0.5060	0.327	0.0048	-0.0140	0.062	0.3333	0.0071	840
Ensemble (weighted	0.4381	0.132	0.0069	-0.1100	0.109	0.4345	0.2000	840
Ensemble (mean pro	0.4393	-0.071	-0.0033	-0.1169	-0.056	0.4097	0.1714	840
Logistic (sklearn,	0.3583	-0.309	-0.0229	-0.1802	-0.344	0.4548	0.3952	840
Random forest	0.4774	-0.603	-0.0190	-0.0933	-0.265	0.3600	0.0595	840
MNLogit (statsmode	0.4393	-0.608	-0.0363	-0.1662	-0.545	0.4085	0.2536	840

NZD/USD (NZD_USD) - ternary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MNLogit (statsmodels, PCA) | Ensemble (weighted train acc) | Logistic (sklearn, on global PCA) | Ensemble (mean proba) | MLP neural net

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MNLogit (statsmode	0.3845	0.758	0.0724	-0.1490	0.797	0.4982	0.3250	840
Ensemble (weighted	0.3583	0.638	0.0581	-0.1719	0.691	0.4685	0.3405	840
Logistic (sklearn,	0.3536	0.470	0.0496	-0.1630	0.550	0.4863	0.4357	840
Ensemble (mean pro	0.3607	0.407	0.0347	-0.1610	0.409	0.4627	0.3036	840
MLP neural net	0.4298	0.387	0.0122	-0.0256	0.090	0.5714	0.0167	840
SVM (RBF)	0.3988	0.182	0.0088	-0.0777	0.093	0.4605	0.0905	840
Hist gradient boos	0.3417	-0.033	-0.0039	-0.3282	-0.039	0.4577	0.5202	840
Gradient boosting	0.3738	-0.046	-0.0054	-0.2547	-0.045	0.4662	0.4929	840
Random forest	0.3571	-0.070	-0.0042	-0.0846	-0.044	0.4815	0.1607	840
XGBoost	0.3500	-0.125	-0.0128	-0.2358	-0.137	0.4500	0.4524	840

NZD/USD (NZD_USD) - ternary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MNLogit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | MLP neural net | Ensemble (mean proba) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MNLogit (statsmode	0.3548	0.889	0.1152	-0.2322	1.129	0.5269	0.5083	840
Logistic (sklearn,	0.3536	0.877	0.1118	-0.2336	1.124	0.5253	0.4940	840
MLP neural net	0.3369	-0.230	-0.0115	-0.2125	-0.094	0.5000	0.0881	840
Ensemble (mean pro	0.3429	-0.237	-0.0279	-0.2461	-0.289	0.4609	0.4571	840
Hist gradient boos	0.3238	-0.288	-0.0418	-0.3143	-0.381	0.4743	0.6250	840
Ensemble (weighted	0.3381	-0.303	-0.0367	-0.2580	-0.381	0.4621	0.4869	840
Gradient boosting	0.3286	-0.334	-0.0470	-0.3198	-0.385	0.4776	0.5857	840
XGBoost	0.3333	-0.605	-0.0780	-0.3309	-0.739	0.4515	0.5643	840
SVM (RBF)	0.3179	-0.963	-0.0849	-0.3174	-0.763	0.4545	0.2750	840
Random forest	0.3286	-1.185	-0.1001	-0.3265	-1.106	0.4070	0.3071	840

NZD/USD (NZD_USD) - ternary, lead 5d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MNLogit (statsmodels, PCA) | Hist gradient boosting | XGBoost | Logistic (sklearn, on global PCA) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MNLogit (statsmode	0.3405	0.417	0.0512	-0.3039	0.501	0.5344	0.5012	840
Hist gradient boos	0.3190	0.305	0.0483	-0.4031	0.423	0.5102	0.6417	840
XGBoost	0.3262	0.195	0.0300	-0.3568	0.268	0.5078	0.6119	840
Logistic (sklearn,	0.3250	0.183	0.0210	-0.3039	0.219	0.5296	0.4631	840
Ensemble (mean pro	0.3369	0.126	0.0156	-0.2814	0.144	0.5239	0.4976	840
Ensemble (weighted	0.3286	-0.057	-0.0073	-0.3130	-0.066	0.5191	0.5298	840
Gradient boosting	0.3476	-0.121	-0.0186	-0.3010	-0.167	0.5072	0.6619	840
SVM (RBF)	0.3048	-0.214	-0.0258	-0.3786	-0.229	0.5071	0.4202	840
Random forest	0.3333	-0.348	-0.0373	-0.3343	-0.335	0.4883	0.3560	840
MLP neural net	0.3274	-1.756	-0.1124	-0.3519	-1.194	0.3659	0.1464	840

NZD/USD (NZD_USD) - ternary, lead 6d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Hist gradient boosting | Gradient boosting (sklearn) | MLP neural net | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.3488	1.050	0.1838	-0.3396	1.593	0.5269	0.5964	840
Hist gradient boos	0.3321	0.263	0.0465	-0.3332	0.391	0.4845	0.6536	840
Gradient boosting	0.3381	0.172	0.0296	-0.4273	0.252	0.4807	0.6488	840
MLP neural net	0.3452	-0.019	-0.0017	-0.2085	-0.015	0.4690	0.1726	840
Ensemble (weighted	0.3321	-0.192	-0.0261	-0.3553	-0.246	0.4896	0.5155	840
Logistic (sklearn,	0.3179	-0.222	-0.0287	-0.3849	-0.290	0.4977	0.5167	840
Ensemble (mean pro	0.3357	-0.237	-0.0314	-0.3801	-0.300	0.4893	0.5012	840
MNLogit (statsmode	0.3119	-0.265	-0.0365	-0.3823	-0.339	0.5021	0.5690	840
Random forest	0.3333	-0.387	-0.0444	-0.3135	-0.420	0.4610	0.3512	840
SVM (RBF)	0.3000	-1.321	-0.1597	-0.5099	-1.409	0.4450	0.4548	840

NZD/USD (NZD_USD) - ternary, lead 7d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | Logistic (sklearn, on global PCA) | XGBoost | Ensemble (mean proba) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.3607	1.078	0.2488	-0.3185	1.595	0.5209	0.6833	840
Logistic (sklearn,	0.3548	0.954	0.1709	-0.3860	1.373	0.5130	0.5036	840
XGBoost	0.3607	0.951	0.2126	-0.3569	1.308	0.5260	0.6881	840
Ensemble (mean pro	0.3607	0.862	0.1510	-0.4102	1.259	0.5076	0.5488	840
Ensemble (weighted	0.3631	0.622	0.1123	-0.4045	0.755	0.5145	0.5738	840
MNLogit (statsmode	0.3393	0.586	0.1057	-0.5214	0.887	0.4959	0.5857	840
Random forest	0.3369	0.525	0.0756	-0.3594	0.562	0.5130	0.4131	840
MLP neural net	0.3155	0.307	0.0366	-0.2328	0.233	0.5251	0.2131	840
Gradient boosting	0.3202	-0.316	-0.0628	-0.5858	-0.375	0.4856	0.7012	840
SVM (RBF)	0.3143	-0.538	-0.0698	-0.4355	-0.585	0.4856	0.4536	840

NZD/USD (NZD_USD) - ternary, lead 8d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | XGBoost | Ensemble (mean proba) | Ensemble (weighted train acc) | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.3298	1.281	0.2926	-0.2916	1.785	0.5309	0.6929	840
XGBoost	0.3393	0.835	0.1851	-0.4768	1.122	0.5232	0.6940	840
Ensemble (mean pro	0.3357	0.664	0.1438	-0.4858	0.909	0.5094	0.5726	840
Ensemble (weighted	0.3310	0.414	0.0816	-0.4720	0.530	0.5092	0.5821	840
Gradient boosting	0.3429	0.407	0.0879	-0.4840	0.517	0.5141	0.6738	840
Random forest	0.3107	0.074	0.0118	-0.5814	0.076	0.5078	0.4571	840
MLP neural net	0.3345	0.068	0.0125	-0.5268	0.060	0.4779	0.2964	840
Logistic (sklearn,	0.3060	-0.111	-0.0206	-0.5897	-0.152	0.4836	0.5071	840
MNLogit (statsmode	0.3060	-0.416	-0.0801	-0.6392	-0.579	0.4743	0.6024	840
SVM (RBF)	0.3393	-1.259	-0.1882	-0.6596	-1.365	0.4835	0.5417	840

NZD/USD (NZD_USD) - ternary, lead 9d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | MLP neural net | Random forest | Hist gradient boosting | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.3393	0.106	0.0203	-0.6297	0.130	0.5009	0.6560	840
MLP neural net	0.3452	-0.152	-0.0286	-0.5169	-0.154	0.4655	0.3274	840
Random forest	0.3345	-0.162	-0.0251	-0.6979	-0.159	0.4975	0.4857	840
Hist gradient boos	0.3071	-0.356	-0.0664	-0.6763	-0.455	0.4792	0.6881	840
Gradient boosting	0.2893	-0.579	-0.1021	-0.6745	-0.804	0.4598	0.6964	840
SVM (RBF)	0.3238	-0.588	-0.0999	-0.6519	-0.652	0.5022	0.5476	840
Ensemble (weighted	0.3190	-0.665	-0.1100	-0.7062	-0.791	0.4677	0.5905	840
Ensemble (mean pro	0.3131	-0.688	-0.1109	-0.7155	-0.797	0.4688	0.5714	840
Logistic (sklearn,	0.2786	-0.744	-0.1210	-0.7321	-0.999	0.4621	0.5179	840
MNLogit (statsmode	0.2869	-1.349	-0.2228	-0.7632	-1.861	0.4469	0.5940	840

NZD/USD (NZD_USD) - ternary, lead 10d

Rows used: 1000 | Effective end: 2021-12-18

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | Random forest | Gradient boosting (sklearn) | XGBoost | SVM (RBF)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.3679	0.051	0.0099	-0.4672	0.049	0.5094	0.3179	840
Random forest	0.3274	-0.574	-0.0979	-0.7392	-0.647	0.4836	0.5071	840
Gradient boosting	0.3036	-0.887	-0.1828	-0.7672	-1.075	0.4717	0.7143	840
XGBoost	0.3119	-0.983	-0.1872	-0.6974	-1.280	0.4550	0.6750	840
SVM (RBF)	0.3143	-0.988	-0.1697	-0.7239	-1.145	0.4727	0.5893	840
Hist gradient boos	0.3179	-1.159	-0.2173	-0.7102	-1.498	0.4553	0.6798	840
Logistic (sklearn,	0.2821	-1.801	-0.2748	-0.8090	-2.380	0.4355	0.6095	840
Ensemble (weighted	0.2988	-1.817	-0.2940	-0.7825	-2.168	0.4343	0.5893	840
Ensemble (mean pro	0.3000	-2.056	-0.3223	-0.7973	-2.422	0.4236	0.5762	840
MNLogit (statsmode	0.2893	-2.072	-0.3252	-0.8600	-2.727	0.4404	0.6595	840

USD/CAD (USD_CAD) - binary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | Ensemble (mean proba) | Probit (statsmodels, PCA) | Logit (statsmodels, PCA) | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.5345	0.824	0.0516	-0.0728	1.182	0.5351	0.9500	840
Ensemble (mean pro	0.5190	0.585	0.0366	-0.0747	0.842	0.5256	0.9536	840
Probit (statsmodel	0.5202	0.462	0.0289	-0.1000	0.654	0.5237	0.9548	840
Logit (statsmodels	0.5250	0.401	0.0249	-0.0987	0.571	0.5218	0.9560	840
Gradient boosting	0.5119	0.345	0.0218	-0.0817	0.481	0.5102	0.9917	840
MLP neural net	0.5310	0.118	0.0070	-0.0766	0.158	0.5245	0.8762	840
SVM (RBF)	0.5060	0.073	0.0042	-0.0923	0.091	0.5087	0.8190	840
XGBoost	0.5119	0.064	0.0040	-0.0723	0.091	0.5139	0.9821	840
Hist gradient boos	0.5119	0.019	0.0012	-0.0711	0.027	0.5115	0.9798	840
Ensemble (weighted	0.5155	-0.016	-0.0010	-0.1138	-0.021	0.5161	0.9595	840
Random forest	0.5083	-0.298	-0.0181	-0.1329	-0.390	0.5094	0.9536	840

USD/CAD (USD_CAD) - binary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | SVM (RBF) | Hist gradient boosting | Logit (statsmodels, PCA) | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.5488	1.329	0.1197	-0.1030	1.952	0.5572	0.8952	840
SVM (RBF)	0.5405	1.156	0.1071	-0.1780	1.515	0.5422	0.9595	840
Hist gradient boos	0.5179	1.107	0.1037	-0.1724	1.602	0.5186	0.9940	840
Logit (statsmodels	0.5131	1.107	0.1016	-0.1260	1.757	0.5168	0.9583	840
Gradient boosting	0.5226	1.045	0.0978	-0.1481	1.501	0.5193	0.9881	840
Ensemble (weighted	0.5155	0.882	0.0813	-0.1415	1.241	0.5171	0.9738	840
Probit (statsmodel	0.5167	0.865	0.0795	-0.1274	1.330	0.5112	0.9595	840
Random forest	0.5202	0.821	0.0746	-0.1547	1.055	0.5218	0.9536	840
Ensemble (mean pro	0.5060	0.775	0.0687	-0.1453	1.160	0.5086	0.9667	840
Logistic (sklearn,	0.5000	0.744	0.0673	-0.1264	1.216	0.4950	0.9619	840
XGBoost	0.4964	0.138	0.0124	-0.1913	0.192	0.4982	0.9845	840

USD/CAD (USD_CAD) - binary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | MLP neural net | Ensemble (weighted train acc) | Random forest | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.5226	0.756	0.0849	-0.2062	1.044	0.5216	0.9905	840
MLP neural net	0.4988	0.670	0.0727	-0.2027	1.064	0.5033	0.9083	840
Ensemble (weighted	0.5190	0.594	0.0660	-0.2349	0.807	0.5189	0.9750	840
Random forest	0.5107	0.531	0.0583	-0.1757	0.719	0.5137	0.9548	840
Hist gradient boos	0.5000	0.484	0.0538	-0.2398	0.686	0.5012	0.9929	840
Gradient boosting	0.5190	0.424	0.0469	-0.1538	0.569	0.5211	0.9869	840
Ensemble (mean pro	0.5071	0.364	0.0400	-0.2559	0.496	0.5085	0.9762	840
SVM (RBF)	0.5262	0.234	0.0255	-0.2317	0.310	0.5263	0.9750	840
Logistic (sklearn,	0.4905	0.051	0.0055	-0.2365	0.079	0.4891	0.9786	840
Logit (statsmodels	0.4917	-0.094	-0.0100	-0.2314	-0.146	0.4872	0.9750	840
Probit (statsmodel	0.4893	-0.130	-0.0139	-0.2315	-0.202	0.4866	0.9786	840

USD/CAD (USD_CAD) - binary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Probit (statsmodels, PCA) | Logit (statsmodels, PCA) | Random forest | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.5369	1.822	0.2451	-0.1584	2.912	0.5385	0.9750	840
Probit (statsmodel	0.5119	0.657	0.0827	-0.2146	1.031	0.5135	0.9714	840
Logit (statsmodels	0.5155	0.651	0.0820	-0.2146	1.021	0.5153	0.9750	840
Random forest	0.5179	0.637	0.0772	-0.2362	0.915	0.5160	0.9643	840
Logistic (sklearn,	0.5060	0.487	0.0607	-0.2380	0.767	0.5061	0.9762	840
Hist gradient boos	0.4940	0.413	0.0514	-0.2011	0.591	0.4909	0.9845	840
Ensemble (mean pro	0.4940	0.300	0.0370	-0.2175	0.478	0.4933	0.9845	840
Ensemble (weighted	0.4893	0.228	0.0279	-0.2498	0.366	0.4860	0.9750	840
XGBoost	0.5012	0.212	0.0262	-0.1746	0.303	0.5012	0.9929	840
MLP neural net	0.5262	-0.028	-0.0034	-0.5127	-0.033	0.5309	0.9060	840
Gradient boosting	0.4869	-0.491	-0.0581	-0.3107	-0.694	0.4880	0.9952	840

USD/CAD (USD_CAD) - binary, lead 5d

Rows used: 1000 | Effective end: 2021-12-25

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Hist gradient boosting | XGBoost | Random forest | MLP neural net

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.5393	2.363	0.3597	-0.1973	4.015	0.5392	0.9560	840
Hist gradient boos	0.5298	1.493	0.2226	-0.2036	2.289	0.5287	0.9952	840
XGBoost	0.5321	1.258	0.1832	-0.1887	1.880	0.5313	0.9881	840
Random forest	0.5250	1.231	0.1762	-0.1887	1.866	0.5257	0.9714	840
MLP neural net	0.5238	1.120	0.1592	-0.2419	1.632	0.5281	0.9107	840
Ensemble (weighted	0.5083	0.647	0.0905	-0.2472	0.915	0.5121	0.9810	840
Gradient boosting	0.5119	0.579	0.0813	-0.2364	0.858	0.5150	0.9940	840
Logistic (sklearn,	0.5012	0.509	0.0703	-0.3512	0.703	0.5012	0.9833	840
Ensemble (mean pro	0.5107	0.470	0.0637	-0.2920	0.651	0.5110	0.9714	840
Logit (statsmodels	0.5083	0.470	0.0650	-0.3431	0.635	0.5067	0.9726	840
Probit (statsmodel	0.5083	0.442	0.0611	-0.3431	0.600	0.5067	0.9774	840

USD/CAD (USD_CAD) - binary, lead 6d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Random forest | Hist gradient boosting | Gradient boosting (sklearn) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.5750	2.704	0.4726	-0.2177	4.410	0.5768	0.9845	840
Random forest	0.5631	2.434	0.4103	-0.2142	3.810	0.5683	0.9679	840
Hist gradient boos	0.5560	2.036	0.3459	-0.2264	3.120	0.5566	0.9881	840
Gradient boosting	0.5548	1.535	0.2533	-0.2014	2.175	0.5538	0.9952	840
Ensemble (weighted	0.5560	1.204	0.1921	-0.3448	1.589	0.5526	0.9845	840
Ensemble (mean pro	0.5500	1.152	0.1830	-0.3727	1.521	0.5504	0.9798	840
MLP neural net	0.5226	0.853	0.1304	-0.3634	1.121	0.5256	0.9060	840
SVM (RBF)	0.5369	0.842	0.1317	-0.3851	1.103	0.5377	0.9786	840
Logistic (sklearn,	0.5167	-0.055	-0.0081	-0.4552	-0.073	0.5157	0.9833	840
Logit (statsmodels	0.5167	-0.062	-0.0092	-0.4770	-0.083	0.5199	0.9869	840
Probit (statsmodel	0.5155	-0.230	-0.0333	-0.4881	-0.304	0.5163	0.9845	840

USD/CAD (USD_CAD) - binary, lead 7d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | Ensemble (weighted train acc) | Ensemble (mean proba) | XGBoost | SVM (RBF)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.5345	1.430	0.2537	-0.4432	1.937	0.5361	0.9905	840
Ensemble (weighted	0.5274	1.174	0.2047	-0.5051	1.556	0.5291	0.9810	840
Ensemble (mean pro	0.5226	1.149	0.2002	-0.4986	1.528	0.5267	0.9810	840
XGBoost	0.5274	1.138	0.1981	-0.4215	1.563	0.5271	0.9893	840
SVM (RBF)	0.5298	1.072	0.1857	-0.4549	1.415	0.5319	0.9893	840
Gradient boosting	0.5131	1.005	0.1733	-0.3912	1.408	0.5150	0.9940	840
Random forest	0.5226	0.980	0.1674	-0.3325	1.429	0.5147	0.9714	840
MLP neural net	0.5226	0.861	0.1424	-0.5324	1.086	0.5270	0.9262	840
Probit (statsmodel	0.5143	0.633	0.1058	-0.4750	0.841	0.5128	0.9774	840
Logit (statsmodels	0.5143	0.566	0.0941	-0.4795	0.750	0.5115	0.9798	840
Logistic (sklearn,	0.4952	-0.117	-0.0183	-0.5721	-0.157	0.4933	0.9821	840

USD/CAD (USD_CAD) - binary, lead 8d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | Gradient boosting (sklearn) | Hist gradient boosting | XGBoost | SVM (RBF)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.5131	1.185	0.2180	-0.2772	1.857	0.5153	0.9750	840
Gradient boosting	0.5024	0.937	0.1720	-0.3319	1.552	0.5018	0.9988	840
Hist gradient boos	0.5357	0.919	0.1683	-0.4965	1.217	0.5341	0.9940	840
XGBoost	0.5179	0.826	0.1495	-0.3597	1.266	0.5174	0.9917	840
SVM (RBF)	0.5250	0.814	0.1471	-0.5116	1.057	0.5279	0.9810	840
Ensemble (weighted	0.5262	0.724	0.1296	-0.5324	0.940	0.5286	0.9798	840
Ensemble (mean pro	0.5274	0.708	0.1265	-0.5297	0.920	0.5291	0.9810	840
MLP neural net	0.5345	0.542	0.0941	-0.5544	0.665	0.5352	0.9298	840
Logistic (sklearn,	0.5167	-0.272	-0.0450	-0.5924	-0.351	0.5176	0.9821	840
Logit (statsmodels	0.5155	-0.284	-0.0467	-0.5344	-0.366	0.5127	0.9821	840
Probit (statsmodel	0.5143	-0.313	-0.0513	-0.5355	-0.404	0.5109	0.9857	840

USD/CAD (USD_CAD) - binary, lead 9d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | MLP neural net | Gradient boosting (sklearn) | Ensemble (weighted train acc) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.5393	0.915	0.1727	-0.3145	1.351	0.5408	0.9619	840
MLP neural net	0.5524	0.808	0.1510	-0.5271	0.950	0.5483	0.9357	840
Gradient boosting	0.5417	0.612	0.1155	-0.3776	0.819	0.5436	0.9964	840
Ensemble (weighted	0.5417	0.302	0.0553	-0.5169	0.374	0.5409	0.9905	840
Ensemble (mean pro	0.5429	0.259	0.0470	-0.5211	0.319	0.5399	0.9857	840
XGBoost	0.5429	0.247	0.0451	-0.5793	0.297	0.5413	0.9940	840
Probit (statsmodel	0.5345	0.047	0.0084	-0.5411	0.059	0.5341	0.9762	840
Logit (statsmodels	0.5214	-0.122	-0.0215	-0.5582	-0.153	0.5255	0.9786	840
Logistic (sklearn,	0.5310	-0.180	-0.0315	-0.5731	-0.226	0.5301	0.9881	840
SVM (RBF)	0.5214	-0.377	-0.0651	-0.6346	-0.468	0.5175	0.9869	840
Hist gradient boos	0.5190	-0.562	-0.0953	-0.6747	-0.674	0.5180	0.9905	840

USD/CAD (USD_CAD) - binary, lead 10d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | SVM (RBF) | Ensemble (weighted train acc) | Ensemble (mean proba) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.5190	0.387	0.0745	-0.5696	0.617	0.5195	0.9762	840
SVM (RBF)	0.5095	0.230	0.0435	-0.5505	0.417	0.5103	0.9845	840
Ensemble (weighted	0.5440	0.105	0.0198	-0.6250	0.131	0.5430	0.9821	840
Ensemble (mean pro	0.5417	0.086	0.0162	-0.6294	0.107	0.5412	0.9833	840
Hist gradient boos	0.5333	0.046	0.0087	-0.5120	0.058	0.5341	0.9964	840
Logistic (sklearn,	0.5202	-0.319	-0.0583	-0.6198	-0.405	0.5229	0.9881	840
Gradient boosting	0.5024	-0.341	-0.0621	-0.5944	-0.438	0.5042	0.9964	840
Probit (statsmodel	0.5226	-0.386	-0.0698	-0.6287	-0.487	0.5237	0.9798	840
Logit (statsmodels	0.5250	-0.444	-0.0798	-0.6331	-0.560	0.5212	0.9845	840
XGBoost	0.4952	-0.550	-0.0953	-0.6207	-0.826	0.4982	0.9869	840
MLP neural net	0.5012	-1.237	-0.1990	-0.6977	-1.470	0.5000	0.9143	840

USD/CAD (USD_CAD) - ternary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MNLogit (statsmodels, PCA) | Gradient boosting (sklearn) | Ensemble (mean proba) | Ensemble (weighted train acc) | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MNLogit (statsmode	0.6619	1.017	0.0158	-0.0200	0.505	0.5733	0.0893	840
Gradient boosting	0.7310	0.981	0.0242	-0.0413	0.619	0.5917	0.1429	840
Ensemble (mean pro	0.7048	0.685	0.0069	-0.0086	0.235	0.6429	0.0333	840
Ensemble (weighted	0.7083	0.503	0.0051	-0.0086	0.182	0.6071	0.0333	840
Logistic (sklearn,	0.4155	0.400	0.0193	-0.0721	0.427	0.5126	0.5202	840
Hist gradient boos	0.7250	-0.016	-0.0004	-0.0318	-0.008	0.5391	0.1524	840
XGBoost	0.7833	-0.169	-0.0022	-0.0259	-0.062	0.5510	0.0583	840
Random forest	0.8286	NA	0.0000	0.0000	NA	NA	0.0000	840
MLP neural net	0.8250	NA	0.0000	0.0000	NA	NA	0.0000	840
SVM (RBF)	0.8298	NA	0.0000	0.0000	NA	NA	0.0000	840

USD/CAD (USD_CAD) - ternary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Ensemble (mean proba) | Ensemble (weighted train acc) | MNLogit (statsmodels, PCA) | SVM (RBF) | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Ensemble (mean pro	0.6464	1.178	0.0402	-0.0271	0.679	0.6154	0.0929	840
Ensemble (weighted	0.6429	1.150	0.0395	-0.0288	0.694	0.5882	0.1012	840
MNLogit (statsmode	0.6238	1.041	0.0413	-0.0736	0.633	0.5565	0.1476	840
SVM (RBF)	0.7107	0.964	0.0068	-0.0010	NA	0.8333	0.0071	840
XGBoost	0.6381	0.855	0.0283	-0.0471	0.512	0.5575	0.1345	840
Hist gradient boos	0.5881	0.838	0.0427	-0.0708	0.662	0.5211	0.2536	840
Random forest	0.6845	0.794	0.0140	-0.0159	0.207	0.5769	0.0310	840
Gradient boosting	0.5619	0.677	0.0344	-0.0626	0.584	0.5325	0.2750	840
Logistic (sklearn,	0.4131	0.481	0.0360	-0.2247	0.465	0.5152	0.5107	840
MLP neural net	0.7095	NA	0.0000	0.0000	NA	NA	0.0000	840

USD/CAD (USD_CAD) - ternary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Ensemble (mean proba) | XGBoost | Hist gradient boosting | Ensemble (weighted train acc) | MNLogit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Ensemble (mean pro	0.5464	2.323	0.1060	-0.0321	1.957	0.6497	0.1869	840
XGBoost	0.5226	2.310	0.1243	-0.0411	2.440	0.6262	0.2548	840
Hist gradient boos	0.5000	2.281	0.1289	-0.0497	2.334	0.6094	0.3048	840
Ensemble (weighted	0.5440	2.228	0.1042	-0.0321	1.851	0.6503	0.1940	840
MNLogit (statsmode	0.5274	1.232	0.0659	-0.1073	0.932	0.5440	0.2298	840
Random forest	0.5714	1.164	0.0344	-0.0213	0.744	0.5606	0.0786	840
Gradient boosting	0.5012	1.027	0.0818	-0.1275	0.985	0.5463	0.3988	840
Logistic (sklearn,	0.3952	0.910	0.0735	-0.1951	0.909	0.5160	0.4821	840
SVM (RBF)	0.5917	0.366	0.0086	-0.0579	0.157	0.5111	0.0536	840
MLP neural net	0.6226	NA	0.0000	0.0000	NA	NA	0.0000	840

USD/CAD (USD_CAD) - ternary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Ensemble (mean proba) | Gradient boosting (sklearn) | Ensemble (weighted train acc) | MNLogit (statsmodels, PCA) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Ensemble (mean pro	0.4417	1.141	0.0650	-0.1247	1.030	0.5421	0.2548	840
Gradient boosting	0.4107	0.909	0.0838	-0.1092	0.859	0.5172	0.4488	840
Ensemble (weighted	0.4369	0.899	0.0550	-0.1261	0.762	0.5365	0.2774	840
MNLogit (statsmode	0.4583	0.747	0.0655	-0.1592	0.678	0.5234	0.3048	840
Hist gradient boos	0.4202	0.740	0.0601	-0.1233	0.619	0.5419	0.3976	840
Logistic (sklearn,	0.3655	0.738	0.0755	-0.1996	0.749	0.5175	0.5107	840
SVM (RBF)	0.4988	0.733	0.0247	-0.0560	0.533	0.5417	0.0857	840
XGBoost	0.4393	0.623	0.0550	-0.1764	0.472	0.5452	0.3690	840
MLP neural net	0.5155	0.602	0.0096	-0.0219	0.113	0.6667	0.0179	840
Random forest	0.4738	-0.039	-0.0016	-0.1468	-0.024	0.5045	0.1321	840

USD/CAD (USD_CAD) - ternary, lead 5d

Rows used: 1000 | Effective end: 2021-12-25

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | SVM (RBF) | XGBoost | Ensemble (weighted train acc) | MLP neural net

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.4298	2.070	0.0984	-0.0934	1.516	0.6294	0.1702	840
SVM (RBF)	0.4607	1.798	0.0879	-0.1109	1.582	0.5839	0.1631	840
XGBoost	0.4060	1.529	0.1347	-0.1137	1.446	0.5559	0.3833	840
Ensemble (weighted	0.4107	0.884	0.0777	-0.1378	0.672	0.5556	0.3321	840
MLP neural net	0.4750	0.883	0.0092	-0.0047	0.353	0.6667	0.0071	840
Gradient boosting	0.3810	0.863	0.0829	-0.1226	0.820	0.5347	0.4631	840
Hist gradient boos	0.4393	0.797	0.0872	-0.2527	0.693	0.5403	0.4583	840
Ensemble (mean pro	0.4179	0.757	0.0671	-0.1580	0.565	0.5451	0.3167	840
Logistic (sklearn,	0.3738	0.090	0.0101	-0.3876	0.085	0.5124	0.5274	840
MNLogit (statsmode	0.4190	-0.196	-0.0200	-0.3963	-0.163	0.4804	0.3940	840

USD/CAD (USD_CAD) - ternary, lead 6d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | Ensemble (weighted train acc) | XGBoost | SVM (RBF) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.4107	2.895	0.3476	-0.1317	3.702	0.6026	0.5393	840
Ensemble (weighted	0.4179	2.415	0.2477	-0.1499	2.768	0.5875	0.4012	840
XGBoost	0.4036	1.992	0.2223	-0.1322	2.456	0.5620	0.4702	840
SVM (RBF)	0.4452	1.826	0.1267	-0.0922	1.772	0.5895	0.2262	840
Ensemble (mean pro	0.4095	1.823	0.1792	-0.1431	1.951	0.5652	0.3833	840
Random forest	0.4369	1.526	0.1230	-0.1508	1.509	0.5439	0.2845	840
Gradient boosting	0.3821	0.629	0.0594	-0.1693	0.669	0.5373	0.4786	840
Logistic (sklearn,	0.3345	0.072	0.0090	-0.4865	0.072	0.5221	0.5929	840
MNLogit (statsmode	0.3833	0.013	0.0015	-0.4982	0.012	0.5172	0.4857	840
MLP neural net	0.4393	-0.045	-0.0019	-0.0879	-0.015	0.5135	0.0440	840

USD/CAD (USD_CAD) - ternary, lead 7d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Random forest | Ensemble (weighted train acc) | Ensemble (mean proba) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.4167	2.366	0.1914	-0.0934	2.647	0.5854	0.2440	840
Random forest	0.4083	2.172	0.1877	-0.1656	1.685	0.6195	0.2440	840
Ensemble (weighted	0.3655	1.876	0.2002	-0.1559	1.961	0.5609	0.3714	840
Ensemble (mean pro	0.3690	1.806	0.1954	-0.1634	1.886	0.5570	0.3762	840
Hist gradient boos	0.3857	1.656	0.2078	-0.1497	2.024	0.5518	0.5286	840
Gradient boosting	0.3714	1.592	0.1828	-0.2413	1.837	0.5396	0.4810	840
XGBoost	0.3738	1.274	0.1400	-0.2432	1.289	0.5492	0.4357	840
MLP neural net	0.4000	0.989	0.0466	-0.1129	0.453	0.6102	0.0702	840
Logistic (sklearn,	0.3119	0.600	0.0880	-0.5203	0.631	0.5182	0.6548	840
MNLogit (statsmode	0.3286	0.434	0.0609	-0.5154	0.424	0.5165	0.5762	840

USD/CAD (USD_CAD) - ternary, lead 8d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | MLP neural net | SVM (RBF) | Ensemble (weighted train acc) | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.3774	2.130	0.3249	-0.1592	2.362	0.5634	0.5536	840
MLP neural net	0.3893	1.919	0.1566	-0.0712	2.406	0.5985	0.1571	840
SVM (RBF)	0.3631	1.611	0.1273	-0.1608	1.736	0.5556	0.2571	840
Ensemble (weighted	0.3548	1.442	0.1788	-0.3070	1.263	0.5523	0.4440	840
Gradient boosting	0.3738	1.414	0.1753	-0.3079	1.766	0.5379	0.5024	840
XGBoost	0.3536	1.406	0.1883	-0.2208	1.397	0.5383	0.4976	840
Ensemble (mean pro	0.3571	1.106	0.1346	-0.3349	0.914	0.5503	0.4262	840
Random forest	0.3952	1.031	0.1076	-0.2332	0.684	0.5696	0.2821	840
Logistic (sklearn,	0.2964	0.035	0.0050	-0.5384	0.033	0.5347	0.6345	840
MNLogit (statsmode	0.3226	-0.030	-0.0041	-0.5399	-0.027	0.5262	0.5905	840

USD/CAD (USD_CAD) - ternary, lead 9d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | XGBoost | SVM (RBF) | Ensemble (weighted train acc) | Random forest

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.3786	1.997	0.2737	-0.2254	1.896	0.5899	0.5429	840
XGBoost	0.3607	1.222	0.1346	-0.1510	1.180	0.5586	0.4774	840
SVM (RBF)	0.3738	1.107	0.0929	-0.1433	0.916	0.5833	0.2714	840
Ensemble (weighted	0.3595	0.842	0.0963	-0.2936	0.612	0.5706	0.3881	840
Random forest	0.3548	0.780	0.0669	-0.1502	0.470	0.5971	0.2452	840
Gradient boosting	0.3619	0.735	0.1134	-0.3936	0.717	0.5748	0.6048	840
Ensemble (mean pro	0.3571	0.511	0.0597	-0.3476	0.353	0.5674	0.3798	840
MLP neural net	0.3929	-0.145	-0.0129	-0.3164	-0.053	0.6094	0.1524	840
MNLogit (statsmode	0.3238	-0.145	-0.0208	-0.5449	-0.120	0.5427	0.5571	840
Logistic (sklearn,	0.3155	-0.471	-0.0664	-0.5422	-0.408	0.5298	0.6000	840

USD/CAD (USD_CAD) - ternary, lead 10d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Hist gradient boosting | XGBoost | Ensemble (mean proba) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.3321	1.223	0.1023	-0.1118	1.115	0.5628	0.2940	840
Hist gradient boos	0.3369	1.161	0.1645	-0.4205	1.345	0.5717	0.6393	840
XGBoost	0.3250	0.987	0.1289	-0.4275	1.069	0.5491	0.5571	840
Ensemble (mean pro	0.3226	0.791	0.0937	-0.3271	0.745	0.5541	0.4512	840
Ensemble (weighted	0.3274	0.726	0.0889	-0.3832	0.694	0.5561	0.4774	840
Random forest	0.3464	0.511	0.0436	-0.2486	0.470	0.5455	0.3012	840
Gradient boosting	0.3310	-0.017	-0.0023	-0.5271	-0.017	0.5335	0.5869	840
MLP neural net	0.3536	-0.298	-0.0234	-0.2744	-0.105	0.5952	0.1000	840
Logistic (sklearn,	0.3024	-0.846	-0.1260	-0.5664	-0.790	0.5009	0.6512	840
MNLogit (statsmode	0.3048	-0.914	-0.1351	-0.5672	-0.821	0.4942	0.6119	840

EUR/JPY (EUR_JPY) - binary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Logistic (sklearn, on global PCA) | Gradient boosting (sklearn) | Logit (statsmodels, PCA) | Probit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.5274	0.758	0.0456	-0.0884	0.961	0.5274	0.9119	840
Logistic (sklearn,	0.4940	0.170	0.0105	-0.1331	0.258	0.4926	0.9714	840
Gradient boosting	0.4976	0.014	0.0009	-0.0908	0.021	0.4964	0.9905	840
Logit (statsmodels	0.4798	0.011	0.0007	-0.1159	0.017	0.4787	0.9798	840
Probit (statsmodel	0.4798	0.004	0.0002	-0.1159	0.006	0.4782	0.9810	840
Random forest	0.5012	-0.228	-0.0138	-0.1518	-0.307	0.4975	0.9500	840
XGBoost	0.4940	-0.395	-0.0242	-0.2073	-0.585	0.4922	0.9869	840
Ensemble (mean pro	0.4893	-0.549	-0.0331	-0.1837	-0.800	0.4895	0.9679	840
Ensemble (weighted	0.4857	-0.550	-0.0332	-0.1841	-0.792	0.4932	0.9679	840
MLP neural net	0.4988	-0.563	-0.0330	-0.2120	-0.780	0.4987	0.9214	840
Hist gradient boos	0.4869	-0.589	-0.0356	-0.1822	-0.883	0.4855	0.9881	840

EUR/JPY (EUR_JPY) - binary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | SVM (RBF) | Ensemble (mean proba) | Ensemble (weighted train acc) | MLP neural net

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.5167	0.535	0.0479	-0.2063	0.789	0.5199	0.9869	840
SVM (RBF)	0.5060	0.224	0.0193	-0.1530	0.319	0.5013	0.9357	840
Ensemble (mean pro	0.5107	0.149	0.0128	-0.2651	0.213	0.5111	0.9643	840
Ensemble (weighted	0.5107	0.139	0.0121	-0.2764	0.198	0.5131	0.9560	840
MLP neural net	0.5060	0.063	0.0054	-0.2244	0.088	0.4954	0.9107	840
XGBoost	0.5048	-0.086	-0.0075	-0.2769	-0.125	0.5030	0.9869	840
Probit (statsmodel	0.4810	-0.206	-0.0177	-0.2390	-0.299	0.4846	0.9655	840
Random forest	0.5107	-0.206	-0.0174	-0.2503	-0.295	0.5100	0.9548	840
Hist gradient boos	0.4821	-0.213	-0.0184	-0.2902	-0.306	0.4837	0.9845	840
Logit (statsmodels	0.4833	-0.256	-0.0219	-0.2340	-0.373	0.4809	0.9679	840
Logistic (sklearn,	0.4810	-0.372	-0.0315	-0.2295	-0.544	0.4828	0.9667	840

EUR/JPY (EUR_JPY) - binary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | SVM (RBF) | Ensemble (mean proba) | Ensemble (weighted train acc) | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.5393	1.499	0.1732	-0.1246	2.408	0.5403	0.9893	840
SVM (RBF)	0.5476	1.301	0.1370	-0.1830	1.930	0.5529	0.9000	840
Ensemble (mean pro	0.5262	0.710	0.0788	-0.2140	1.066	0.5291	0.9833	840
Ensemble (weighted	0.5298	0.696	0.0766	-0.2230	1.043	0.5280	0.9762	840
XGBoost	0.5262	0.469	0.0511	-0.2447	0.688	0.5291	0.9833	840
Random forest	0.5107	0.268	0.0286	-0.2227	0.398	0.5169	0.9536	840
Hist gradient boos	0.5143	0.223	0.0242	-0.2646	0.334	0.5145	0.9881	840
Logit (statsmodels	0.5190	0.136	0.0146	-0.2532	0.198	0.5172	0.9714	840
Probit (statsmodel	0.5202	0.082	0.0088	-0.2611	0.120	0.5154	0.9679	840
Logistic (sklearn,	0.5036	-0.207	-0.0217	-0.2704	-0.301	0.5074	0.9667	840
MLP neural net	0.5095	-0.423	-0.0426	-0.3055	-0.602	0.5104	0.9190	840

EUR/JPY (EUR_JPY) - binary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Probit (statsmodels, PCA) | Logit (statsmodels, PCA) | XGBoost | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.5238	0.331	0.0412	-0.2685	0.493	0.5239	0.9726	840
Probit (statsmodel	0.5167	-0.085	-0.0103	-0.3443	-0.122	0.5153	0.9726	840
Logit (statsmodels	0.5167	-0.155	-0.0187	-0.3443	-0.223	0.5128	0.9798	840
XGBoost	0.5071	-0.215	-0.0260	-0.3638	-0.318	0.5054	0.9917	840
Ensemble (mean pro	0.5036	-0.393	-0.0470	-0.3638	-0.567	0.5055	0.9702	840
Logistic (sklearn,	0.5071	-0.420	-0.0500	-0.3443	-0.610	0.5049	0.9738	840
Ensemble (weighted	0.4976	-0.540	-0.0638	-0.3638	-0.781	0.4982	0.9774	840
Hist gradient boos	0.5012	-0.577	-0.0686	-0.4214	-0.828	0.5000	0.9952	840
MLP neural net	0.4988	-0.614	-0.0717	-0.4177	-0.880	0.4937	0.9500	840
Gradient boosting	0.4833	-0.677	-0.0800	-0.4566	-1.001	0.4821	0.9952	840
Random forest	0.4869	-0.991	-0.1133	-0.4476	-1.442	0.4859	0.9702	840

EUR/JPY (EUR_JPY) - binary, lead 5d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | XGBoost | Hist gradient boosting | Logit (statsmodels, PCA) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.5464	0.774	0.1080	-0.2911	1.118	0.5442	0.9690	840
XGBoost	0.5155	0.253	0.0346	-0.4250	0.372	0.5156	0.9905	840
Hist gradient boos	0.5250	0.214	0.0292	-0.4333	0.309	0.5253	0.9881	840
Logit (statsmodels	0.5179	-0.124	-0.0165	-0.3769	-0.178	0.5157	0.9857	840
Ensemble (weighted	0.5119	-0.137	-0.0182	-0.3914	-0.197	0.5096	0.9881	840
Probit (statsmodel	0.5179	-0.159	-0.0211	-0.3839	-0.228	0.5140	0.9750	840
Ensemble (mean pro	0.5083	-0.174	-0.0231	-0.3823	-0.251	0.5085	0.9810	840
Gradient boosting	0.5060	-0.205	-0.0272	-0.4291	-0.301	0.5066	0.9964	840
Logistic (sklearn,	0.5000	-0.305	-0.0401	-0.3653	-0.445	0.5018	0.9821	840
Random forest	0.4988	-0.329	-0.0430	-0.4005	-0.470	0.4994	0.9774	840
MLP neural net	0.5048	-0.732	-0.0926	-0.5182	-1.022	0.5069	0.9464	840

EUR/JPY (EUR_JPY) - binary, lead 6d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | SVM (RBF) | Logistic (sklearn, on global PCA) | XGBoost | Random forest

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.5333	0.447	0.0671	-0.4750	0.649	0.5302	0.9857	840
SVM (RBF)	0.5274	0.372	0.0553	-0.3262	0.565	0.5302	0.9857	840
Logistic (sklearn,	0.5238	0.249	0.0370	-0.4049	0.372	0.5239	0.9952	840
XGBoost	0.5238	0.104	0.0151	-0.4858	0.148	0.5241	0.9881	840
Random forest	0.5286	0.096	0.0138	-0.4829	0.136	0.5301	0.9702	840
Ensemble (mean pro	0.5238	0.027	0.0039	-0.4124	0.039	0.5242	0.9833	840
Ensemble (weighted	0.5190	-0.032	-0.0047	-0.4229	-0.046	0.5232	0.9738	840
Logit (statsmodels	0.5107	-0.145	-0.0208	-0.4049	-0.212	0.5139	0.9869	840
Probit (statsmodel	0.5083	-0.312	-0.0444	-0.4151	-0.458	0.5096	0.9881	840
Gradient boosting	0.5167	-0.350	-0.0497	-0.4499	-0.517	0.5138	0.9917	840
MLP neural net	0.5048	-0.666	-0.0910	-0.4317	-0.944	0.5068	0.9655	840

EUR/JPY (EUR_JPY) - binary, lead 7d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Hist gradient boosting | MLP neural net | Ensemble (mean proba) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.5702	1.088	0.1831	-0.3803	1.583	0.5714	0.9833	840
Hist gradient boos	0.5464	0.691	0.1131	-0.3268	1.025	0.5461	0.9940	840
MLP neural net	0.5476	0.690	0.1115	-0.4607	0.985	0.5561	0.9548	840
Ensemble (mean pro	0.5345	0.246	0.0389	-0.4501	0.360	0.5361	0.9881	840
Ensemble (weighted	0.5357	0.232	0.0367	-0.4467	0.341	0.5362	0.9857	840
Logit (statsmodels	0.5321	0.199	0.0313	-0.4425	0.294	0.5302	0.9857	840
Probit (statsmodel	0.5321	0.191	0.0301	-0.4425	0.284	0.5300	0.9905	840
Logistic (sklearn,	0.5250	0.158	0.0246	-0.4338	0.236	0.5291	0.9810	840
Gradient boosting	0.5071	0.103	0.0162	-0.4702	0.152	0.5060	0.9952	840
Random forest	0.5238	-0.341	-0.0511	-0.5236	-0.496	0.5226	0.9726	840
XGBoost	0.5083	-0.556	-0.0825	-0.5123	-0.820	0.5060	0.9857	840

EUR/JPY (EUR_JPY) - binary, lead 8d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | Ensemble (mean proba) | Probit (statsmodels, PCA) | SVM (RBF) | Logit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.5143	0.127	0.0212	-0.4786	0.193	0.5167	0.9952	840
Ensemble (mean pro	0.5321	-0.136	-0.0221	-0.4989	-0.201	0.5315	0.9833	840
Probit (statsmodel	0.5250	-0.157	-0.0255	-0.4986	-0.231	0.5257	0.9964	840
SVM (RBF)	0.5310	-0.212	-0.0341	-0.5004	-0.305	0.5320	0.9845	840
Logit (statsmodels	0.5226	-0.238	-0.0383	-0.5078	-0.351	0.5234	0.9917	840
MLP neural net	0.5155	-0.255	-0.0401	-0.5840	-0.374	0.5083	0.9369	840
Ensemble (weighted	0.5286	-0.339	-0.0541	-0.5330	-0.499	0.5268	0.9786	840
Logistic (sklearn,	0.5071	-0.457	-0.0722	-0.5056	-0.695	0.5084	0.9905	840
Gradient boosting	0.5107	-0.467	-0.0736	-0.5635	-0.703	0.5096	0.9905	840
XGBoost	0.4964	-0.697	-0.1080	-0.5961	-1.058	0.4976	0.9857	840
Random forest	0.4905	-1.161	-0.1709	-0.7086	-1.711	0.4870	0.9583	840

EUR/JPY (EUR_JPY) - binary, lead 9d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | MLP neural net | Logit (statsmodels, PCA) | Hist gradient boosting | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.5357	0.222	0.0390	-0.5599	0.312	0.5383	0.9798	840
MLP neural net	0.5512	0.041	0.0070	-0.6046	0.056	0.5532	0.9512	840
Logit (statsmodels	0.5238	-0.336	-0.0563	-0.5933	-0.483	0.5240	0.9905	840
Hist gradient boos	0.4667	-0.417	-0.0697	-0.5939	-0.639	0.4677	0.9952	840
Ensemble (mean pro	0.4988	-0.471	-0.0780	-0.5928	-0.681	0.4976	0.9857	840
Ensemble (weighted	0.4964	-0.475	-0.0785	-0.5907	-0.686	0.4958	0.9845	840
Probit (statsmodel	0.5179	-0.501	-0.0828	-0.6077	-0.724	0.5157	0.9857	840
Logistic (sklearn,	0.5083	-0.516	-0.0849	-0.5889	-0.754	0.5079	0.9845	840
Gradient boosting	0.4774	-0.732	-0.1190	-0.5482	-1.075	0.4785	0.9976	840
Random forest	0.4762	-1.135	-0.1769	-0.7244	-1.636	0.4775	0.9774	840
XGBoost	0.4738	-1.254	-0.1935	-0.7075	-1.830	0.4709	0.9833	840

EUR/JPY (EUR_JPY) - binary, lead 10d

Rows used: 1000 | Effective end: 2021-12-18

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | Logit (statsmodels, PCA) | Hist gradient boosting | Probit (statsmodels, PCA) | MLP neural net

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.5214	0.572	0.1084	-0.5161	0.857	0.5216	0.9929	840
Logit (statsmodels	0.5262	0.469	0.0878	-0.5150	0.681	0.5271	0.9869	840
Hist gradient boos	0.5095	0.427	0.0799	-0.4607	0.639	0.5096	0.9929	840
Probit (statsmodel	0.5262	0.422	0.0785	-0.5174	0.612	0.5253	0.9881	840
MLP neural net	0.5226	0.335	0.0610	-0.5895	0.478	0.5245	0.9488	840
XGBoost	0.5060	0.139	0.0254	-0.5000	0.210	0.5060	0.9881	840
Gradient boosting	0.5226	0.127	0.0232	-0.5538	0.183	0.5227	0.9952	840
Ensemble (mean pro	0.5048	-0.012	-0.0021	-0.5891	-0.018	0.5054	0.9869	840
Ensemble (weighted	0.5036	-0.045	-0.0080	-0.5897	-0.066	0.5036	0.9857	840
SVM (RBF)	0.5131	-0.356	-0.0617	-0.6349	-0.513	0.5091	0.9845	840
Random forest	0.5012	-0.401	-0.0692	-0.6451	-0.580	0.4982	0.9917	840

EUR/JPY (EUR_JPY) - ternary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | MNLogit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | XGBoost | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.6821	-0.222	-0.0082	-0.1038	-0.164	0.5000	0.2214	840
MNLogit (statsmode	0.7536	-0.335	-0.0110	-0.0865	-0.189	0.4902	0.1214	840
Logistic (sklearn,	0.4071	-0.482	-0.0210	-0.1108	-0.502	0.4725	0.4762	840
XGBoost	0.7988	-0.501	-0.0102	-0.0440	-0.178	0.4872	0.0464	840
Hist gradient boos	0.7476	-0.568	-0.0152	-0.0628	-0.320	0.4381	0.1250	840
Ensemble (mean pro	0.8119	-0.573	-0.0086	-0.0424	-0.157	0.4706	0.0202	840
Ensemble (weighted	0.8107	-0.635	-0.0097	-0.0350	-0.168	0.4444	0.0214	840
Random forest	0.8321	NA	0.0000	0.0000	NA	NA	0.0000	840
MLP neural net	0.8298	NA	0.0000	0.0000	NA	NA	0.0000	840
SVM (RBF)	0.8321	NA	0.0000	0.0000	NA	NA	0.0000	840

EUR/JPY (EUR_JPY) - ternary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Ensemble (mean proba) | XGBoost | Random forest | Ensemble (weighted train acc) | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Ensemble (mean pro	0.6298	0.253	0.0079	-0.0770	0.113	0.5316	0.0940	840
XGBoost	0.6262	0.152	0.0057	-0.0978	0.101	0.5040	0.1488	840
Random forest	0.6631	0.095	0.0016	-0.0458	0.029	0.4800	0.0298	840
Ensemble (weighted	0.6333	0.087	0.0027	-0.0626	0.042	0.5185	0.0964	840
Gradient boosting	0.5369	-0.190	-0.0107	-0.1609	-0.158	0.5018	0.3226	840
Hist gradient boos	0.5560	-0.248	-0.0124	-0.1596	-0.192	0.4914	0.2762	840
Logistic (sklearn,	0.4226	-0.492	-0.0308	-0.2226	-0.476	0.4694	0.4286	840
MNLogit (statsmode	0.6012	-0.694	-0.0289	-0.1812	-0.408	0.4317	0.1655	840
MLP neural net	0.6869	NA	0.0000	0.0000	NA	NA	0.0000	840
SVM (RBF)	0.6869	NA	0.0000	0.0000	NA	NA	0.0000	840

EUR/JPY (EUR_JPY) - ternary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | SVM (RBF) | XGBoost | Gradient boosting (sklearn) | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.4869	1.005	0.0715	-0.1396	0.987	0.5425	0.3643	840
SVM (RBF)	0.5762	0.943	0.0061	0.0000	NA	1.0000	0.0036	840
XGBoost	0.5000	0.403	0.0231	-0.1427	0.308	0.5094	0.2524	840
Gradient boosting	0.4083	0.047	0.0035	-0.2161	0.048	0.4935	0.4560	840
Logistic (sklearn,	0.3548	-0.069	-0.0055	-0.2445	-0.066	0.5161	0.4429	840
Random forest	0.5429	-0.135	-0.0047	-0.1128	-0.056	0.4478	0.0798	840
MNLogit (statsmode	0.5131	-0.267	-0.0177	-0.2568	-0.181	0.4948	0.2310	840
Ensemble (weighted	0.5071	-0.491	-0.0246	-0.1821	-0.337	0.4625	0.1905	840
Ensemble (mean pro	0.5167	-0.611	-0.0322	-0.2192	-0.395	0.4706	0.1821	840
MLP neural net	0.5810	-0.748	-0.0039	-0.0128	-0.099	0.0000	0.0024	840

EUR/JPY (EUR_JPY) - ternary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | Gradient boosting (sklearn) | MLP neural net | SVM (RBF) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.3405	-0.359	-0.0333	-0.3060	-0.350	0.4961	0.4583	840
Gradient boosting	0.4119	-0.489	-0.0423	-0.2693	-0.439	0.5069	0.4298	840
MLP neural net	0.4833	-0.503	-0.0212	-0.1426	-0.133	0.4516	0.0369	840
SVM (RBF)	0.4595	-0.621	-0.0282	-0.1844	-0.206	0.4923	0.0774	840
Ensemble (weighted	0.4155	-0.959	-0.0716	-0.3015	-0.675	0.4595	0.2643	840
MNLogit (statsmode	0.4274	-0.997	-0.0807	-0.3052	-0.806	0.4627	0.3036	840
XGBoost	0.4095	-1.000	-0.0804	-0.3236	-0.806	0.4778	0.3488	840
Ensemble (mean pro	0.4214	-1.036	-0.0754	-0.3057	-0.708	0.4567	0.2476	840
Hist gradient boos	0.3833	-1.489	-0.1278	-0.4003	-1.430	0.4381	0.4619	840
Random forest	0.4667	-1.586	-0.0940	-0.3468	-0.760	0.4135	0.1238	840

EUR/JPY (EUR_JPY) - ternary, lead 5d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | XGBoost | Hist gradient boosting | MLP neural net | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.3774	-0.108	-0.0112	-0.3662	-0.104	0.5408	0.5107	840
XGBoost	0.4071	-0.328	-0.0312	-0.3830	-0.289	0.5077	0.3869	840
Hist gradient boos	0.3845	-0.457	-0.0476	-0.3941	-0.466	0.4938	0.4774	840
MLP neural net	0.4464	-0.583	-0.0395	-0.3073	-0.269	0.5049	0.1226	840
Ensemble (weighted	0.4060	-0.685	-0.0621	-0.3886	-0.554	0.4982	0.3250	840
MNLogit (statsmode	0.3750	-0.751	-0.0730	-0.3661	-0.688	0.4788	0.4202	840
SVM (RBF)	0.4083	-0.765	-0.0495	-0.3157	-0.385	0.5352	0.1690	840
Logistic (sklearn,	0.3393	-0.805	-0.0813	-0.4429	-0.802	0.4794	0.4917	840
Ensemble (mean pro	0.4036	-0.828	-0.0737	-0.3886	-0.656	0.4811	0.3143	840
Random forest	0.4167	-0.923	-0.0675	-0.3873	-0.535	0.4759	0.1976	840

EUR/JPY (EUR_JPY) - ternary, lead 6d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | Gradient boosting (sklearn) | Hist gradient boosting | XGBoost | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.4048	0.847	0.0621	-0.2029	0.424	0.5809	0.1619	840
Gradient boosting	0.3631	0.705	0.0777	-0.3530	0.740	0.5271	0.5262	840
Hist gradient boos	0.3643	0.251	0.0300	-0.4088	0.272	0.5126	0.5667	840
XGBoost	0.3690	0.201	0.0227	-0.4477	0.198	0.5073	0.4881	840
Ensemble (weighted	0.3726	-0.165	-0.0174	-0.4218	-0.150	0.4905	0.4393	840
Ensemble (mean pro	0.3738	-0.379	-0.0388	-0.4124	-0.336	0.4798	0.4119	840
MNLogit (statsmode	0.3321	-0.459	-0.0487	-0.4087	-0.446	0.4916	0.4964	840
Logistic (sklearn,	0.3488	-0.541	-0.0611	-0.4042	-0.580	0.4799	0.5631	840
Random forest	0.3833	-0.583	-0.0455	-0.3876	-0.345	0.5111	0.2143	840
SVM (RBF)	0.3667	-0.985	-0.0719	-0.3308	-0.579	0.5126	0.2369	840

EUR/JPY (EUR_JPY) - ternary, lead 7d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | MNLogit (statsmodels, PCA) | XGBoost | Gradient boosting (sklearn) | SVM (RBF)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.3940	-0.166	-0.0145	-0.3844	-0.105	0.5532	0.2238	840
MNLogit (statsmode	0.3238	-0.459	-0.0531	-0.4457	-0.475	0.5170	0.5595	840
XGBoost	0.3524	-0.549	-0.0594	-0.4679	-0.524	0.5117	0.4583	840
Gradient boosting	0.3548	-0.590	-0.0663	-0.5130	-0.568	0.5194	0.4905	840
SVM (RBF)	0.3274	-0.734	-0.0684	-0.3644	-0.546	0.5520	0.3321	840
Logistic (sklearn,	0.3119	-0.867	-0.0991	-0.4590	-0.928	0.5020	0.5905	840
Hist gradient boos	0.3274	-0.909	-0.1014	-0.5101	-0.900	0.5071	0.5048	840
Random forest	0.3310	-0.961	-0.0872	-0.4709	-0.694	0.5063	0.2845	840
Ensemble (mean pro	0.3298	-0.972	-0.1015	-0.4693	-0.892	0.5000	0.4190	840
Ensemble (weighted	0.3345	-1.183	-0.1218	-0.5008	-1.071	0.4943	0.4190	840

EUR/JPY (EUR_JPY) - ternary, lead 8d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | Hist gradient boosting | Ensemble (weighted train acc) | Ensemble (mean proba) | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.3548	0.527	0.0691	-0.3727	0.564	0.5497	0.6107	840
Hist gradient boos	0.3488	0.470	0.0650	-0.4569	0.543	0.5515	0.6583	840
Ensemble (weighted	0.3595	0.064	0.0080	-0.4698	0.066	0.5414	0.5321	840
Ensemble (mean pro	0.3560	0.010	0.0012	-0.4698	0.010	0.5379	0.5333	840
Logistic (sklearn,	0.3536	-0.087	-0.0114	-0.4974	-0.098	0.5376	0.6333	840
MNLogit (statsmode	0.3536	-0.269	-0.0355	-0.5003	-0.306	0.5327	0.6369	840
XGBoost	0.3310	-0.291	-0.0364	-0.4701	-0.323	0.5060	0.6000	840
Random forest	0.3250	-0.337	-0.0352	-0.4271	-0.270	0.5267	0.3571	840
MLP neural net	0.3583	-0.338	-0.0342	-0.4437	-0.256	0.5354	0.3024	840
SVM (RBF)	0.3476	-0.480	-0.0497	-0.4261	-0.382	0.5565	0.4000	840

EUR/JPY (EUR_JPY) - ternary, lead 9d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | Logistic (sklearn, on global PCA) | MNLogit (statsmodels, PCA) | Ensemble (mean proba) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.3429	0.653	0.0954	-0.3067	0.804	0.5245	0.6560	840
Logistic (sklearn,	0.3643	0.629	0.0973	-0.4824	0.750	0.5407	0.6869	840
MNLogit (statsmode	0.3464	0.619	0.0939	-0.4824	0.707	0.5511	0.6524	840
Ensemble (mean pro	0.3643	0.195	0.0263	-0.4811	0.196	0.5228	0.5214	840
Ensemble (weighted	0.3643	0.097	0.0130	-0.4886	0.097	0.5181	0.5262	840
SVM (RBF)	0.3607	0.063	0.0076	-0.4239	0.053	0.5678	0.4214	840
Gradient boosting	0.3452	0.015	0.0021	-0.4885	0.016	0.5093	0.6405	840
XGBoost	0.3500	-0.090	-0.0123	-0.4742	-0.102	0.4888	0.5869	840
Random forest	0.3548	-0.162	-0.0187	-0.3742	-0.139	0.5125	0.3810	840
MLP neural net	0.3310	-0.943	-0.0866	-0.5473	-0.549	0.5132	0.2250	840

EUR/JPY (EUR_JPY) - ternary, lead 10d

Rows used: 1000 | Effective end: 2021-12-18

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MNLogit (statsmodels, PCA) | SVM (RBF) | MLP neural net | Logistic (sklearn, on global PCA) | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MNLogit (statsmode	0.3643	0.737	0.1197	-0.5233	0.864	0.5437	0.6679	840
SVM (RBF)	0.3643	0.712	0.1047	-0.5012	0.677	0.5714	0.5000	840
MLP neural net	0.3917	0.580	0.0428	-0.3266	0.440	0.5547	0.1524	840
Logistic (sklearn,	0.3774	0.423	0.0662	-0.5165	0.489	0.5325	0.6595	840
Gradient boosting	0.3786	0.327	0.0528	-0.5947	0.390	0.5359	0.7286	840
Ensemble (weighted	0.3726	-0.059	-0.0087	-0.5811	-0.065	0.5132	0.5845	840
Ensemble (mean pro	0.3738	-0.063	-0.0094	-0.5925	-0.070	0.5153	0.5845	840
Random forest	0.3679	-0.111	-0.0147	-0.4924	-0.102	0.5163	0.4381	840
Hist gradient boos	0.3929	-0.160	-0.0255	-0.6000	-0.198	0.5009	0.6798	840
XGBoost	0.3726	-0.751	-0.1087	-0.6290	-0.879	0.4773	0.6286	840

EUR/GBP (EUR_GBP) - binary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Logistic (sklearn, on global PCA) | Random forest | Logit (statsmodels, PCA) | Probit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.4929	0.342	0.0229	-0.0748	0.488	0.4927	0.9810	840
Logistic (sklearn,	0.5167	0.332	0.0223	-0.0927	0.436	0.5154	0.9679	840
Random forest	0.4893	0.322	0.0212	-0.1472	0.451	0.4956	0.9536	840
Logit (statsmodels	0.5155	-0.040	-0.0026	-0.1287	-0.052	0.5111	0.9690	840
Probit (statsmodel	0.5155	-0.042	-0.0027	-0.1287	-0.054	0.5111	0.9667	840
MLP neural net	0.4976	-0.216	-0.0136	-0.1762	-0.280	0.4920	0.8905	840
Ensemble (mean pro	0.4929	-0.272	-0.0176	-0.1826	-0.369	0.4926	0.9619	840
Hist gradient boos	0.4869	-0.351	-0.0231	-0.1628	-0.470	0.4892	0.9929	840
Ensemble (weighted	0.4869	-0.392	-0.0257	-0.1826	-0.539	0.4872	0.9774	840
Gradient boosting	0.4964	-0.398	-0.0261	-0.2046	-0.538	0.4940	0.9905	840
SVM (RBF)	0.4917	-0.732	-0.0437	-0.1607	-0.906	0.4956	0.8167	840

EUR/GBP (EUR_GBP) - binary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | XGBoost | Gradient boosting (sklearn) | Ensemble (weighted train acc) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.5226	0.946	0.0935	-0.1393	1.385	0.5228	0.9929	840
XGBoost	0.5155	0.932	0.0917	-0.1028	1.401	0.5169	0.9857	840
Gradient boosting	0.5179	0.508	0.0492	-0.2510	0.675	0.5168	0.9929	840
Ensemble (weighted	0.5119	0.484	0.0463	-0.1877	0.693	0.5097	0.9810	840
Ensemble (mean pro	0.5048	0.466	0.0448	-0.1932	0.673	0.5048	0.9857	840
Probit (statsmodel	0.4893	0.334	0.0319	-0.1305	0.495	0.4884	0.9750	840
Logit (statsmodels	0.4881	0.300	0.0286	-0.1309	0.445	0.4859	0.9702	840
Logistic (sklearn,	0.4917	0.237	0.0224	-0.1572	0.349	0.4890	0.9762	840
SVM (RBF)	0.4964	0.086	0.0080	-0.2288	0.121	0.4911	0.9405	840
Random forest	0.4976	0.074	0.0069	-0.2867	0.107	0.4975	0.9667	840
MLP neural net	0.4821	-0.200	-0.0180	-0.2046	-0.289	0.4838	0.9202	840

EUR/GBP (EUR_GBP) - binary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | Gradient boosting (sklearn) | SVM (RBF) | Logistic (sklearn, on global PCA) | Probit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.5214	0.893	0.1067	-0.2699	1.282	0.5198	0.9917	840
Gradient boosting	0.5369	0.800	0.0956	-0.3256	1.143	0.5337	0.9881	840
SVM (RBF)	0.4964	0.602	0.0698	-0.2469	0.917	0.4982	0.9702	840
Logistic (sklearn,	0.4964	0.544	0.0637	-0.2866	0.806	0.5000	0.9714	840
Probit (statsmodel	0.4893	0.453	0.0524	-0.2744	0.669	0.4926	0.9714	840
Logit (statsmodels	0.4940	0.437	0.0505	-0.2747	0.647	0.4951	0.9762	840
Random forest	0.5179	0.334	0.0384	-0.4093	0.446	0.5190	0.9702	840
XGBoost	0.5190	0.283	0.0328	-0.4135	0.383	0.5187	0.9893	840
Ensemble (weighted	0.5143	0.233	0.0266	-0.3712	0.316	0.5110	0.9714	840
Ensemble (mean pro	0.5107	0.233	0.0267	-0.3583	0.320	0.5133	0.9833	840
MLP neural net	0.4988	-0.523	-0.0566	-0.4286	-0.650	0.5026	0.9310	840

EUR/GBP (EUR_GBP) - binary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | Logistic (sklearn, on global PCA) | Probit (statsmodels, PCA) | Logit (statsmodels, PCA) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.5107	0.815	0.1114	-0.3124	1.186	0.5108	0.9905	840
Logistic (sklearn,	0.5167	0.723	0.0980	-0.3506	1.094	0.5146	0.9786	840
Probit (statsmodel	0.5000	0.536	0.0719	-0.3438	0.810	0.5036	0.9833	840
Logit (statsmodels	0.5036	0.504	0.0675	-0.3451	0.761	0.5024	0.9786	840
Ensemble (weighted	0.5024	0.344	0.0456	-0.3467	0.506	0.4994	0.9845	840
Ensemble (mean pro	0.4952	0.249	0.0328	-0.3459	0.365	0.4982	0.9750	840
XGBoost	0.4988	0.147	0.0192	-0.3986	0.204	0.4994	0.9869	840
Gradient boosting	0.4857	0.082	0.0107	-0.3532	0.122	0.4844	0.9929	840
SVM (RBF)	0.4905	-0.162	-0.0184	-0.3775	-0.239	0.4867	0.9393	840
Random forest	0.4929	-0.197	-0.0248	-0.3623	-0.274	0.4914	0.9714	840
MLP neural net	0.5000	-0.234	-0.0294	-0.3923	-0.322	0.4994	0.9512	840

EUR/GBP (EUR_GBP) - binary, lead 5d

Rows used: 1000 | Effective end: 2021-12-25

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Probit (statsmodels, PCA) | Hist gradient boosting | Logit (statsmodels, PCA) | XGBoost | MLP neural net

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Probit (statsmodel	0.5298	0.322	0.0483	-0.3989	0.439	0.5292	0.9786	840
Hist gradient boos	0.5143	0.237	0.0355	-0.4350	0.343	0.5127	0.9845	840
Logit (statsmodels	0.5238	0.171	0.0255	-0.4044	0.234	0.5225	0.9798	840
XGBoost	0.5321	0.136	0.0203	-0.4300	0.180	0.5337	0.9905	840
MLP neural net	0.4940	0.048	0.0067	-0.3527	0.074	0.4936	0.9286	840
Random forest	0.5214	0.023	0.0034	-0.4068	0.032	0.5225	0.9774	840
Logistic (sklearn,	0.5083	-0.011	-0.0016	-0.4188	-0.015	0.5091	0.9798	840
Ensemble (weighted	0.5131	-0.070	-0.0103	-0.4176	-0.096	0.5134	0.9786	840
Ensemble (mean pro	0.5107	-0.086	-0.0126	-0.4236	-0.118	0.5128	0.9774	840
Gradient boosting	0.4952	-0.414	-0.0591	-0.4504	-0.559	0.4976	0.9905	840
SVM (RBF)	0.4988	-0.773	-0.1074	-0.5972	-1.013	0.4988	0.9833	840

EUR/GBP (EUR_GBP) - binary, lead 6d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | XGBoost | Ensemble (weighted train acc) | Gradient boosting (sklearn) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.5202	0.687	0.1181	-0.4571	0.955	0.5191	0.9952	840
XGBoost	0.5179	0.660	0.1126	-0.4519	0.948	0.5174	0.9917	840
Ensemble (weighted	0.5119	0.460	0.0772	-0.4346	0.665	0.5121	0.9857	840
Gradient boosting	0.5060	0.454	0.0762	-0.3824	0.733	0.5054	0.9940	840
Ensemble (mean pro	0.5143	0.446	0.0739	-0.4247	0.637	0.5139	0.9821	840
Logistic (sklearn,	0.5107	0.354	0.0592	-0.4326	0.482	0.5090	0.9869	840
MLP neural net	0.4893	0.325	0.0515	-0.4795	0.518	0.4872	0.9286	840
Logit (statsmodels	0.5083	0.209	0.0342	-0.4951	0.279	0.5115	0.9869	840
Probit (statsmodel	0.5071	0.103	0.0168	-0.5003	0.137	0.5097	0.9857	840
SVM (RBF)	0.5024	-0.074	-0.0120	-0.4823	-0.110	0.5006	0.9798	840
Random forest	0.5060	-0.160	-0.0249	-0.5106	-0.218	0.5055	0.9774	840

EUR/GBP (EUR_GBP) - binary, lead 7d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | Logit (statsmodels, PCA) | Ensemble (mean proba) | Ensemble (weighted train acc) | SVM (RBF)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.5095	1.010	0.1930	-0.3927	1.559	0.5084	0.9881	840
Logit (statsmodels	0.5107	0.816	0.1538	-0.4380	1.224	0.5127	0.9845	840
Ensemble (mean pro	0.5024	0.738	0.1392	-0.3918	1.157	0.5024	0.9833	840
Ensemble (weighted	0.5012	0.725	0.1368	-0.3905	1.144	0.5006	0.9917	840
SVM (RBF)	0.5202	0.640	0.1199	-0.4856	0.941	0.5247	0.9869	840
MLP neural net	0.5060	0.625	0.1141	-0.4179	0.986	0.5051	0.9357	840
Probit (statsmodel	0.5071	0.593	0.1105	-0.4387	0.884	0.5066	0.9917	840
Hist gradient boos	0.5167	0.418	0.0769	-0.4419	0.644	0.5138	0.9940	840
Gradient boosting	0.5190	0.307	0.0556	-0.4683	0.450	0.5180	0.9905	840
XGBoost	0.4929	0.013	0.0023	-0.5139	0.019	0.4934	0.9940	840
Random forest	0.4833	-0.153	-0.0260	-0.5155	-0.220	0.4830	0.9786	840

EUR/GBP (EUR_GBP) - binary, lead 8d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | SVM (RBF) | Logistic (sklearn, on global PCA) | Probit (statsmodels, PCA) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.5250	1.159	0.2362	-0.4639	1.769	0.5319	0.9131	840
SVM (RBF)	0.5452	1.082	0.2273	-0.4513	1.557	0.5432	0.9774	840
Logistic (sklearn,	0.5000	0.621	0.1254	-0.4108	0.889	0.4982	0.9893	840
Probit (statsmodel	0.5012	0.406	0.0803	-0.4722	0.578	0.5006	0.9940	840
Ensemble (mean pro	0.5000	0.405	0.0799	-0.4567	0.623	0.5024	0.9810	840
Logit (statsmodels	0.5000	0.388	0.0766	-0.4641	0.552	0.5006	0.9940	840
XGBoost	0.4857	0.365	0.0720	-0.4685	0.579	0.4886	0.9893	840
Ensemble (weighted	0.4976	0.352	0.0692	-0.4567	0.545	0.4982	0.9845	840
Hist gradient boos	0.5107	0.285	0.0551	-0.4557	0.431	0.5090	0.9917	840
Gradient boosting	0.4833	0.183	0.0355	-0.5430	0.279	0.4814	0.9940	840
Random forest	0.4893	-0.040	-0.0074	-0.5150	-0.060	0.4921	0.9750	840

EUR/GBP (EUR_GBP) - binary, lead 9d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Logistic (sklearn, on global PCA) | Gradient boosting (sklearn) | MLP neural net | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.5369	0.910	0.2030	-0.4859	1.325	0.5350	0.9857	840
Logistic (sklearn,	0.5179	0.834	0.1847	-0.4556	1.142	0.5173	0.9964	840
Gradient boosting	0.5310	0.780	0.1717	-0.6349	1.052	0.5329	0.9940	840
MLP neural net	0.5095	0.504	0.1023	-0.4785	0.718	0.5154	0.9262	840
Hist gradient boos	0.4964	0.310	0.0649	-0.5582	0.436	0.4976	0.9952	840
XGBoost	0.4988	0.269	0.0562	-0.6392	0.366	0.4988	0.9905	840
Logit (statsmodels	0.5095	0.252	0.0524	-0.6083	0.329	0.5090	0.9893	840
Ensemble (weighted	0.5095	0.064	0.0131	-0.6922	0.083	0.5097	0.9810	840
Probit (statsmodel	0.5048	0.059	0.0121	-0.6292	0.077	0.5048	0.9952	840
Ensemble (mean pro	0.5107	-0.048	-0.0096	-0.7061	-0.061	0.5085	0.9810	840
Random forest	0.4917	-0.781	-0.1464	-0.7499	-0.988	0.4903	0.9810	840

EUR/GBP (EUR_GBP) - binary, lead 10d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | MLP neural net | XGBoost | Gradient boosting (sklearn) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.5226	0.128	0.0269	-0.5444	0.184	0.5242	0.9583	840
MLP neural net	0.5405	0.023	0.0049	-0.7453	0.027	0.5418	0.9405	840
XGBoost	0.4988	-0.026	-0.0057	-0.6240	-0.035	0.5000	0.9929	840
Gradient boosting	0.4988	-0.123	-0.0261	-0.6299	-0.165	0.4994	0.9893	840
Hist gradient boos	0.5060	-0.287	-0.0594	-0.5985	-0.374	0.5072	0.9905	840
Ensemble (mean pro	0.5024	-0.417	-0.0854	-0.6949	-0.540	0.5000	0.9857	840
Ensemble (weighted	0.4952	-0.462	-0.0943	-0.6956	-0.602	0.4958	0.9869	840
Logistic (sklearn,	0.5060	-0.535	-0.1073	-0.7206	-0.691	0.5024	0.9810	840
Probit (statsmodel	0.4976	-0.574	-0.1155	-0.7529	-0.743	0.4988	0.9905	840
Logit (statsmodels	0.4929	-0.719	-0.1429	-0.7613	-0.931	0.4946	0.9917	840
Random forest	0.4810	-1.008	-0.1939	-0.7591	-1.289	0.4848	0.9821	840

EUR/GBP (EUR_GBP) - ternary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | Hist gradient boosting | MNLogit (statsmodels, PCA) | XGBoost | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.8393	0.548	0.0004	0.0000	NA	1.0000	0.0012	840
Hist gradient boos	0.7607	0.341	0.0060	-0.0223	0.212	0.4719	0.1060	840
MNLogit (statsmode	0.7488	0.324	0.0066	-0.0376	0.173	0.5319	0.1119	840
XGBoost	0.8274	0.211	0.0018	-0.0152	0.079	0.4000	0.0179	840
Ensemble (weighted	0.8321	0.154	0.0007	-0.0069	0.025	0.5000	0.0071	840
Ensemble (mean pro	0.8262	0.154	0.0007	-0.0069	0.025	0.5000	0.0071	840
Logistic (sklearn,	0.4524	-0.085	-0.0037	-0.0827	-0.083	0.4872	0.4643	840
Gradient boosting	0.6714	-0.208	-0.0062	-0.0557	-0.135	0.4780	0.2167	840
Random forest	0.8417	NA	0.0000	0.0000	NA	NA	0.0000	840
SVM (RBF)	0.8429	NA	0.0000	0.0000	NA	NA	0.0000	840

EUR/GBP (EUR_GBP) - ternary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | Hist gradient boosting | Ensemble (weighted train acc) | MNLogit (statsmodels, PCA) | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.6774	0.563	0.0078	-0.0245	0.160	0.5000	0.0214	840
Hist gradient boos	0.5536	0.118	0.0056	-0.0882	0.101	0.4575	0.2524	840
Ensemble (weighted	0.6369	0.031	0.0007	-0.0497	0.013	0.4603	0.0750	840
MNLogit (statsmode	0.6417	-0.083	-0.0024	-0.0904	-0.034	0.5122	0.0976	840
Logistic (sklearn,	0.3738	-0.093	-0.0060	-0.1413	-0.094	0.4750	0.4762	840
Ensemble (mean pro	0.6345	-0.196	-0.0039	-0.0516	-0.075	0.4151	0.0631	840
MLP neural net	0.6905	-0.742	-0.0045	-0.0149	-0.089	0.0000	0.0024	840
XGBoost	0.6119	-1.171	-0.0372	-0.1580	-0.641	0.3805	0.1345	840
Gradient boosting	0.5429	-1.842	-0.0980	-0.3234	-1.310	0.4017	0.2786	840
SVM (RBF)	0.6952	NA	0.0000	0.0000	NA	NA	0.0000	840

EUR/GBP (EUR_GBP) - ternary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | Logistic (sklearn, on global PCA) | Gradient boosting (sklearn) | SVM (RBF) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.4738	-0.165	-0.0114	-0.1918	-0.123	0.5139	0.3429	840
Logistic (sklearn,	0.3774	-0.436	-0.0357	-0.2543	-0.376	0.5039	0.4583	840
Gradient boosting	0.4429	-0.437	-0.0326	-0.2130	-0.394	0.4869	0.4083	840
SVM (RBF)	0.5583	-0.545	-0.0138	-0.0664	-0.193	0.4314	0.0607	840
Ensemble (weighted	0.4798	-0.665	-0.0396	-0.1920	-0.427	0.4785	0.2488	840
MLP neural net	0.5976	-0.680	-0.0112	-0.0510	-0.085	0.2857	0.0083	840
Random forest	0.5357	-0.841	-0.0236	-0.1017	-0.383	0.3816	0.0905	840
XGBoost	0.4845	-0.863	-0.0531	-0.2265	-0.629	0.4628	0.2881	840
Ensemble (mean pro	0.4798	-1.077	-0.0608	-0.2315	-0.652	0.4531	0.2286	840
MNLogit (statsmode	0.4702	-1.297	-0.0849	-0.3031	-0.857	0.4609	0.2738	840

EUR/GBP (EUR_GBP) - ternary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | XGBoost | Ensemble (weighted train acc) | Ensemble (mean proba) | Gradient boosting (sklearn)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.4262	0.588	0.0590	-0.1461	0.567	0.5231	0.4643	840
XGBoost	0.4405	0.326	0.0310	-0.2191	0.287	0.5101	0.4131	840
Ensemble (weighted	0.4345	-0.040	-0.0032	-0.1937	-0.031	0.4963	0.3238	840
Ensemble (mean pro	0.4274	-0.061	-0.0049	-0.2032	-0.046	0.4865	0.3083	840
Gradient boosting	0.4024	-0.281	-0.0262	-0.2733	-0.280	0.4822	0.5012	840
SVM (RBF)	0.4798	-0.351	-0.0160	-0.1682	-0.185	0.4381	0.1250	840
Logistic (sklearn,	0.3440	-0.385	-0.0393	-0.3163	-0.440	0.4721	0.5548	840
Random forest	0.4548	-0.592	-0.0280	-0.1531	-0.359	0.4453	0.1631	840
MLP neural net	0.5214	-0.941	-0.0274	-0.0983	-0.181	0.4333	0.0357	840
MNLogit (statsmode	0.4155	-1.326	-0.1124	-0.4073	-1.061	0.4554	0.3607	840

EUR/GBP (EUR_GBP) - ternary, lead 5d

Rows used: 1000 | Effective end: 2021-12-25

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Ensemble (weighted train acc) | Hist gradient boosting | MLP neural net | Random forest

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.4048	0.701	0.0738	-0.2798	0.628	0.5699	0.4429	840
Ensemble (weighted	0.4131	0.317	0.0304	-0.2797	0.264	0.5469	0.3679	840
Hist gradient boos	0.4155	0.285	0.0321	-0.2852	0.248	0.5569	0.4810	840
MLP neural net	0.4583	0.222	0.0078	-0.0735	0.076	0.5938	0.0762	840
Random forest	0.4095	-0.001	-0.0001	-0.1719	-0.001	0.5440	0.2167	840
Ensemble (mean pro	0.4071	-0.035	-0.0033	-0.3060	-0.029	0.5304	0.3524	840
Gradient boosting	0.3833	-0.079	-0.0089	-0.3209	-0.078	0.5347	0.5143	840
Logistic (sklearn,	0.3440	-0.128	-0.0145	-0.3718	-0.144	0.5011	0.5464	840
MNLogit (statsmode	0.3702	-0.226	-0.0244	-0.3416	-0.230	0.5000	0.4571	840
SVM (RBF)	0.4417	-0.793	-0.0484	-0.2714	-0.462	0.4694	0.1750	840

EUR/GBP (EUR_GBP) - ternary, lead 6d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | Hist gradient boosting | Ensemble (weighted train acc) | XGBoost | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.3738	0.912	0.0609	-0.0560	0.710	0.5697	0.1964	840
Hist gradient boos	0.3607	0.791	0.1138	-0.3048	0.795	0.5493	0.5917	840
Ensemble (weighted	0.3560	0.567	0.0609	-0.2236	0.531	0.5258	0.3917	840
XGBoost	0.3595	0.475	0.0548	-0.2319	0.448	0.5342	0.4702	840
Ensemble (mean pro	0.3643	0.469	0.0493	-0.2461	0.429	0.5224	0.3714	840
Logistic (sklearn,	0.3429	0.266	0.0347	-0.3548	0.271	0.5089	0.5357	840
Gradient boosting	0.3690	0.142	0.0165	-0.3472	0.136	0.5182	0.4893	840
SVM (RBF)	0.4024	0.065	0.0041	-0.1788	0.058	0.4615	0.1857	840
MNLogit (statsmode	0.3821	-0.095	-0.0113	-0.3120	-0.087	0.4959	0.4369	840
MLP neural net	0.4607	-0.143	-0.0083	-0.1680	-0.055	0.3571	0.0500	840

EUR/GBP (EUR_GBP) - ternary, lead 7d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Gradient boosting (sklearn) | Ensemble (weighted train acc) | Ensemble (mean proba) | MLP neural net

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.4060	0.499	0.0383	-0.1918	0.423	0.5213	0.2238	840
Gradient boosting	0.3512	0.065	0.0094	-0.4839	0.066	0.5083	0.5714	840
Ensemble (weighted	0.3417	-0.100	-0.0134	-0.4153	-0.088	0.4831	0.4238	840
Ensemble (mean pro	0.3536	-0.105	-0.0141	-0.4026	-0.092	0.4826	0.4095	840
MLP neural net	0.4548	-0.171	-0.0161	-0.3366	-0.064	0.5349	0.1024	840
XGBoost	0.3500	-0.270	-0.0385	-0.4861	-0.264	0.4734	0.5155	840
Logistic (sklearn,	0.3393	-0.296	-0.0451	-0.4661	-0.316	0.4804	0.6369	840
Hist gradient boos	0.3440	-0.384	-0.0575	-0.4321	-0.425	0.4660	0.6131	840
MNLogit (statsmode	0.3738	-0.393	-0.0557	-0.4845	-0.370	0.4725	0.5190	840
Random forest	0.3631	-0.462	-0.0427	-0.2959	-0.405	0.4430	0.2821	840

EUR/GBP (EUR_GBP) - ternary, lead 8d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | Random forest | Logistic (sklearn, on global PCA) | Ensemble (weighted train acc) | Hist gradient boosting

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.3952	0.211	0.0217	-0.4093	0.145	0.4412	0.1619	840
Random forest	0.3250	-0.364	-0.0447	-0.4167	-0.331	0.4726	0.3905	840
Logistic (sklearn,	0.3345	-0.630	-0.1031	-0.6021	-0.699	0.4791	0.6560	840
Ensemble (weighted	0.3393	-0.640	-0.0972	-0.5699	-0.702	0.4685	0.5667	840
Hist gradient boos	0.3167	-0.641	-0.1014	-0.5638	-0.800	0.4590	0.6976	840
XGBoost	0.3298	-0.660	-0.1021	-0.5207	-0.803	0.4607	0.6202	840
Gradient boosting	0.3286	-0.737	-0.1158	-0.5478	-0.771	0.4539	0.6321	840
Ensemble (mean pro	0.3393	-0.740	-0.1099	-0.6020	-0.795	0.4635	0.5548	840
MNLogit (statsmode	0.3429	-0.790	-0.1245	-0.6074	-0.828	0.4781	0.5976	840
SVM (RBF)	0.3774	-0.865	-0.0849	-0.4155	-0.765	0.4427	0.3119	840

EUR/GBP (EUR_GBP) - ternary, lead 9d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | XGBoost | Ensemble (weighted train acc) | Gradient boosting (sklearn) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.3321	0.448	0.0419	-0.2837	0.301	0.5101	0.1774	840
XGBoost	0.3417	0.266	0.0483	-0.5410	0.311	0.4888	0.6893	840
Ensemble (weighted	0.3393	0.183	0.0320	-0.4836	0.202	0.4914	0.6202	840
Gradient boosting	0.3417	0.145	0.0266	-0.4841	0.157	0.5192	0.6833	840
Ensemble (mean pro	0.3440	0.035	0.0060	-0.5081	0.038	0.4814	0.6083	840
Hist gradient boos	0.3119	-0.031	-0.0053	-0.3935	-0.037	0.4690	0.7107	840
MNLogit (statsmode	0.3262	-0.061	-0.0107	-0.5444	-0.068	0.4917	0.6488	840
Random forest	0.3250	-0.155	-0.0215	-0.4990	-0.153	0.4753	0.4333	840
Logistic (sklearn,	0.3310	-0.262	-0.0457	-0.5549	-0.296	0.4811	0.6631	840
SVM (RBF)	0.3726	-0.691	-0.0873	-0.5405	-0.679	0.4714	0.4167	840

EUR/GBP (EUR_GBP) - ternary, lead 10d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | Hist gradient boosting | Ensemble (weighted train acc) | XGBoost | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.3714	0.886	0.1891	-0.5177	0.951	0.5508	0.6917	840
Hist gradient boos	0.3405	0.510	0.1039	-0.5043	0.570	0.5100	0.7119	840
Ensemble (weighted	0.3298	0.332	0.0652	-0.5361	0.351	0.5204	0.6429	840
XGBoost	0.3488	0.279	0.0546	-0.5737	0.306	0.5180	0.6940	840
Logistic (sklearn,	0.3488	0.101	0.0201	-0.6224	0.110	0.5068	0.6976	840
Ensemble (mean pro	0.3250	0.032	0.0061	-0.5863	0.033	0.5142	0.6298	840
MNLogit (statsmode	0.3357	-0.150	-0.0291	-0.6252	-0.164	0.4974	0.6917	840
Random forest	0.3179	-0.351	-0.0612	-0.6210	-0.317	0.5125	0.5250	840
SVM (RBF)	0.3429	-0.880	-0.1207	-0.5760	-0.930	0.4737	0.4976	840
MLP neural net	0.3619	-0.961	-0.0979	-0.4830	-0.619	0.4013	0.1810	840

GBP/JPY (GBP_JPY) - binary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | MLP neural net | SVM (RBF) | Logit (statsmodels, PCA) | Probit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.4988	0.069	0.0058	-0.1446	0.094	0.5031	0.9655	840
MLP neural net	0.4893	0.047	0.0037	-0.1992	0.067	0.4919	0.8833	840
SVM (RBF)	0.4940	-0.117	-0.0094	-0.2191	-0.147	0.5058	0.9202	840
Logit (statsmodels	0.5012	-0.180	-0.0147	-0.1302	-0.244	0.5006	0.9536	840
Probit (statsmodel	0.5000	-0.205	-0.0167	-0.1269	-0.280	0.4994	0.9536	840
Hist gradient boos	0.5060	-0.294	-0.0243	-0.1761	-0.393	0.5078	0.9869	840
Random forest	0.5179	-0.336	-0.0266	-0.1827	-0.423	0.5171	0.9417	840
Ensemble (mean pro	0.4929	-0.464	-0.0378	-0.1868	-0.614	0.4975	0.9714	840
Ensemble (weighted	0.4976	-0.487	-0.0396	-0.1849	-0.643	0.5006	0.9702	840
XGBoost	0.4893	-0.726	-0.0589	-0.2122	-0.965	0.4933	0.9821	840
Gradient boosting	0.5000	-0.936	-0.0753	-0.2579	-1.196	0.4988	0.9833	840

GBP/JPY (GBP_JPY) - binary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> XGBoost | Hist gradient boosting | SVM (RBF) | Random forest | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
XGBoost	0.5190	0.694	0.0871	-0.1814	1.017	0.5211	0.9869	840
Hist gradient boos	0.5214	0.491	0.0609	-0.1623	0.692	0.5236	0.9845	840
SVM (RBF)	0.5107	0.421	0.0512	-0.2638	0.579	0.5094	0.9536	840
Random forest	0.4976	-0.199	-0.0235	-0.2957	-0.265	0.4994	0.9655	840
Ensemble (mean pro	0.5012	-0.223	-0.0268	-0.2683	-0.302	0.5000	0.9762	840
Logistic (sklearn,	0.4893	-0.276	-0.0328	-0.3912	-0.376	0.4932	0.9679	840
Ensemble (weighted	0.5000	-0.307	-0.0364	-0.2835	-0.414	0.4988	0.9810	840
Probit (statsmodel	0.4857	-0.412	-0.0486	-0.3650	-0.565	0.4884	0.9774	840
MLP neural net	0.4893	-0.445	-0.0499	-0.3978	-0.580	0.4955	0.9179	840
Logit (statsmodels	0.4917	-0.457	-0.0540	-0.3498	-0.627	0.4866	0.9786	840
Gradient boosting	0.4881	-0.967	-0.1102	-0.4389	-1.269	0.4880	0.9929	840

GBP/JPY (GBP_JPY) - binary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | SVM (RBF) | Hist gradient boosting | Ensemble (mean proba) | Logistic (sklearn, on global PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.5036	0.148	0.0222	-0.4847	0.212	0.5069	0.9440	840
SVM (RBF)	0.5024	0.112	0.0157	-0.3176	0.156	0.4981	0.9298	840
Hist gradient boos	0.4929	-0.534	-0.0773	-0.3813	-0.762	0.4934	0.9869	840
Ensemble (mean pro	0.4821	-0.727	-0.1035	-0.4928	-1.041	0.4801	0.9869	840
Logistic (sklearn,	0.4738	-0.738	-0.1047	-0.5311	-1.051	0.4775	0.9774	840
Random forest	0.4821	-0.752	-0.1061	-0.4693	-1.073	0.4802	0.9619	840
Ensemble (weighted	0.4821	-0.791	-0.1115	-0.4867	-1.122	0.4800	0.9798	840
Logit (statsmodels	0.4774	-0.818	-0.1160	-0.5557	-1.163	0.4757	0.9810	840
Probit (statsmodel	0.4714	-0.920	-0.1290	-0.5691	-1.305	0.4727	0.9798	840
XGBoost	0.4845	-1.150	-0.1594	-0.5317	-1.558	0.4850	0.9940	840
Gradient boosting	0.4690	-1.156	-0.1602	-0.5013	-1.643	0.4695	0.9964	840

GBP/JPY (GBP_JPY) - binary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | MLP neural net | Gradient boosting (sklearn) | Hist gradient boosting | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.5071	0.134	0.0235	-0.3169	0.186	0.5055	0.9750	840
MLP neural net	0.5000	0.112	0.0193	-0.5658	0.151	0.5013	0.9167	840
Gradient boosting	0.5024	-0.289	-0.0496	-0.4143	-0.402	0.5006	0.9940	840
Hist gradient boos	0.4917	-0.410	-0.0699	-0.4274	-0.578	0.4928	0.9976	840
XGBoost	0.4929	-0.535	-0.0898	-0.4869	-0.737	0.4910	0.9917	840
Random forest	0.4881	-0.699	-0.1153	-0.5547	-0.948	0.4854	0.9810	840
Logit (statsmodels	0.4821	-0.864	-0.1410	-0.6050	-1.187	0.4861	0.9869	840
Probit (statsmodel	0.4774	-1.022	-0.1648	-0.6240	-1.411	0.4795	0.9857	840
Logistic (sklearn,	0.4762	-1.065	-0.1703	-0.6257	-1.481	0.4757	0.9810	840
Ensemble (weighted	0.4798	-1.124	-0.1794	-0.6251	-1.504	0.4806	0.9833	840
Ensemble (mean pro	0.4810	-1.176	-0.1872	-0.6367	-1.583	0.4795	0.9881	840

GBP/JPY (GBP_JPY) - binary, lead 5d

Rows used: 1000 | Effective end: 2021-12-25

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | Probit (statsmodels, PCA) | Logit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.4774	-0.191	-0.0348	-0.6934	-0.281	0.4739	0.8893	840
Probit (statsmodel	0.4881	-0.792	-0.1458	-0.6961	-1.083	0.4874	0.9893	840
Logit (statsmodels	0.4845	-0.822	-0.1508	-0.6935	-1.124	0.4861	0.9869	840
Logistic (sklearn,	0.4833	-0.858	-0.1560	-0.6466	-1.204	0.4807	0.9881	840
Ensemble (weighted	0.4845	-1.051	-0.1884	-0.6720	-1.419	0.4813	0.9893	840
Ensemble (mean pro	0.4821	-1.104	-0.1966	-0.6837	-1.491	0.4789	0.9869	840
Hist gradient boos	0.4738	-1.180	-0.2090	-0.6598	-1.610	0.4761	0.9952	840
SVM (RBF)	0.4881	-1.292	-0.2257	-0.7164	-1.646	0.4855	0.9833	840
Gradient boosting	0.4560	-1.350	-0.2341	-0.7052	-1.919	0.4562	0.9917	840
XGBoost	0.4714	-1.424	-0.2463	-0.7441	-1.904	0.4724	0.9929	840
Random forest	0.4679	-1.440	-0.2471	-0.7656	-1.888	0.4661	0.9833	840

GBP/JPY (GBP_JPY) - binary, lead 6d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | Random forest | SVM (RBF) | Logistic (sklearn, on global PCA) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.5286	0.800	0.1878	-0.4142	1.187	0.5265	0.9226	840
Random forest	0.4952	-0.363	-0.0763	-0.6880	-0.514	0.4939	0.9810	840
SVM (RBF)	0.5024	-0.403	-0.0785	-0.6171	-0.509	0.5032	0.9393	840
Logistic (sklearn,	0.4952	-0.515	-0.1065	-0.6898	-0.702	0.4946	0.9893	840
Ensemble (mean pro	0.4929	-0.604	-0.1232	-0.7099	-0.799	0.4982	0.9798	840
Logit (statsmodels	0.5000	-0.729	-0.1481	-0.7437	-0.966	0.4970	0.9893	840
Probit (statsmodel	0.4988	-0.806	-0.1623	-0.7449	-1.070	0.4958	0.9917	840
Gradient boosting	0.4940	-0.907	-0.1804	-0.7413	-1.207	0.4934	0.9893	840
XGBoost	0.4762	-0.938	-0.1852	-0.7415	-1.281	0.4802	0.9917	840
Ensemble (weighted	0.4845	-0.985	-0.1944	-0.7495	-1.314	0.4850	0.9893	840
Hist gradient boos	0.4702	-1.016	-0.1999	-0.6917	-1.396	0.4688	0.9929	840

GBP/JPY (GBP_JPY) - binary, lead 7d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | SVM (RBF) | Logistic (sklearn, on global PCA) | XGBoost | Logit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.5214	0.343	0.0856	-0.6345	0.488	0.5225	0.9798	840
SVM (RBF)	0.5095	-0.171	-0.0400	-0.5449	-0.230	0.5109	0.9786	840
Logistic (sklearn,	0.5048	-0.279	-0.0647	-0.7436	-0.379	0.5030	0.9917	840
XGBoost	0.5167	-0.353	-0.0807	-0.7488	-0.471	0.5157	0.9857	840
Logit (statsmodels	0.5036	-0.412	-0.0936	-0.7532	-0.549	0.5024	0.9905	840
Probit (statsmodel	0.5000	-0.439	-0.1000	-0.7581	-0.586	0.5018	0.9917	840
Gradient boosting	0.4988	-0.445	-0.1009	-0.7278	-0.591	0.5018	0.9869	840
Ensemble (mean pro	0.5107	-0.514	-0.1138	-0.7446	-0.674	0.5111	0.9690	840
Ensemble (weighted	0.5107	-0.515	-0.1155	-0.7331	-0.682	0.5115	0.9798	840
MLP neural net	0.5155	-0.647	-0.1404	-0.8105	-0.801	0.5189	0.9452	840
Hist gradient boos	0.5012	-0.688	-0.1520	-0.7406	-0.941	0.5030	0.9964	840

GBP/JPY (GBP_JPY) - binary, lead 8d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | SVM (RBF) | Gradient boosting (sklearn) | Ensemble (weighted train acc) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.5083	0.803	0.2273	-0.5649	1.189	0.5097	0.9810	840
SVM (RBF)	0.5262	0.484	0.1321	-0.4840	0.667	0.5271	0.9869	840
Gradient boosting	0.5155	0.228	0.0606	-0.6654	0.310	0.5149	0.9964	840
Ensemble (weighted	0.5190	0.188	0.0497	-0.7458	0.253	0.5174	0.9893	840
Ensemble (mean pro	0.5167	0.170	0.0445	-0.7470	0.227	0.5175	0.9845	840
XGBoost	0.5167	0.085	0.0221	-0.7797	0.111	0.5168	0.9905	840
Logistic (sklearn,	0.5107	0.051	0.0131	-0.7303	0.070	0.5108	0.9929	840
Hist gradient boos	0.5083	-0.046	-0.0117	-0.6652	-0.062	0.5072	0.9952	840
Logit (statsmodels	0.5155	-0.158	-0.0399	-0.8146	-0.204	0.5144	0.9929	840
Probit (statsmodel	0.5167	-0.194	-0.0485	-0.8177	-0.250	0.5157	0.9881	840
MLP neural net	0.5143	-0.250	-0.0604	-0.8387	-0.307	0.5203	0.9381	840

GBP/JPY (GBP_JPY) - binary, lead 9d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Random forest | Gradient boosting (sklearn) | MLP neural net | SVM (RBF) | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Random forest	0.5167	0.200	0.0561	-0.7622	0.266	0.5169	0.9881	840
Gradient boosting	0.5179	0.116	0.0322	-0.5921	0.156	0.5174	0.9940	840
MLP neural net	0.5095	0.091	0.0248	-0.8405	0.114	0.5132	0.9464	840
SVM (RBF)	0.5143	0.083	0.0228	-0.5198	0.115	0.5120	0.9905	840
XGBoost	0.4881	-0.655	-0.1638	-0.8260	-0.878	0.4910	0.9917	840
Ensemble (mean pro	0.4845	-0.850	-0.2060	-0.8237	-1.128	0.4891	0.9833	840
Ensemble (weighted	0.4893	-0.861	-0.2100	-0.8304	-1.150	0.4880	0.9881	840
Logistic (sklearn,	0.4655	-0.988	-0.2372	-0.8013	-1.437	0.4659	0.9940	840
Logit (statsmodels	0.4762	-1.197	-0.2791	-0.8357	-1.631	0.4773	0.9976	840
Hist gradient boos	0.4762	-1.401	-0.3186	-0.8637	-1.850	0.4756	0.9988	840
Probit (statsmodel	0.4690	-1.413	-0.3182	-0.8581	-1.924	0.4669	0.9893	840

GBP/JPY (GBP_JPY) - binary, lead 10d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | MLP neural net | Logistic (sklearn, on global PCA) | Probit (statsmodels, PCA) | Logit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.5095	0.269	0.0812	-0.5073	0.398	0.5101	0.9988	840
MLP neural net	0.5155	0.091	0.0263	-0.8540	0.115	0.5140	0.9333	840
Logistic (sklearn,	0.4810	-0.237	-0.0663	-0.8103	-0.348	0.4831	0.9857	840
Probit (statsmodel	0.4905	-0.627	-0.1632	-0.8420	-0.847	0.4892	0.9881	840
Logit (statsmodels	0.4881	-0.697	-0.1807	-0.8431	-0.945	0.4855	0.9881	840
Gradient boosting	0.5179	-1.024	-0.2547	-0.8842	-1.277	0.5156	0.9929	840
Random forest	0.4762	-1.265	-0.3001	-0.9010	-1.616	0.4749	0.9726	840
Ensemble (mean pro	0.4738	-1.367	-0.3266	-0.9041	-1.800	0.4731	0.9940	840
Ensemble (weighted	0.4750	-1.409	-0.3336	-0.9090	-1.846	0.4723	0.9881	840
Hist gradient boos	0.4655	-2.001	-0.4379	-0.9539	-2.584	0.4659	0.9940	840
XGBoost	0.4464	-2.157	-0.4619	-0.9484	-2.801	0.4475	0.9869	840

GBP/JPY (GBP_JPY) - ternary, lead 1d

Rows used: 1000 | Effective end: 2021-12-29

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | MLP neural net | XGBoost | Gradient boosting (sklearn) | MNLogit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.5929	0.050	0.0026	-0.0872	0.036	0.5000	0.2762	840
MLP neural net	0.7381	-0.033	-0.0002	-0.0062	NA	0.5000	0.0024	840
XGBoost	0.6440	-0.277	-0.0112	-0.0662	-0.147	0.4839	0.1476	840
Gradient boosting	0.5905	-0.473	-0.0269	-0.1399	-0.357	0.4760	0.2976	840
MNLogit (statsmode	0.6833	-0.795	-0.0303	-0.1126	-0.310	0.4471	0.1012	840
Ensemble (weighted	0.6833	-0.809	-0.0213	-0.0957	-0.286	0.4394	0.0786	840
Ensemble (mean pro	0.6964	-1.046	-0.0231	-0.0799	-0.322	0.3800	0.0595	840
Logistic (sklearn,	0.3214	-1.224	-0.0727	-0.2599	-1.081	0.4668	0.4667	840
Random forest	0.7262	NA	0.0000	0.0000	NA	NA	0.0000	840
SVM (RBF)	0.7369	NA	0.0000	0.0000	NA	NA	0.0000	840

GBP/JPY (GBP_JPY) - ternary, lead 2d

Rows used: 1000 | Effective end: 2021-12-28

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Hist gradient boosting | Gradient boosting (sklearn) | XGBoost | SVM (RBF) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Hist gradient boos	0.4167	0.222	0.0173	-0.1793	0.187	0.5287	0.4143	840
Gradient boosting	0.4274	-0.297	-0.0269	-0.2337	-0.285	0.4654	0.4476	840
XGBoost	0.4345	-0.414	-0.0267	-0.1572	-0.278	0.4896	0.2869	840
SVM (RBF)	0.5512	-0.517	-0.0050	-0.0283	-0.107	0.4444	0.0214	840
Ensemble (mean pro	0.4536	-0.668	-0.0361	-0.1461	-0.333	0.4744	0.1857	840
MLP neural net	0.5726	-0.729	-0.0018	-0.0061	-0.056	0.0000	0.0036	840
Random forest	0.5012	-0.754	-0.0325	-0.1370	-0.239	0.4328	0.0798	840
Ensemble (weighted	0.4583	-0.762	-0.0429	-0.1664	-0.416	0.4503	0.2036	840
Logistic (sklearn,	0.3310	-0.880	-0.0671	-0.3209	-0.729	0.4731	0.4429	840
MNLogit (statsmode	0.4512	-1.334	-0.0623	-0.2313	-0.708	0.4204	0.1869	840

GBP/JPY (GBP_JPY) - ternary, lead 3d

Rows used: 1000 | Effective end: 2021-12-27

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | MNLogit (statsmodels, PCA) | Logistic (sklearn, on global PCA) | Hist gradient boosting | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.4036	0.458	0.0110	-0.0225	0.157	0.6207	0.0345	840
MNLogit (statsmode	0.3655	-0.520	-0.0507	-0.3895	-0.394	0.5034	0.3548	840
Logistic (sklearn,	0.3310	-0.544	-0.0584	-0.4139	-0.494	0.5024	0.4976	840
Hist gradient boos	0.3631	-0.827	-0.0879	-0.3211	-0.847	0.4682	0.5060	840
XGBoost	0.3607	-0.944	-0.0966	-0.3818	-0.859	0.4690	0.4417	840
Random forest	0.3845	-1.203	-0.0945	-0.3285	-0.791	0.4450	0.2381	840
SVM (RBF)	0.3798	-1.206	-0.0603	-0.2776	-0.637	0.4655	0.1381	840
Ensemble (mean pro	0.3607	-1.331	-0.1136	-0.4058	-1.044	0.4440	0.3298	840
Ensemble (weighted	0.3750	-1.384	-0.1211	-0.4029	-1.116	0.4512	0.3536	840
Gradient boosting	0.3702	-1.512	-0.1675	-0.5146	-1.479	0.4565	0.5607	840

GBP/JPY (GBP_JPY) - ternary, lead 4d

Rows used: 1000 | Effective end: 2021-12-26

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | SVM (RBF) | Hist gradient boosting | XGBoost | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.3548	0.029	0.0011	-0.0804	0.014	0.4643	0.0667	840
SVM (RBF)	0.3417	-0.477	-0.0429	-0.3034	-0.279	0.5049	0.2429	840
Hist gradient boos	0.3298	-0.585	-0.0766	-0.3914	-0.652	0.4698	0.5905	840
XGBoost	0.3536	-0.653	-0.0824	-0.4366	-0.650	0.4843	0.5310	840
Ensemble (weighted	0.3369	-0.694	-0.0796	-0.4088	-0.606	0.4933	0.4464	840
Random forest	0.3429	-0.773	-0.0713	-0.3322	-0.590	0.4839	0.2952	840
Ensemble (mean pro	0.3417	-0.848	-0.0974	-0.4273	-0.726	0.4863	0.4333	840
Logistic (sklearn,	0.3048	-1.050	-0.1271	-0.5381	-1.017	0.4790	0.5667	840
MNLogit (statsmode	0.3060	-1.109	-0.1294	-0.5619	-0.981	0.4833	0.4976	840
Gradient boosting	0.3417	-1.205	-0.1603	-0.4823	-1.245	0.4572	0.5702	840

GBP/JPY (GBP_JPY) - ternary, lead 5d

Rows used: 1000 | Effective end: 2021-12-25

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> MLP neural net | SVM (RBF) | Gradient boosting (sklearn) | Ensemble (weighted train acc) | Ensemble (mean proba)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
MLP neural net	0.3595	0.560	0.0542	-0.2992	0.371	0.5169	0.2119	840
SVM (RBF)	0.3702	-0.560	-0.0697	-0.3690	-0.455	0.5000	0.4310	840
Gradient boosting	0.3369	-0.801	-0.1217	-0.6186	-0.813	0.4960	0.6024	840
Ensemble (weighted	0.3214	-0.993	-0.1410	-0.6394	-0.950	0.4905	0.5631	840
Ensemble (mean pro	0.3202	-1.021	-0.1452	-0.6423	-0.958	0.4936	0.5548	840
MNLogit (statsmode	0.3321	-1.079	-0.1654	-0.6655	-1.106	0.4917	0.6417	840
Hist gradient boos	0.3095	-1.132	-0.1701	-0.6520	-1.251	0.4750	0.6893	840
XGBoost	0.3060	-1.166	-0.1757	-0.6961	-1.179	0.4955	0.6655	840
Logistic (sklearn,	0.3190	-1.213	-0.1781	-0.6377	-1.229	0.4875	0.6202	840
Random forest	0.3333	-1.521	-0.1874	-0.6454	-1.260	0.4629	0.4167	840

GBP/JPY (GBP_JPY) - ternary, lead 6d

Rows used: 1000 | Effective end: 2021-12-23

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Logistic (sklearn, on global PCA) | MLP neural net | Hist gradient boosting | MNLogit (statsmodels, PCA) | XGBoost

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Logistic (sklearn,	0.3429	-0.435	-0.0806	-0.5805	-0.473	0.5144	0.6619	840
MLP neural net	0.3417	-0.682	-0.0883	-0.5031	-0.516	0.4757	0.3179	840
Hist gradient boos	0.3190	-0.787	-0.1407	-0.5940	-0.893	0.4906	0.6964	840
MNLogit (statsmode	0.3286	-0.798	-0.1444	-0.6196	-0.894	0.5000	0.7000	840
XGBoost	0.3357	-0.874	-0.1504	-0.5759	-1.019	0.4732	0.6667	840
SVM (RBF)	0.3488	-0.916	-0.1375	-0.5084	-0.855	0.5000	0.5571	840
Ensemble (mean pro	0.3357	-1.050	-0.1738	-0.6283	-1.087	0.4951	0.6083	840
Ensemble (weighted	0.3298	-1.126	-0.1861	-0.6447	-1.183	0.4894	0.6202	840
Random forest	0.3310	-1.389	-0.1915	-0.5765	-1.203	0.4792	0.4869	840
Gradient boosting	0.3214	-1.873	-0.3045	-0.7815	-2.075	0.4619	0.7036	840

GBP/JPY (GBP_JPY) - ternary, lead 7d

Rows used: 1000 | Effective end: 2021-12-22

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | XGBoost | Logistic (sklearn, on global PCA) | Random forest | Ensemble (weighted train acc)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.3690	0.777	0.1859	-0.5497	0.966	0.5247	0.7714	840
XGBoost	0.3524	-0.263	-0.0549	-0.6512	-0.309	0.5121	0.7393	840
Logistic (sklearn,	0.3393	-0.349	-0.0728	-0.6044	-0.421	0.4856	0.7012	840
Random forest	0.3500	-0.371	-0.0603	-0.4697	-0.384	0.4944	0.5345	840
Ensemble (weighted	0.3548	-0.386	-0.0778	-0.6692	-0.412	0.5246	0.6536	840
SVM (RBF)	0.3488	-0.423	-0.0744	-0.5165	-0.440	0.5087	0.6179	840
Hist gradient boos	0.3619	-0.426	-0.0897	-0.6803	-0.495	0.5235	0.7595	840
MNLogit (statsmode	0.3429	-0.430	-0.0902	-0.7037	-0.502	0.4933	0.7119	840
MLP neural net	0.3762	-0.460	-0.0695	-0.5405	-0.371	0.4691	0.3857	840
Ensemble (mean pro	0.3536	-0.482	-0.0966	-0.6957	-0.507	0.5243	0.6381	840

GBP/JPY (GBP_JPY) - ternary, lead 8d

Rows used: 1000 | Effective end: 2021-12-21

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> Gradient boosting (sklearn) | XGBoost | Logistic (sklearn, on global PCA) | Hist gradient boosting | MNLogit (statsmodels, PCA)

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
Gradient boosting	0.3857	1.336	0.3729	-0.4629	1.797	0.5393	0.7881	840
XGBoost	0.3821	0.530	0.1287	-0.5553	0.668	0.5195	0.7929	840
Logistic (sklearn,	0.3440	0.025	0.0059	-0.6944	0.031	0.5083	0.7143	840
Hist gradient boos	0.3917	0.015	0.0036	-0.7265	0.018	0.5449	0.8214	840
MNLogit (statsmode	0.3690	-0.025	-0.0060	-0.7003	-0.031	0.5168	0.7440	840
SVM (RBF)	0.3821	-0.098	-0.0191	-0.4872	-0.104	0.5295	0.6452	840
Ensemble (weighted	0.3821	-0.127	-0.0286	-0.6516	-0.142	0.5248	0.6964	840
Ensemble (mean pro	0.3762	-0.128	-0.0286	-0.6396	-0.143	0.5242	0.6881	840
MLP neural net	0.3571	-0.382	-0.0614	-0.5984	-0.294	0.5000	0.3976	840
Random forest	0.4012	-0.536	-0.0948	-0.5691	-0.533	0.4958	0.5714	840

GBP/JPY (GBP_JPY) - ternary, lead 9d

Rows used: 1000 | Effective end: 2021-12-20

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Logistic (sklearn, on global PCA) | MNLogit (statsmodels, PCA) | Gradient boosting (sklearn) | Random forest

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.3667	0.364	0.0770	-0.5465	0.392	0.5379	0.6286	840
Logistic (sklearn,	0.3298	0.105	0.0263	-0.6903	0.127	0.5090	0.7250	840
MNLogit (statsmode	0.3571	-0.015	-0.0038	-0.6818	-0.018	0.5083	0.7869	840
Gradient boosting	0.3488	-0.027	-0.0072	-0.7489	-0.033	0.5262	0.8869	840
Random forest	0.3583	-0.504	-0.1022	-0.6393	-0.487	0.5153	0.5821	840
XGBoost	0.3393	-0.807	-0.1859	-0.7807	-0.946	0.5000	0.8381	840
Ensemble (weighted	0.3452	-0.849	-0.1915	-0.7502	-0.961	0.5008	0.7774	840
Ensemble (mean pro	0.3488	-0.901	-0.1987	-0.7509	-1.007	0.4976	0.7560	840
MLP neural net	0.3298	-1.014	-0.1614	-0.7371	-0.771	0.4793	0.4024	840
Hist gradient boos	0.3345	-1.390	-0.3012	-0.8541	-1.648	0.4832	0.8524	840

GBP/JPY (GBP_JPY) - ternary, lead 10d

Rows used: 1000 | Effective end: 2021-12-19

WF folds: 14 | Train / test per window: 140 / 60

Best strategies (top 5 by OOS Sharpe, threshold backtest)

> SVM (RBF) | Gradient boosting (sklearn) | Logistic (sklearn, on global PCA) | Ensemble (mean proba) | MLP neural net

Model	Pooled acc	Sharpe	AnnRet	MaxDD	Sortino	HitRt	%Traded	n_OOS
SVM (RBF)	0.3750	0.761	0.1723	-0.5811	0.826	0.5346	0.5857	840
Gradient boosting	0.3488	-0.157	-0.0406	-0.7650	-0.181	0.5250	0.7869	840
Logistic (sklearn,	0.3476	-0.229	-0.0576	-0.7746	-0.278	0.4974	0.6940	840
Ensemble (mean pro	0.3476	-0.267	-0.0651	-0.7093	-0.294	0.5145	0.6988	840
MLP neural net	0.3452	-0.395	-0.0681	-0.7054	-0.318	0.5130	0.4131	840
Ensemble (weighted	0.3464	-0.409	-0.0969	-0.7450	-0.450	0.5092	0.7155	840
MNLogit (statsmode	0.3429	-0.493	-0.1207	-0.7647	-0.571	0.5024	0.7369	840
Random forest	0.3429	-0.726	-0.1468	-0.7429	-0.665	0.5053	0.5607	840
XGBoost	0.3464	-1.212	-0.2734	-0.8740	-1.452	0.4645	0.8226	840
Hist gradient boos	0.3048	-1.642	-0.3525	-0.8948	-2.008	0.4516	0.8488	840